CE 208
Quantity Surveying



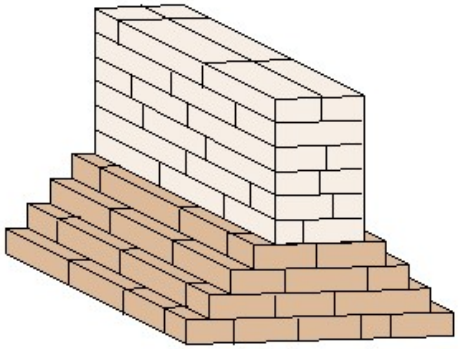
Estimation of a Residential Building

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Mashiat Hossain
Sumaiya Afrose Suma

DEPARTMENT OF CIVIL ENGINEERING
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Brick Wall Footing

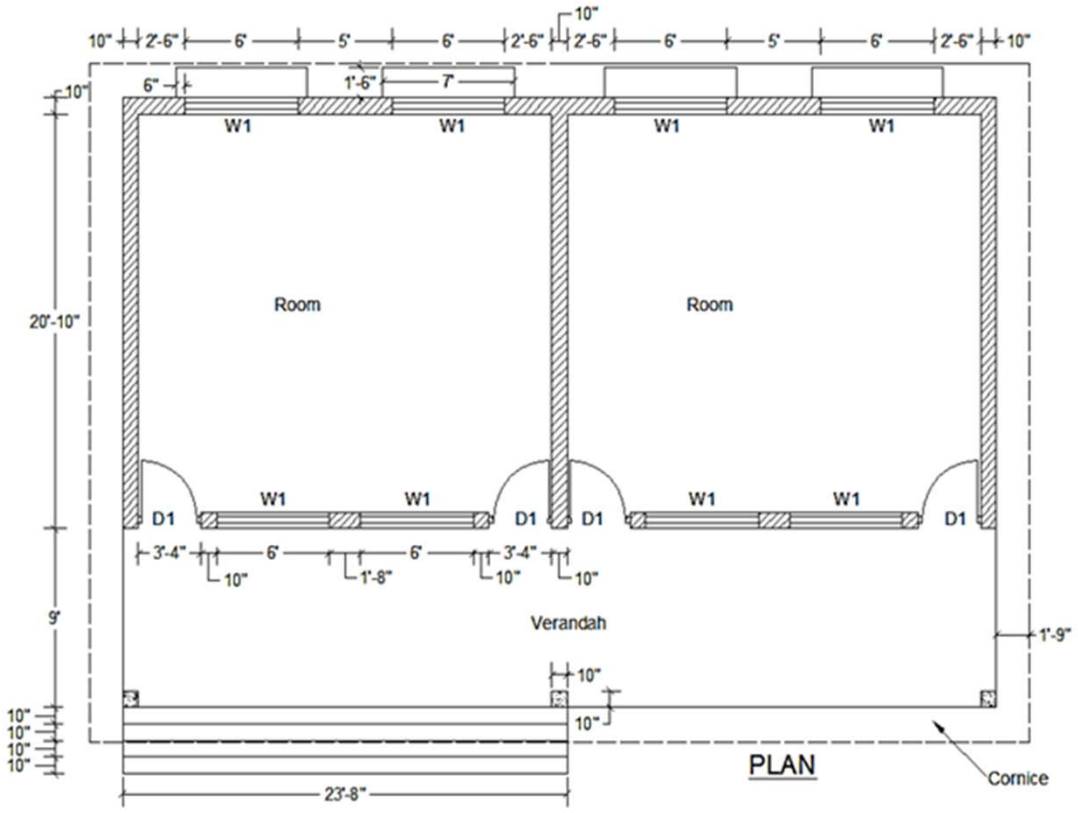
- Wall footing runs along the direction of the wall.
- The size of the footing and the thickness of the foundation wall are specified on the basis of the type of soil at the site.
- The wall footing can be constructed from stone, brick, plain concrete, or reinforced concrete.
- In the case of brick walls, the footing consists of several courses of bricks, the lowest course being usually twice the breadth of the wall above.
- The increased base width of the wall footing is obtained through the provision of 2.5 inch offsets on either side of the wall.



brick or masonry foundation or strip footing

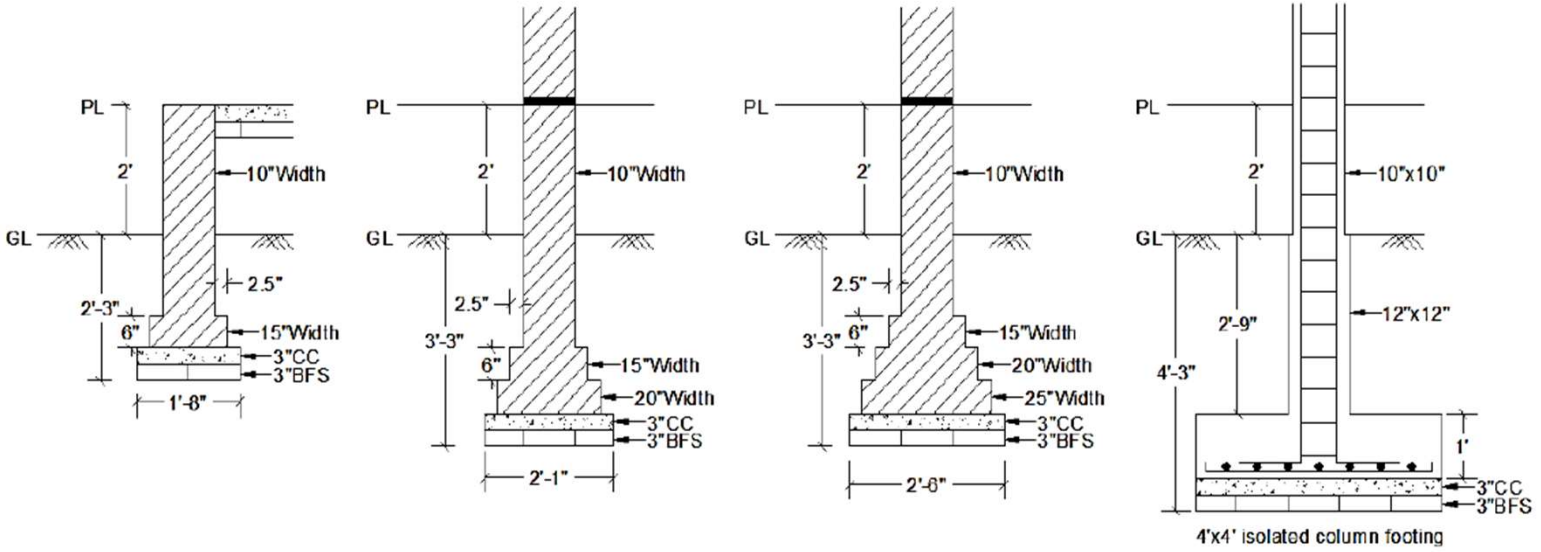


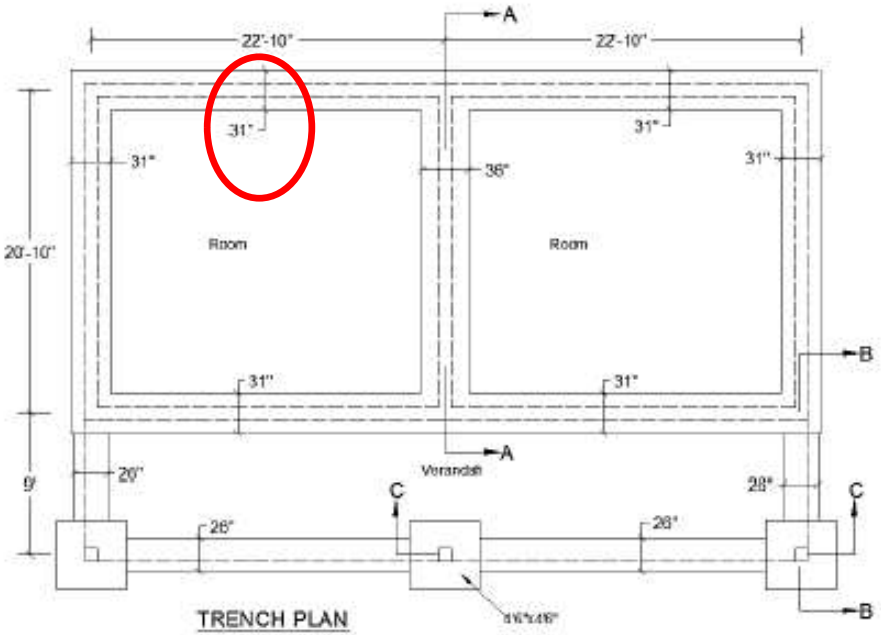
- Estimate the materials required for the following residential building (elementary information is given)



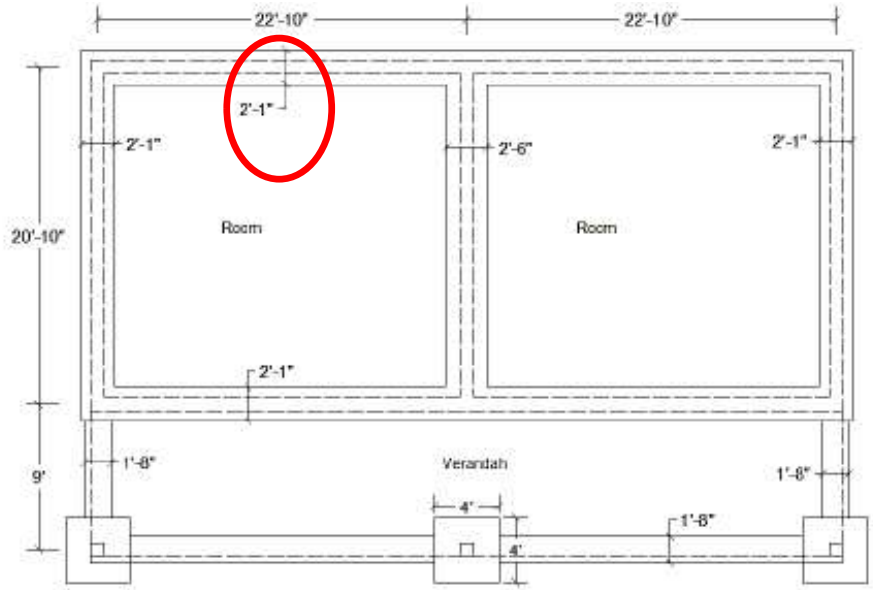
The four types of foundation used and their cross-sections are shown below -

1. 30” wall foundation
2. 25” wall foundation
3. 20” wall foundation
4. Column foundation (Isolated Column Footing)





TRENCH PLAN



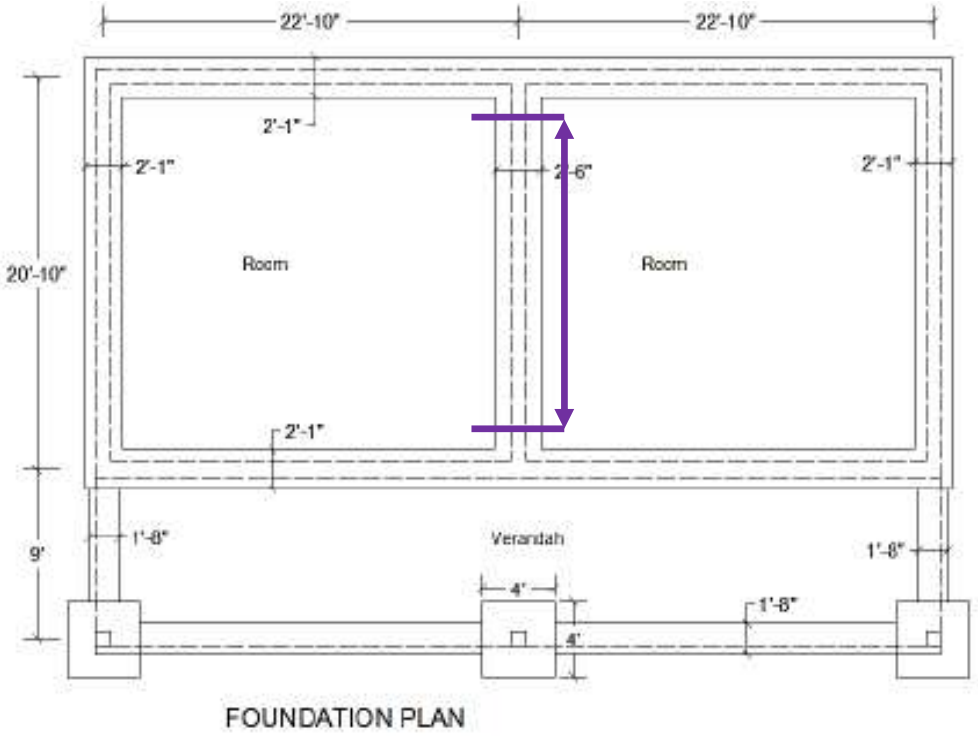
FOUNDATION PLAN



Extra width for excavation

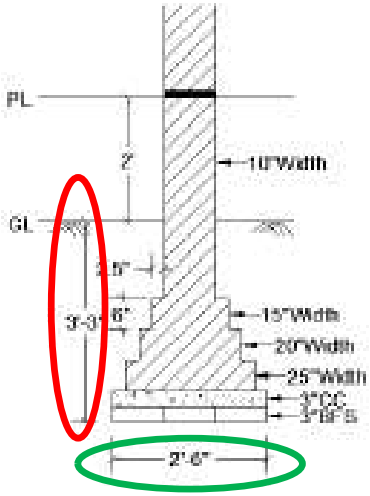
For excavation dimensions, the actual width of foundation is offsetted by **3 inch** on **all sides** (compare the foundation plan and trench plan). If trench plan is given, you can obtain the length of excavation directly from trench plan.

1. Estimation of Earthwork Excavation



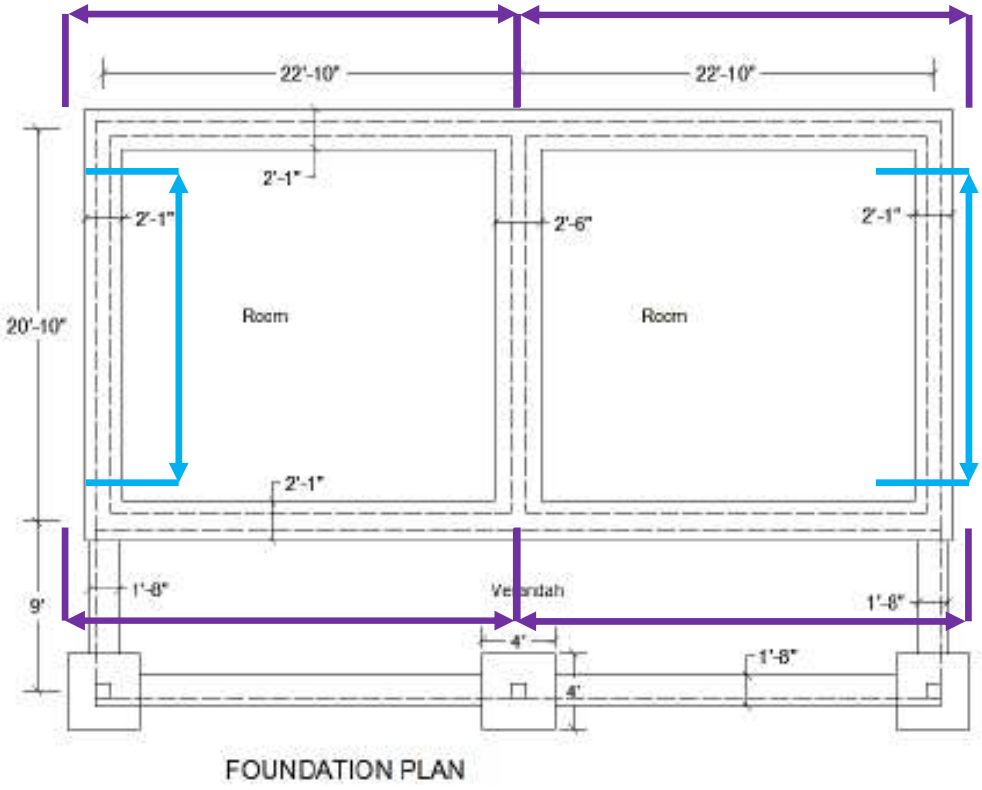
FOUNDATION PLAN

30" wall foundation:



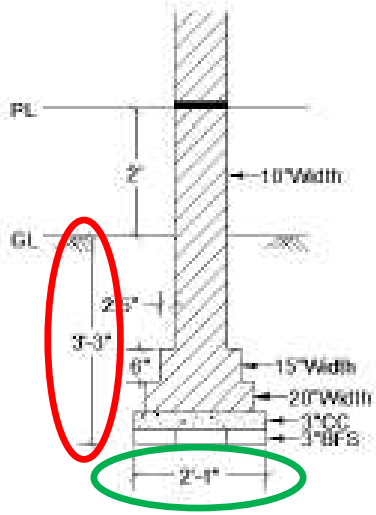
- Height= 3'3"
- Width= (2'6")+3"+3"
- Length= 20'10"-12.5"-3"-12.5"-3"

1. Estimation of Earthwork Excavation



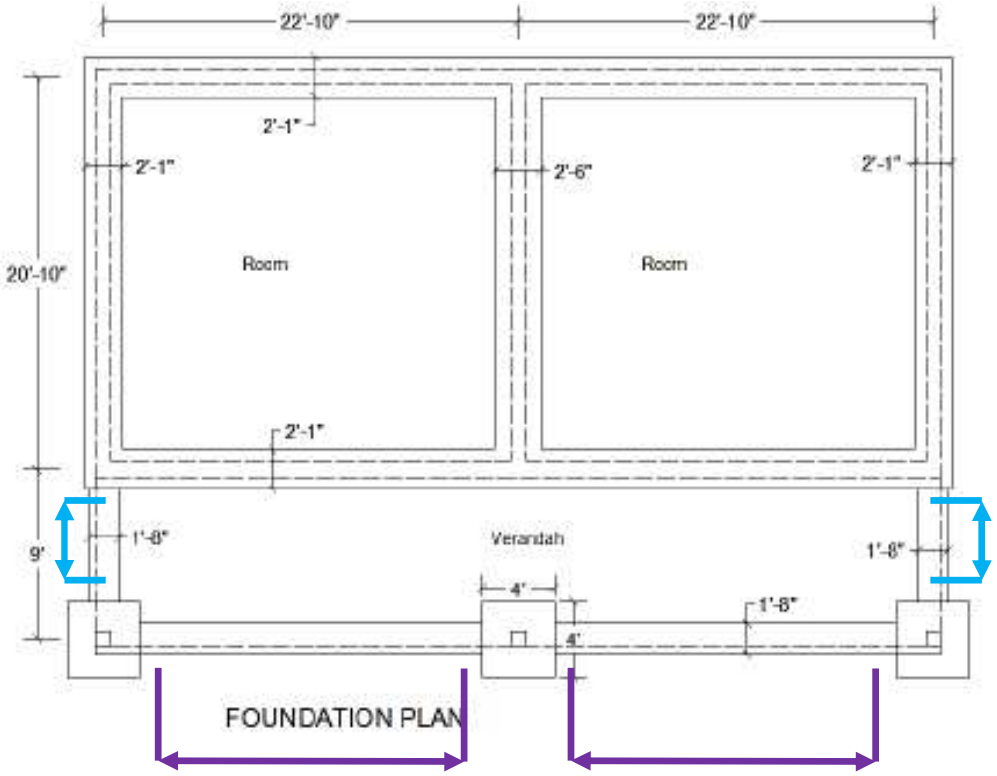
FOUNDATION PLAN

25" wall foundation:

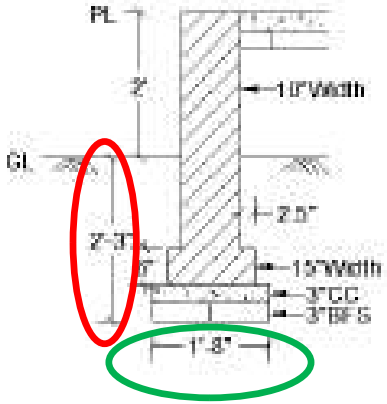


- Height= 3'3"
- Width= (2'1")+3"+3"
- Length=((22'10"+12.5"+3")*4) +((20'10"-25"-6")*2)

1. Estimation of Earthwork Excavation

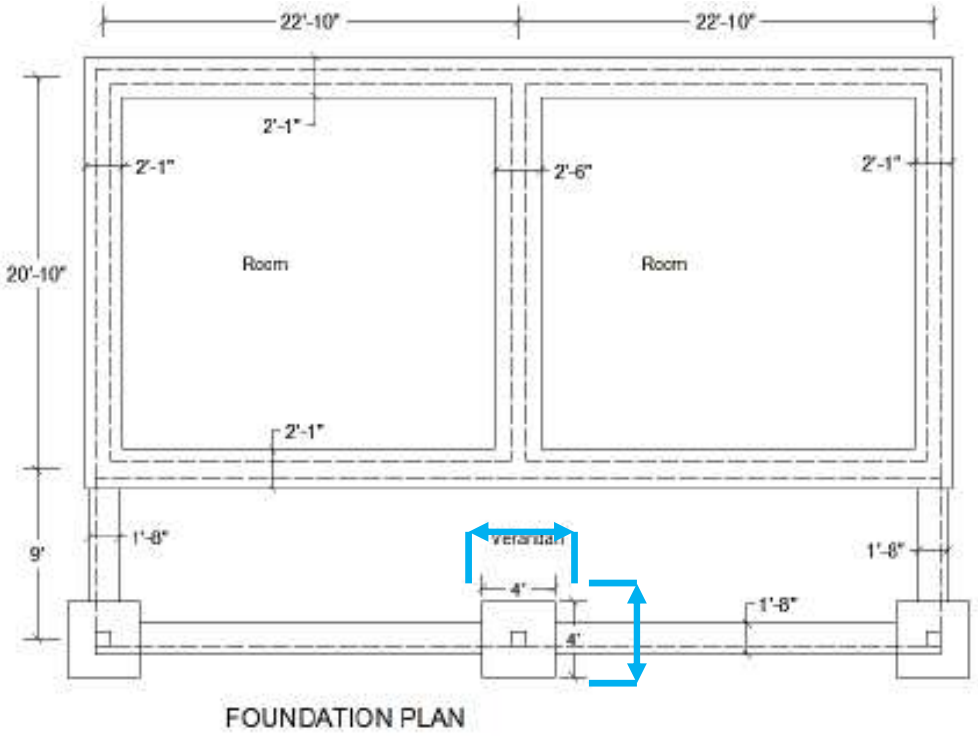


20" wall foundation:



- Height= 2'3"
- Width= (1'8")+3"+3"
- Length=((9'-2'-3"-12.5"-3")*2) + ((22'10"-4'-6")*2)

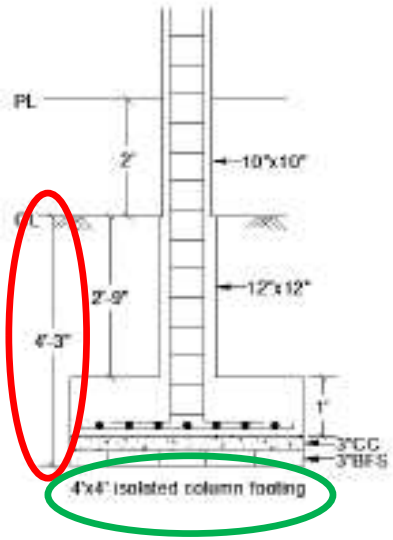
1. Estimation of Earthwork Excavation



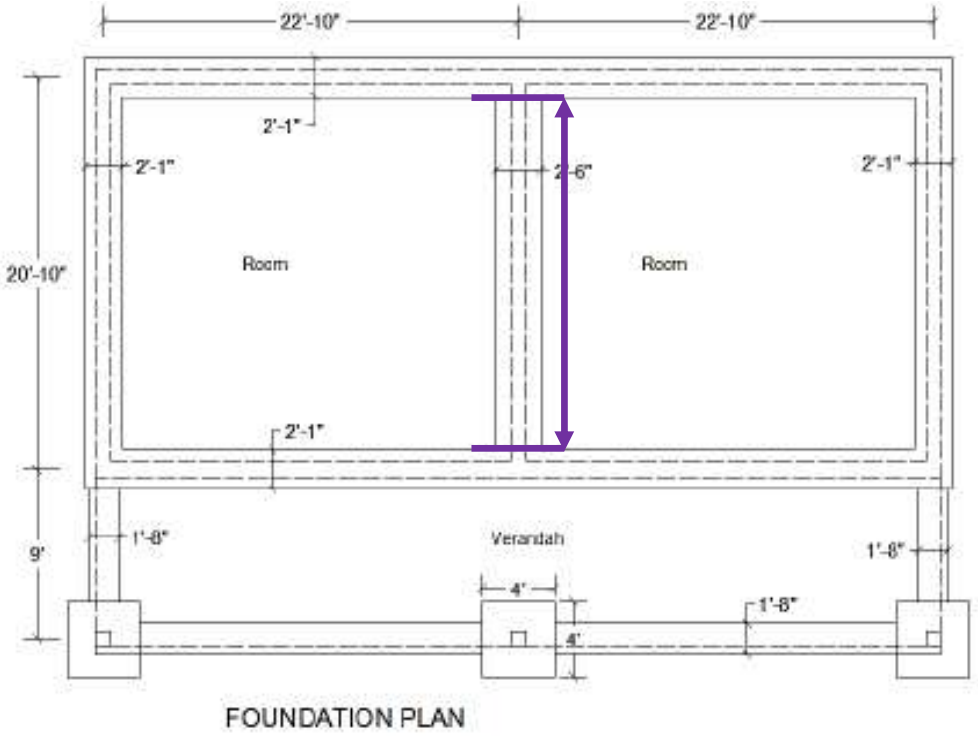
FOUNDATION PLAN

Column foundation:

- Height= 4’3”
- Width= 4’+3”+3”
- Length= 4’+3”+3”



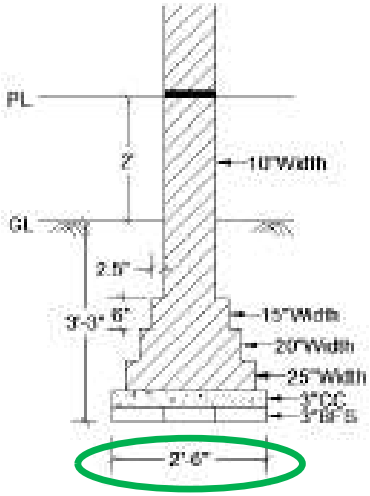
2. Estimation of Brick Flat Soling (one layer)



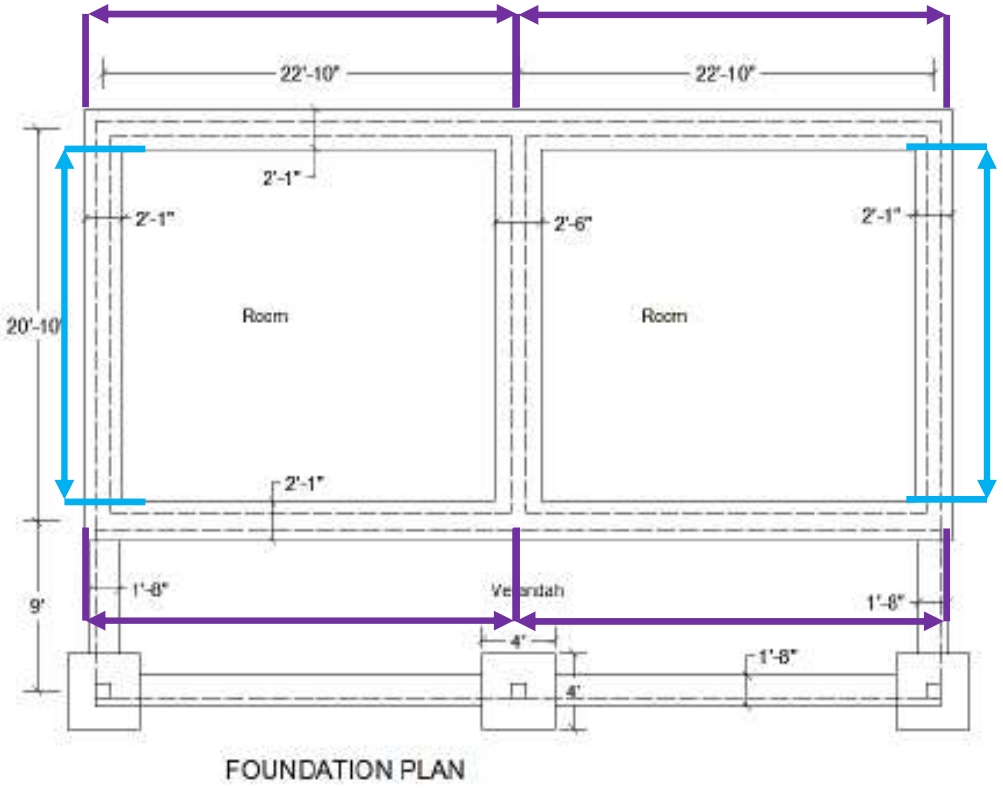
FOUNDATION PLAN

30" wall foundation:

- Width= 30"
- Length= 20'10"-25"



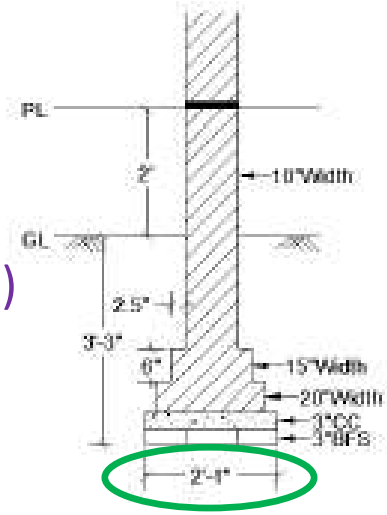
2. Estimation of Brick Flat Soling (one layer)



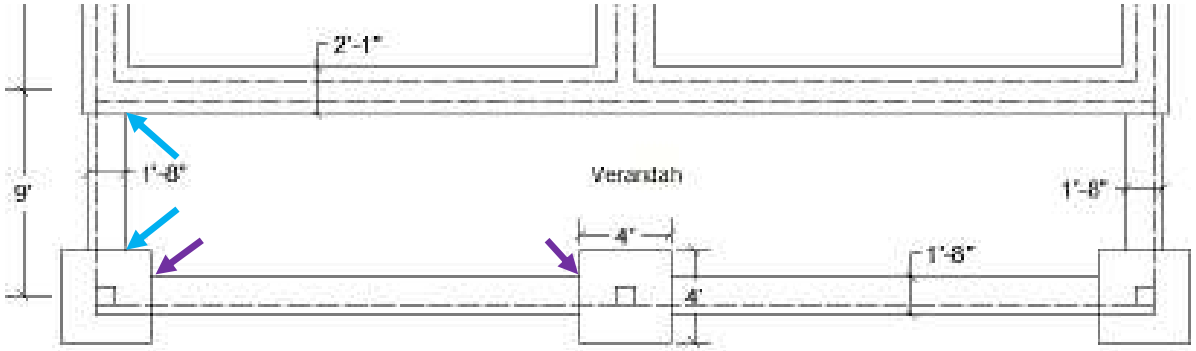
FOUNDATION PLAN

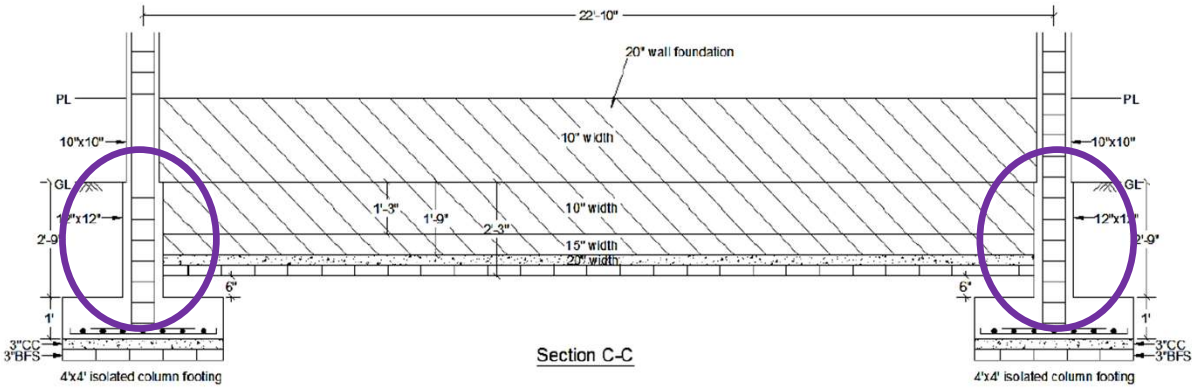
25" wall foundation:

- Width= 25"
- Length=((22'10"+12.5") *4)+((20'10"-25")*2)

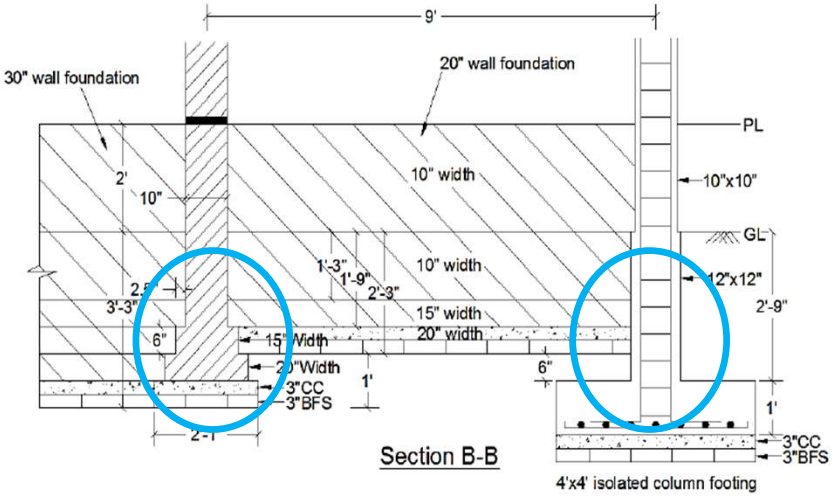


2. Estimation of Brick Flat Soling (one layer)



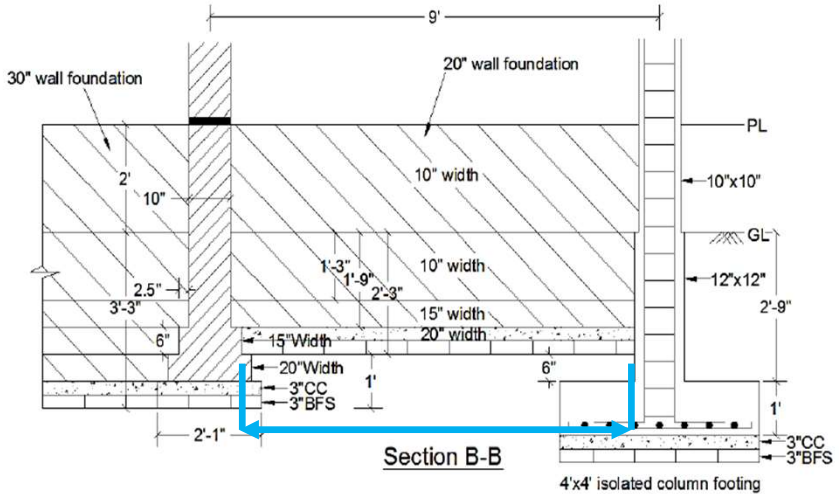


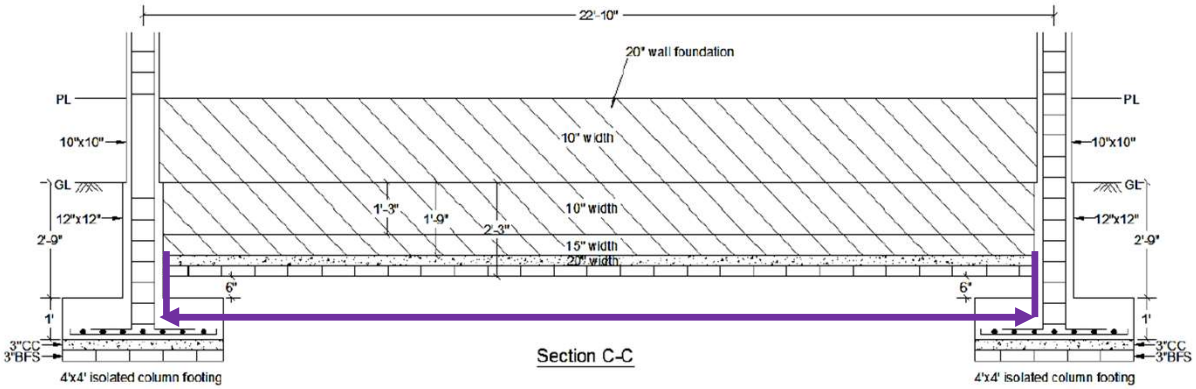
Section C-C



Section B-B

2. Estimation of Brick Flat Soling (one layer)

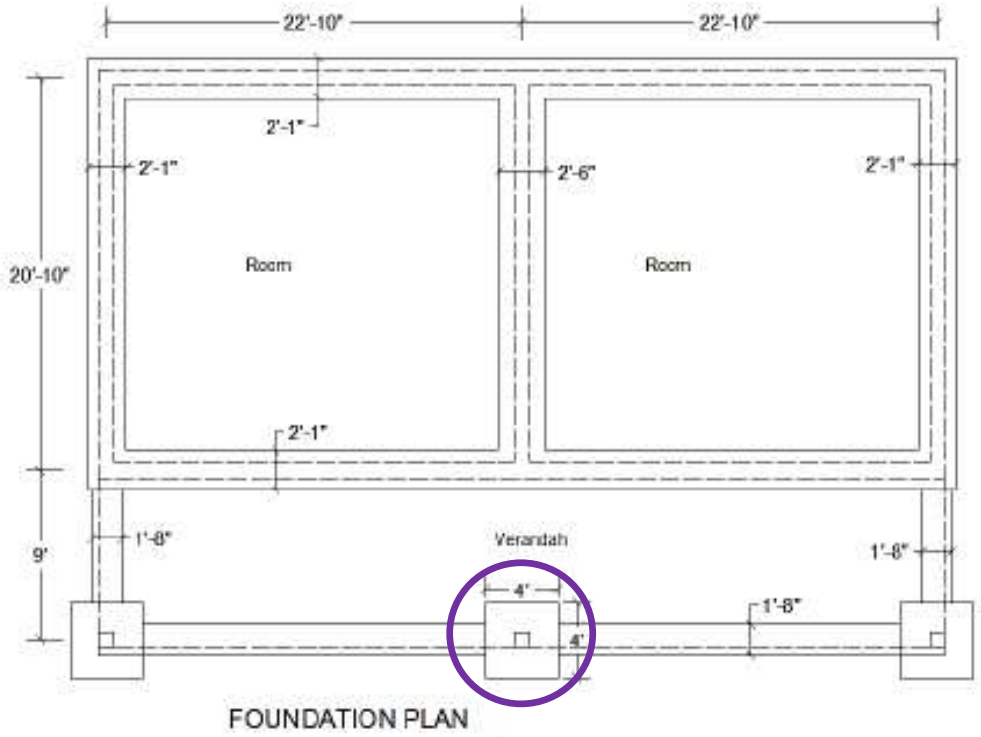




20” wall foundation:

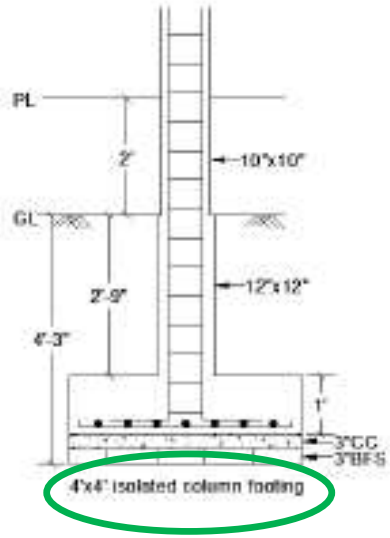
- Width= 20”
- Length=((9’-6”-7.5”)*2) + ((22’10”-12”)*2)

2. Estimation of Brick Flat Soling (one layer)

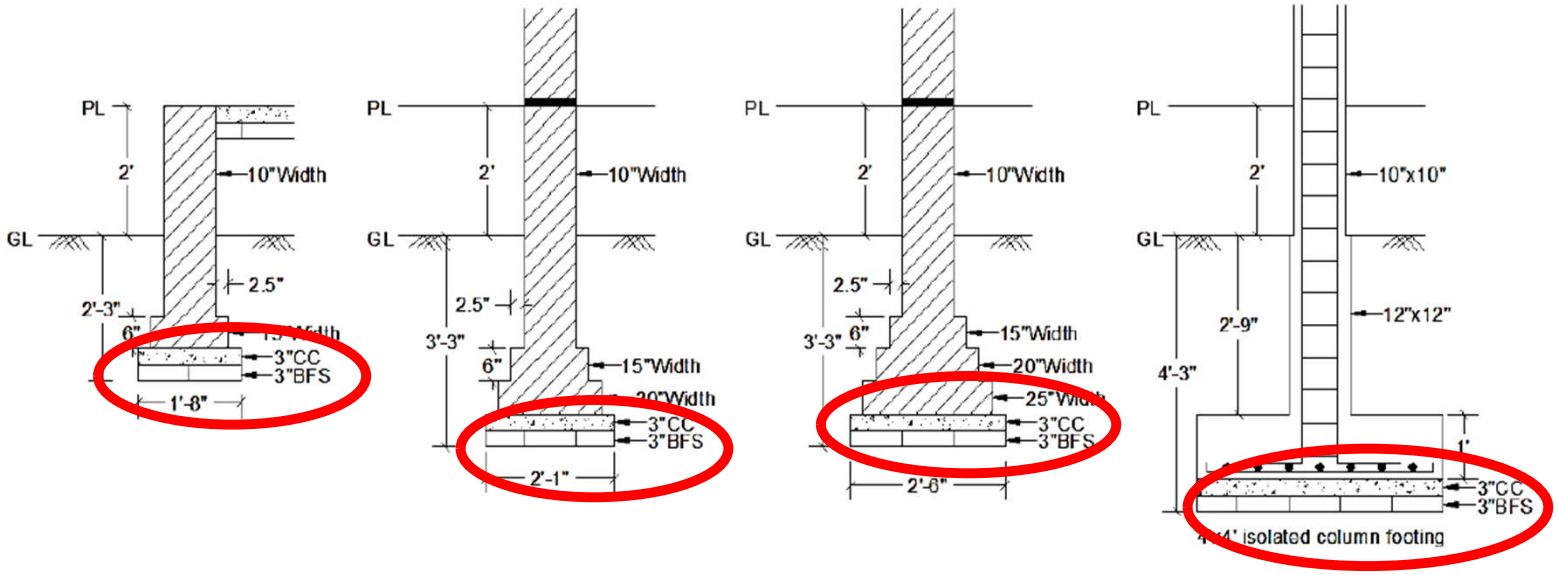


Column foundation:

- Width= 4’
- Length= 4’

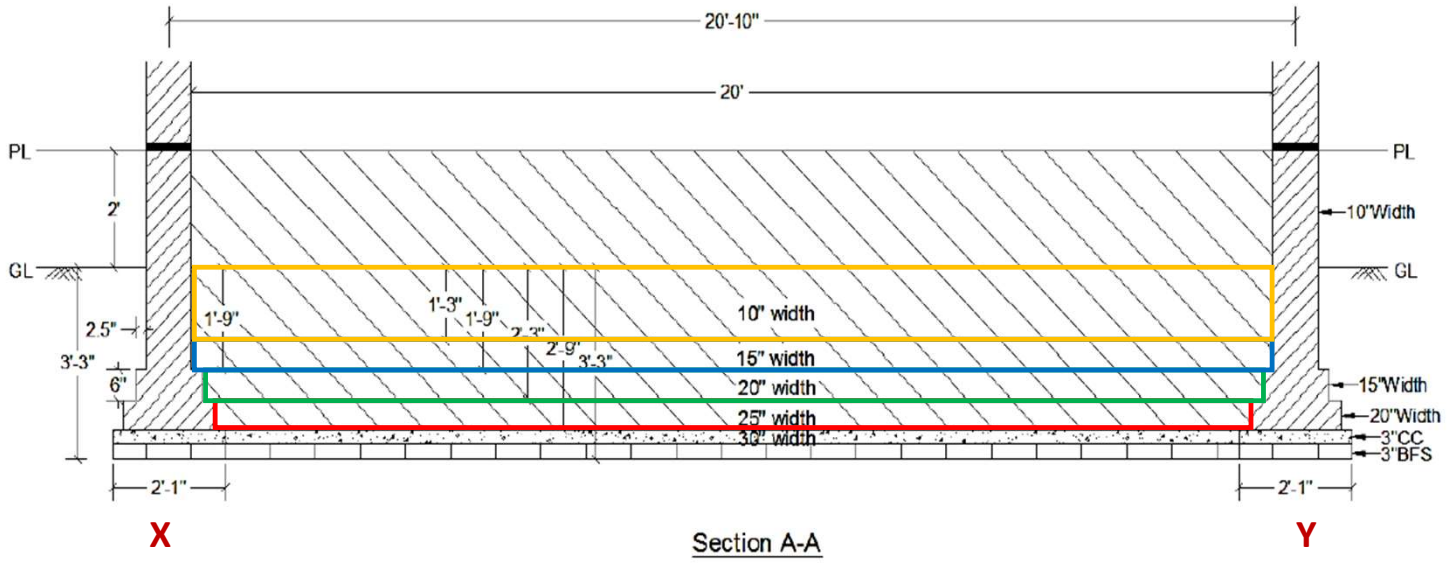


3. Estimation of Cement Concrete (1:3:6) in Foundation



- Length and width **same as BFS**!!!! For cement concrete, 3” thickness is to be considered for volume calculation

4. Estimation of Brickwork up to GL (30” wall foundation)



- The 30” wall foundation has a series of steps. At the bottom, its width is 30” and it gradually reduces to 10” decreasing by 5” in each step. The length along **XY** for each foundation width however increases by 5” as we move upwards

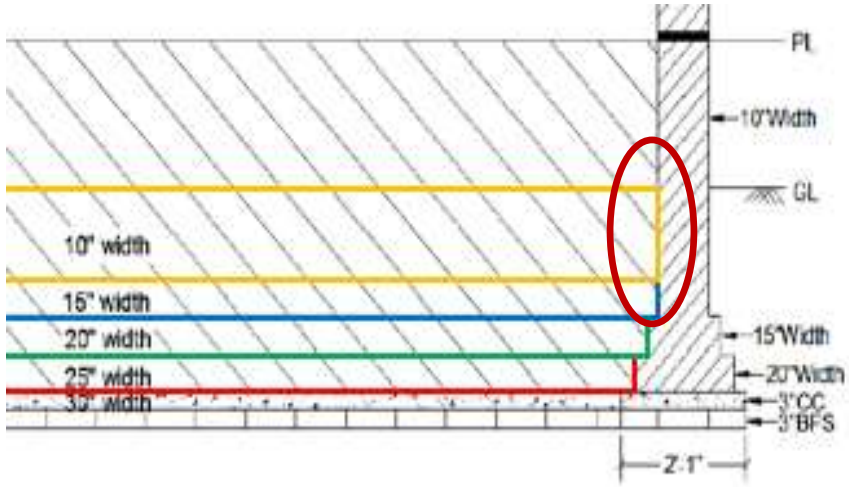
4. Estimation of Brickwork up to GL (30” wall foundation)

25” width:
- Height= 6”
- Width= 25”
- Length= 20’10”-25”+5”

20” width:
- Height= 6”
- Width= 20”
- Length= 20’10”-25”+5”+5”

15” width:
- Height= 6”
- Width= 15”
- Length= 20’10”-25”+5”+5”+5”

10” width:
- Height= 15”
- Width= 10”
- Length= 20’10”-25”+5”+5”+5”



4. Estimation of Brickwork up to GL (25” wall foundation)

Along AE, there is a **2.5”** decrease in length for 20” wall . However, along AD, there is a **5”** increase.

20” width:

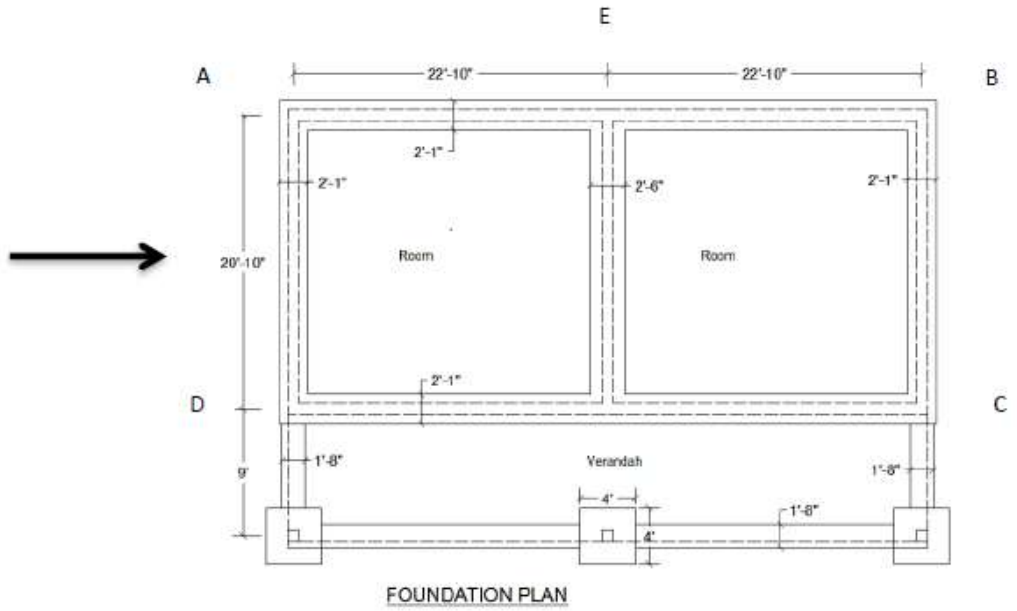
- Height= 6”
- Width= 20”
- Length= ((22’10”+12.5”-2.5”)*4) + ((20’10”-25”+5”)*2)

15” width:

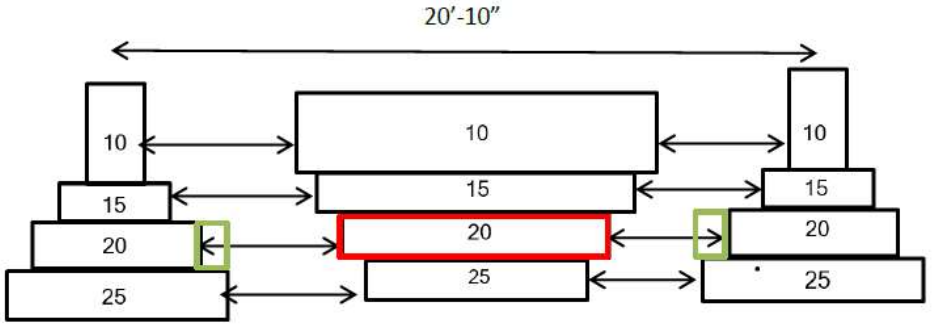
- Height= 6”
- Width= 15”
- Length= = ((22’10”+12.5”-5”)*4) + ((20’10”-25”+10”)*2)

10” width:

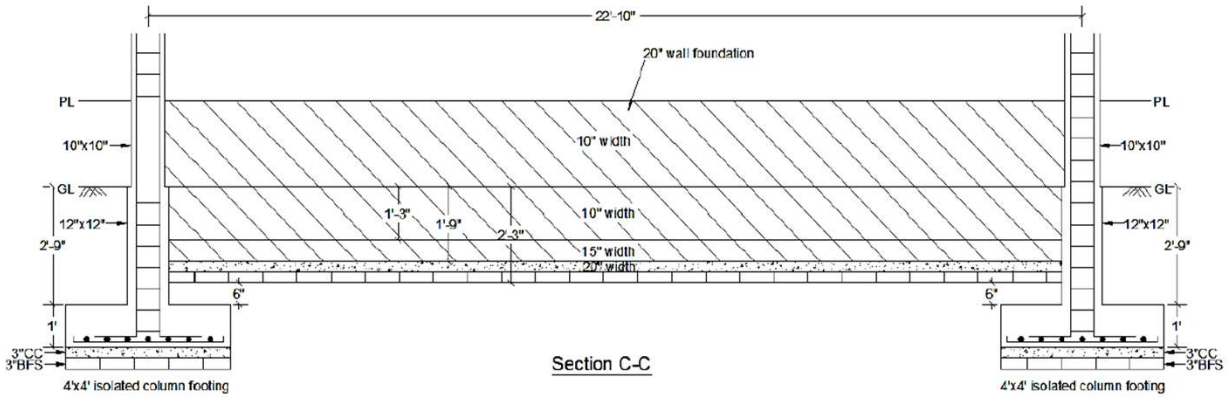
- Height= 21”
- Width= 10”
- Length= = ((22’10”+12.5”-7.5”)*4) + ((20’10”-25”+15”)*2)



FOUNDATION PLAN



4. Estimation of Brickwork up to GL (20” wall foundation)

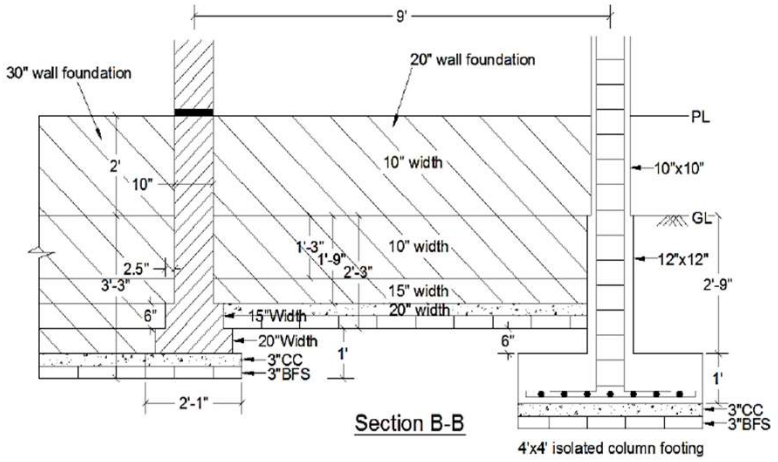


15” width:

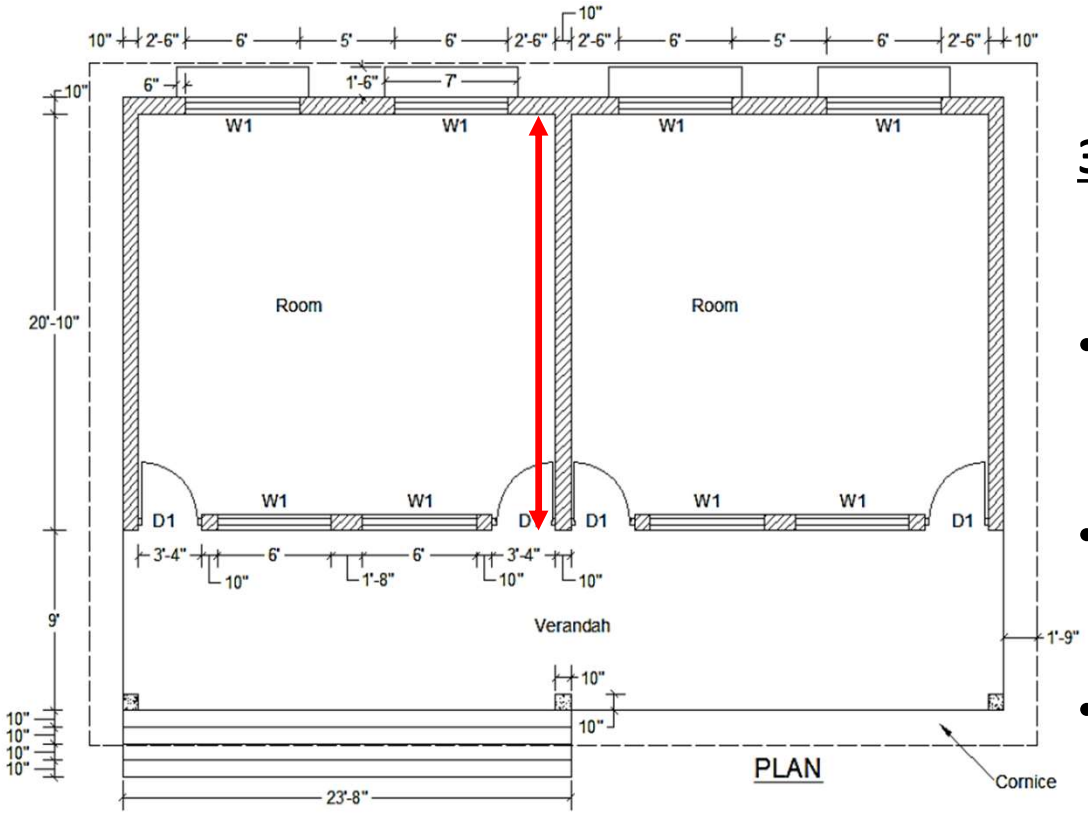
- Height= 6”
- Width= 15”
- Length= = ((9-6”-5”)*2) + ((22’10”-12”)*2)

10” width:

- Height= 15”
- Width= 10”
- Length= = ((9-6”-5”)*2) + ((22’10”-12”)*2)

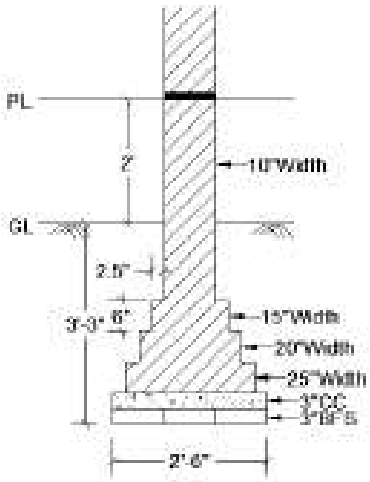


5. Estimation of Brickwork from GL to PL

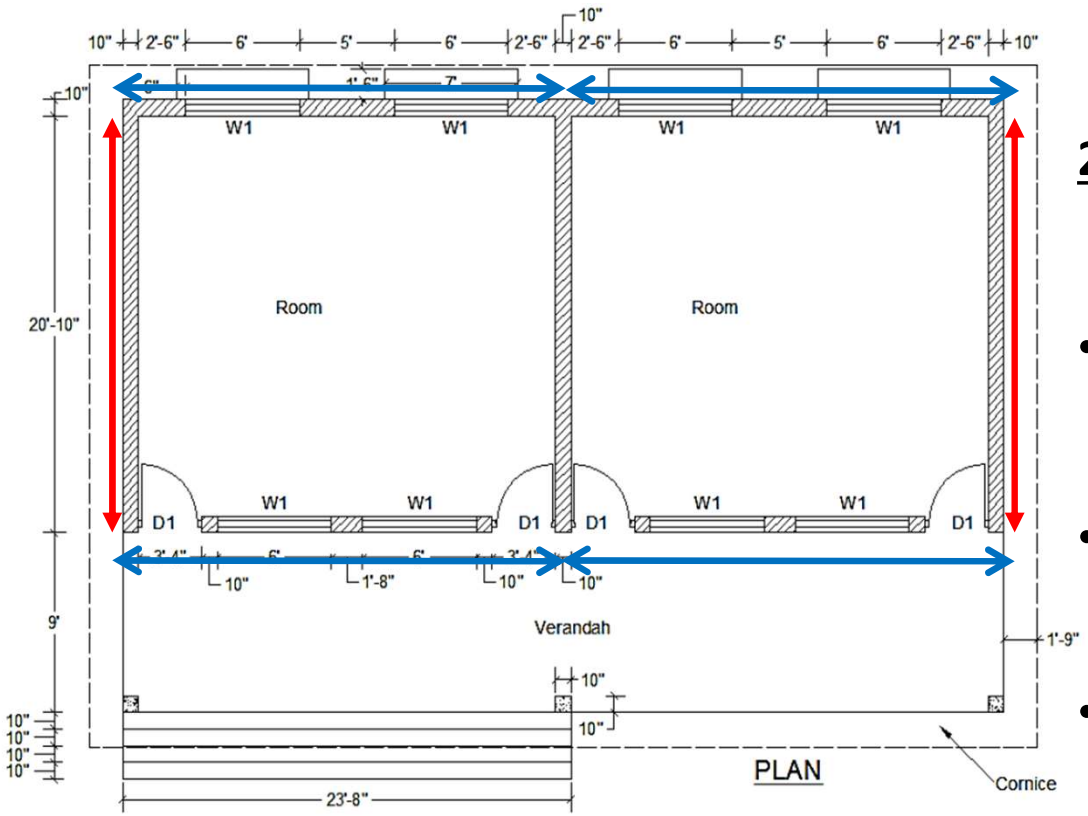


30” wall foundation:

- Height= 2’
- Width= 10”
- Length= 20’10”-5”-5”

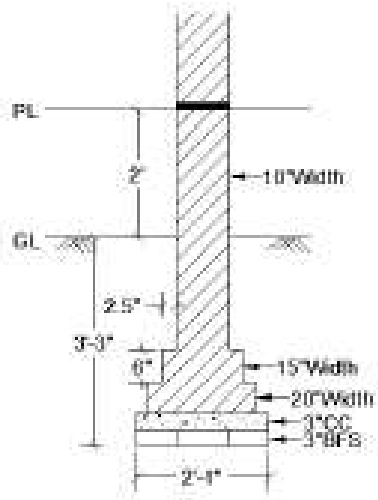


5. Estimation of Brickwork from GL to PL

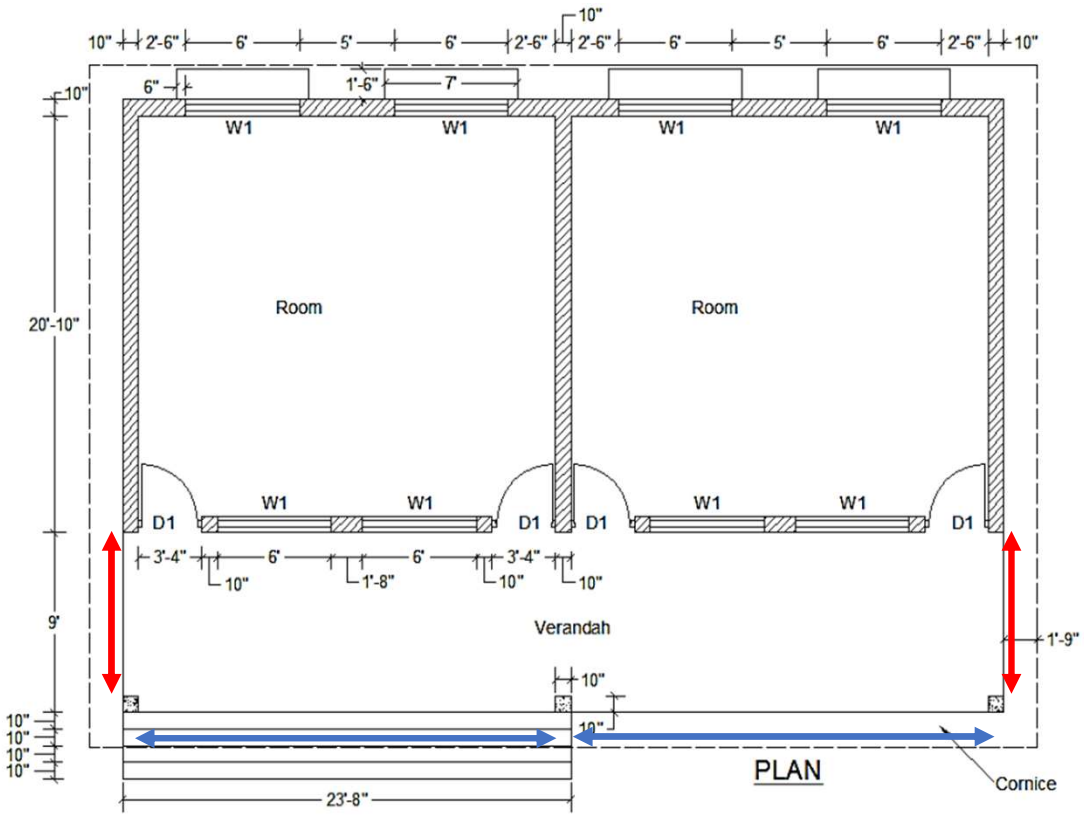


25” wall foundation:

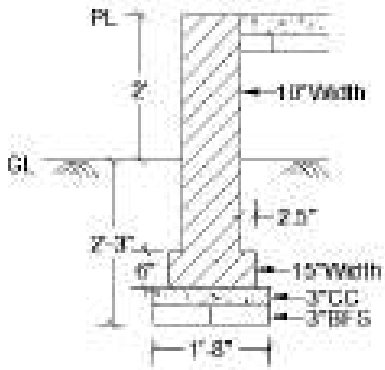
- Height= 2’
- Width= 10”
- Length= (20’10”-5”-5”)*2 +(22’10”+5”)*4



5. Estimation of Brickwork from GL to PL



20” wall foundation:



Height= 2’

Width= 10”

Length= (9’-5”-5”)*2+(22’10”-10”)*2

6. RCC in Column (up to GL)

i) Base Slab

Volume= (4’x4’x1’)*3

i) Column

Volume= (1’x1’x1’9”)*3

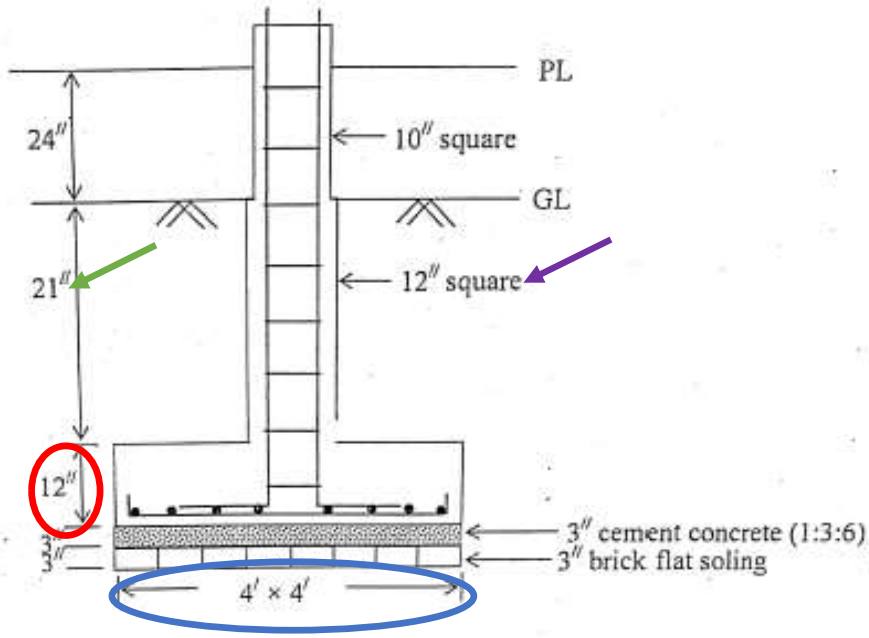


Figure: 4 ft x 4 ft isolated column footing

7. RCC in Column (GL to PL)

- Height= 2’
- Area= 10”x10”
- Volume= (2’x10”x10”)*3

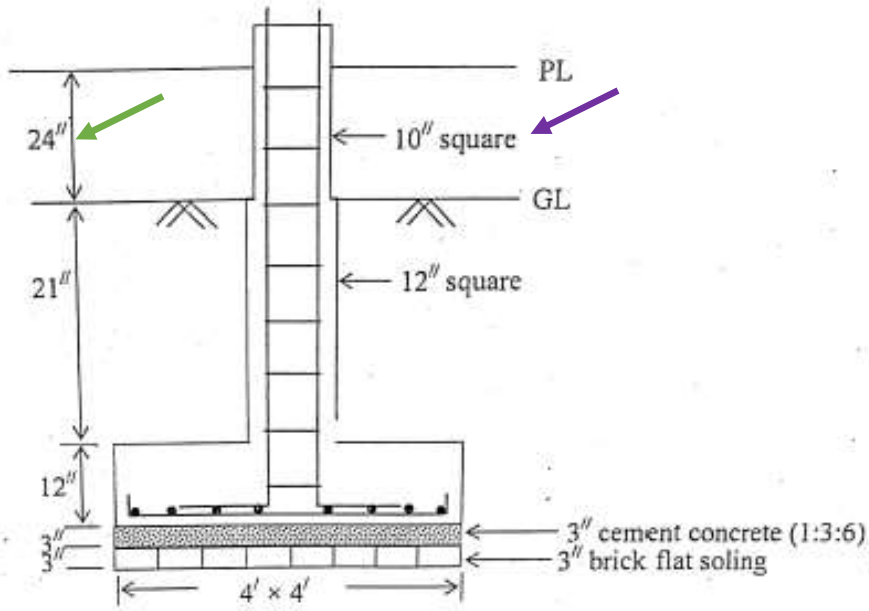


Figure: 4 ft x 4 ft isolated column footing

8. Damp Proof Course (DPC)

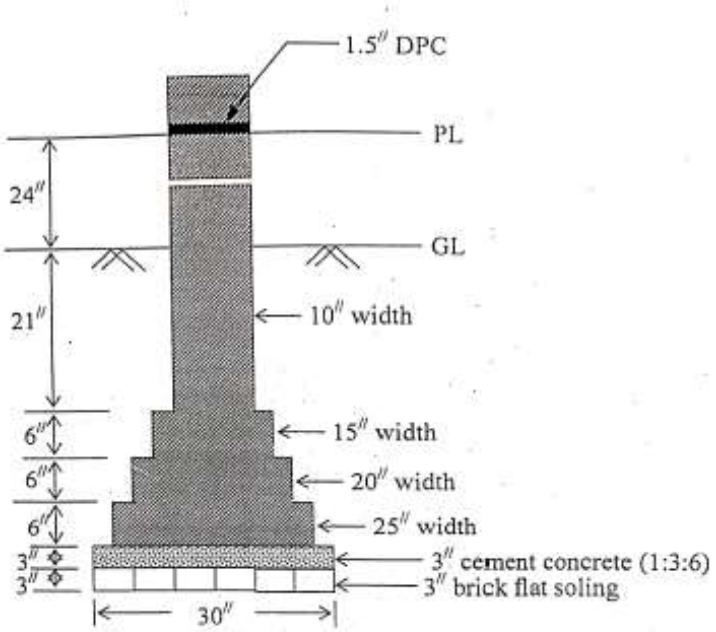


Figure: 30 inch continuous wall foundation

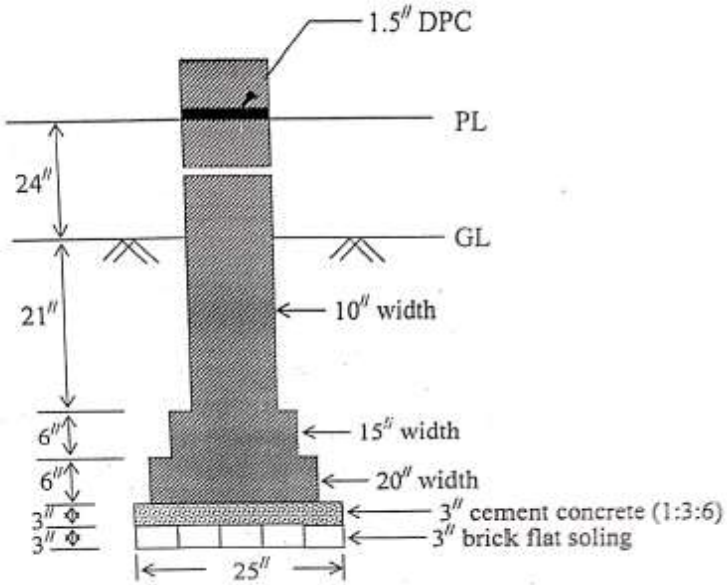


Figure: 25 inch continuous wall foundation

8. Damp Proof Course (DPC)

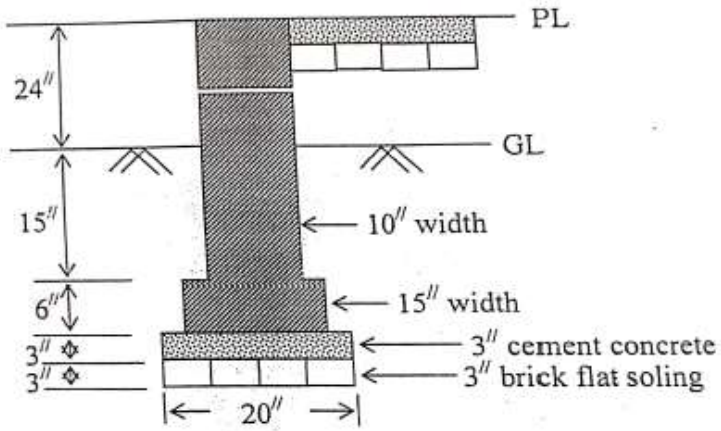


Figure: 20 inch continuous verandah wall foundation

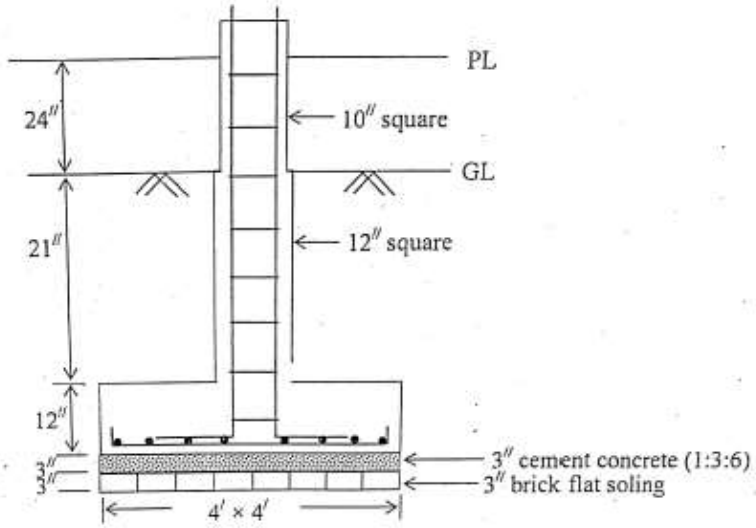
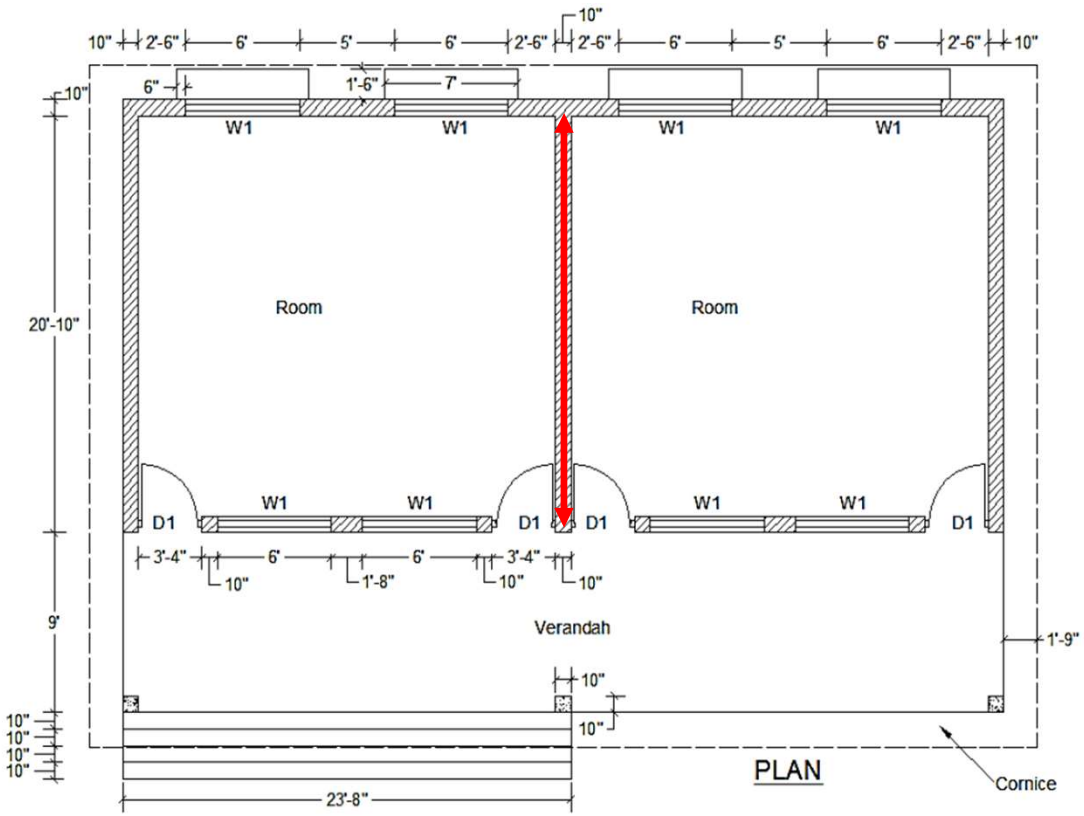


Figure: 4 ft x 4 ft isolated column footing

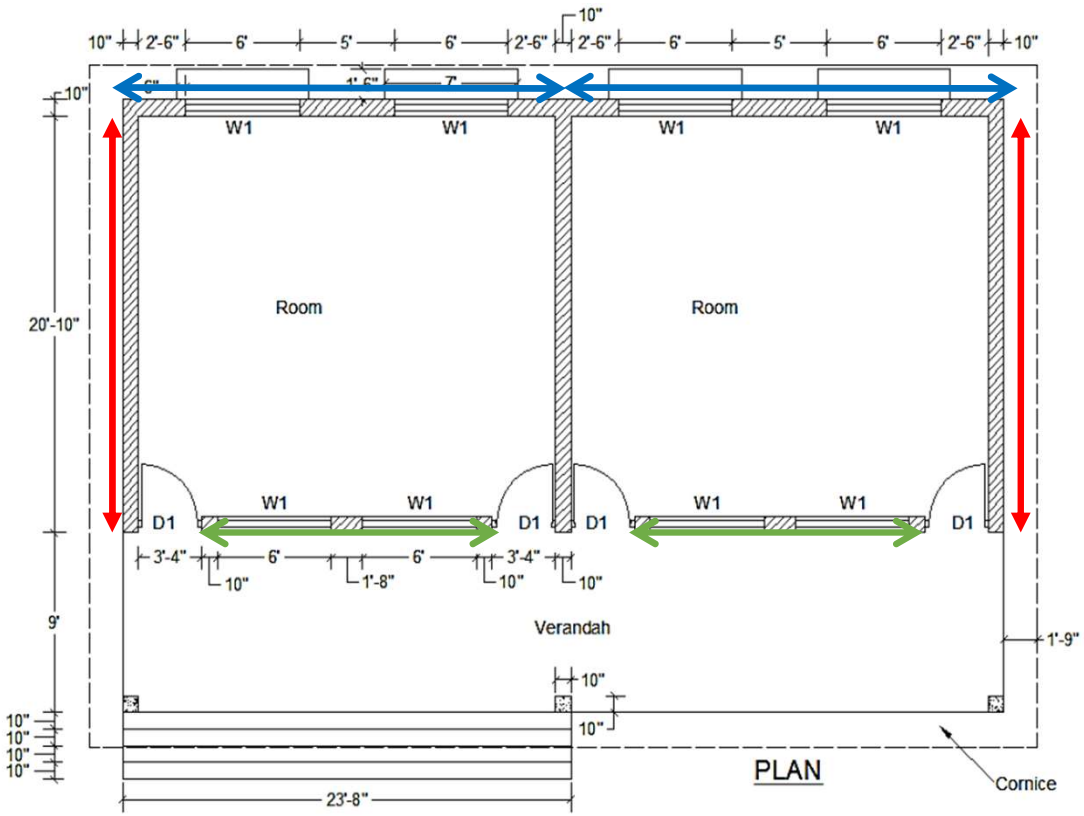
8. Damp Proof Course (DPC)



30" wall foundation:

- Thickness= 1.5"
- Width= 10"
- Length= 20'10"-10"

8. Damp Proof Course (DPC)



25" wall foundation:

- Thickness= 1.5"
- Width= 10"
- Length= 20'*2+(22'10"+5")*2+(15'4")*2

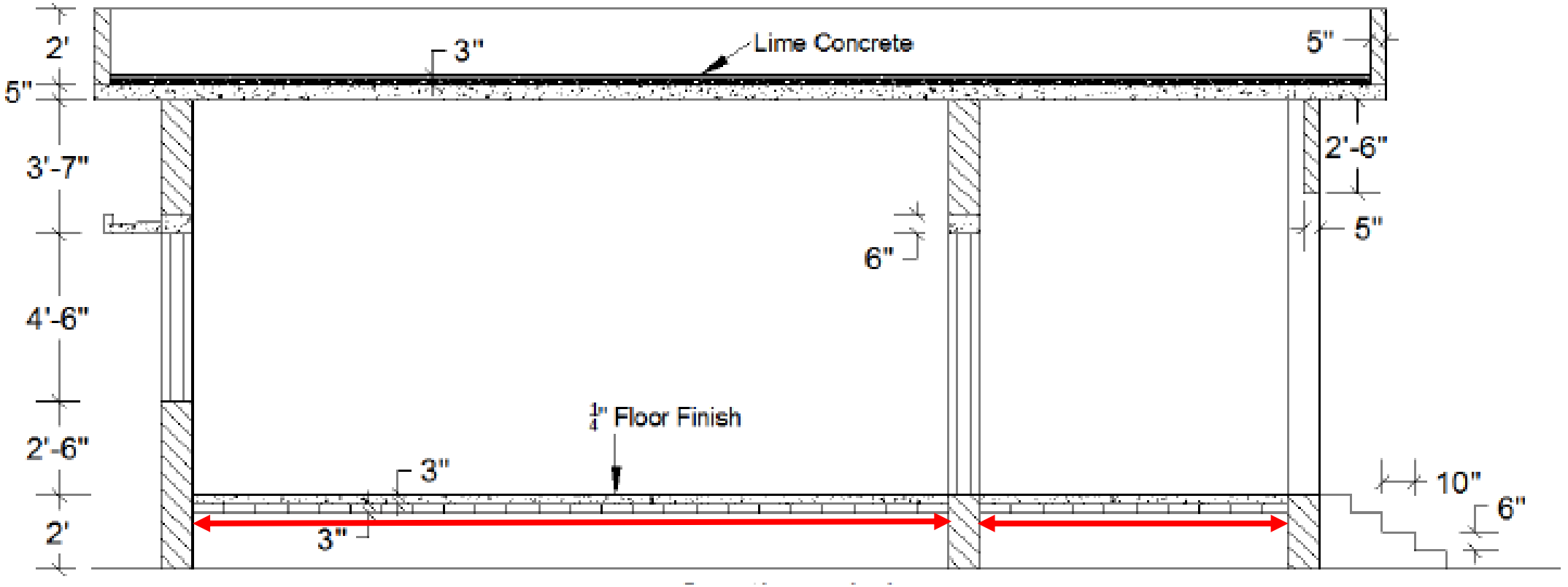
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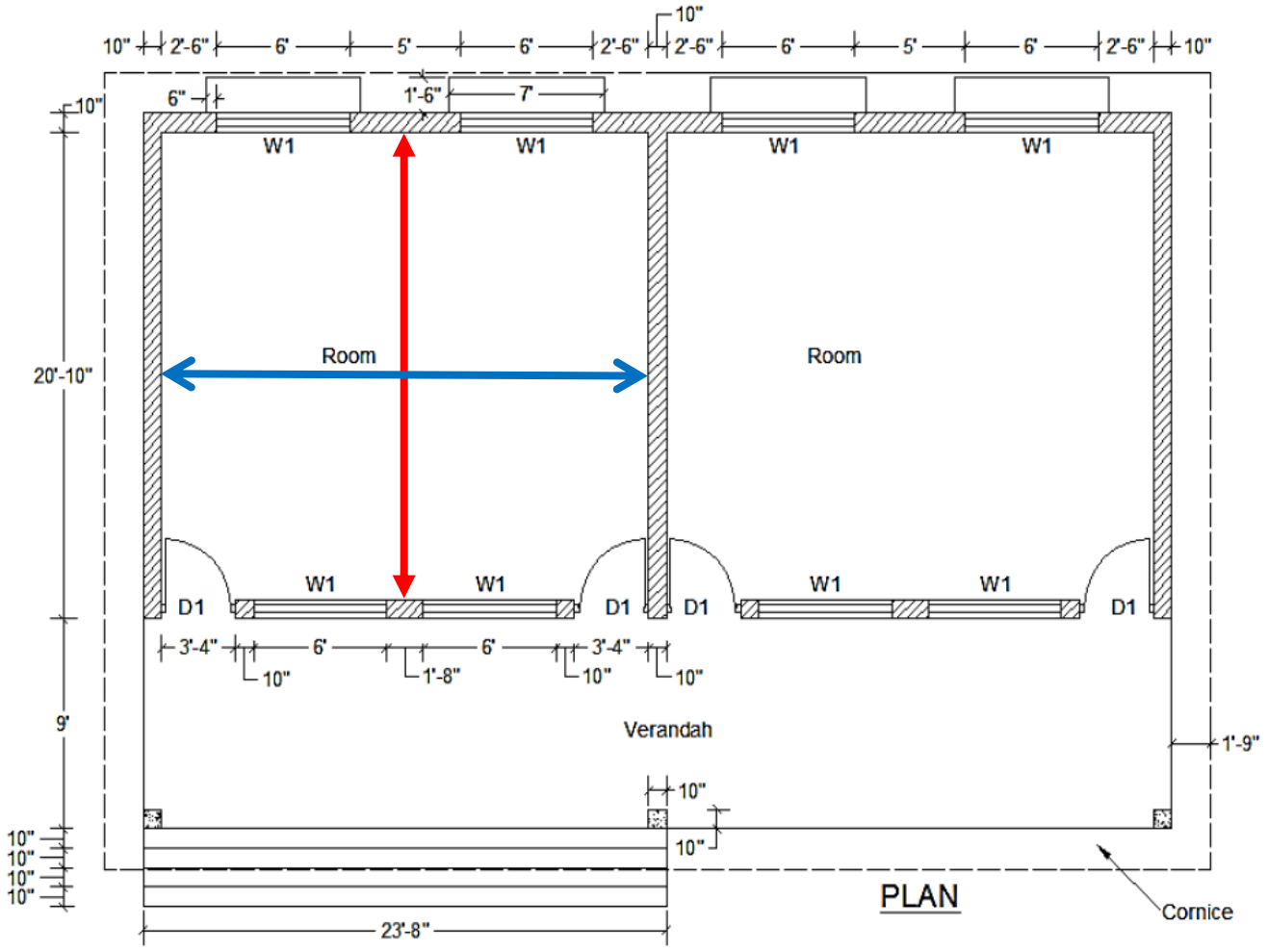


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9. BFS (one layer) in Floors



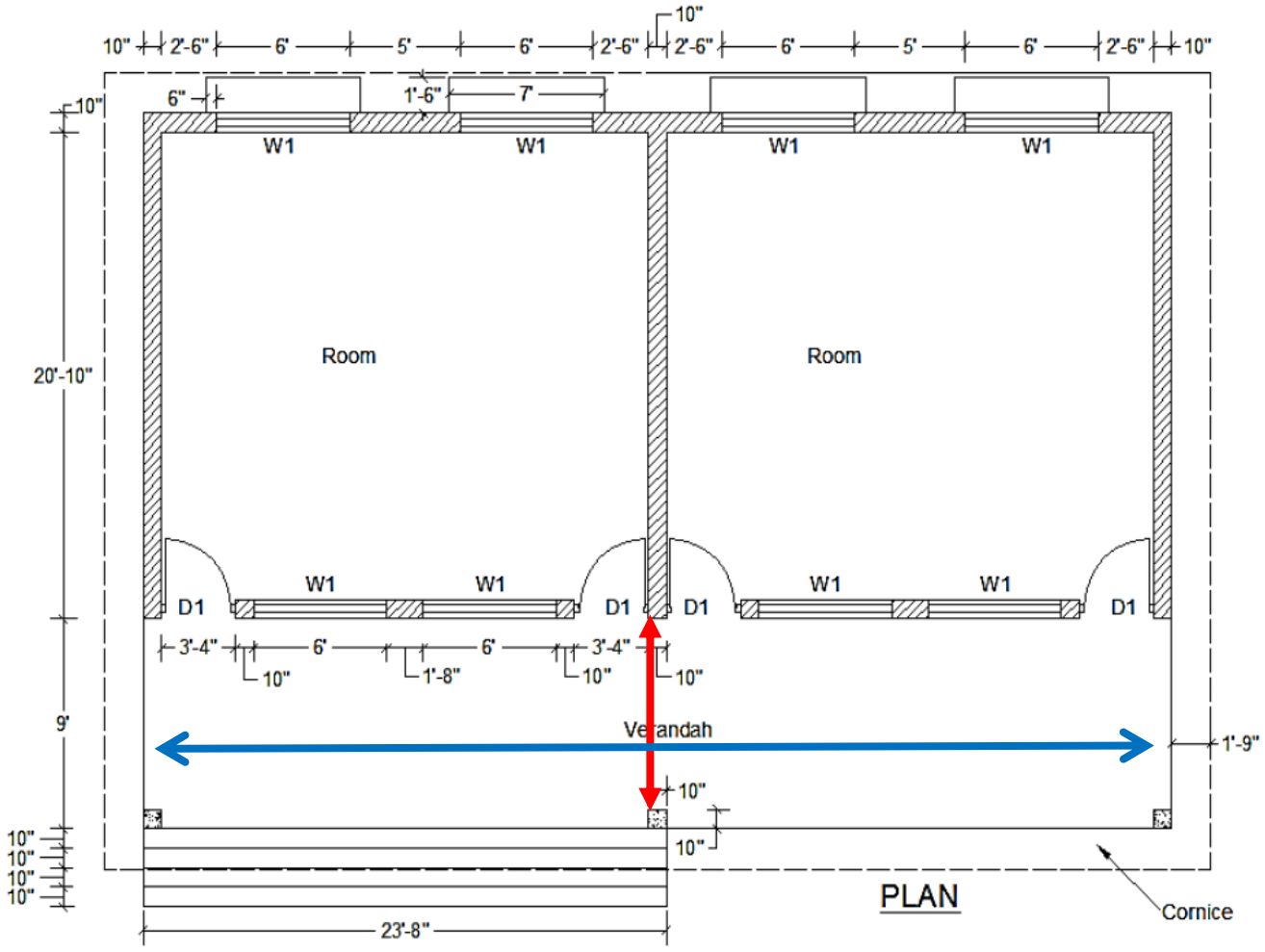
9. BFS (one layer) in Floors



Main Room:

- Length= 22’
- Width= 20’

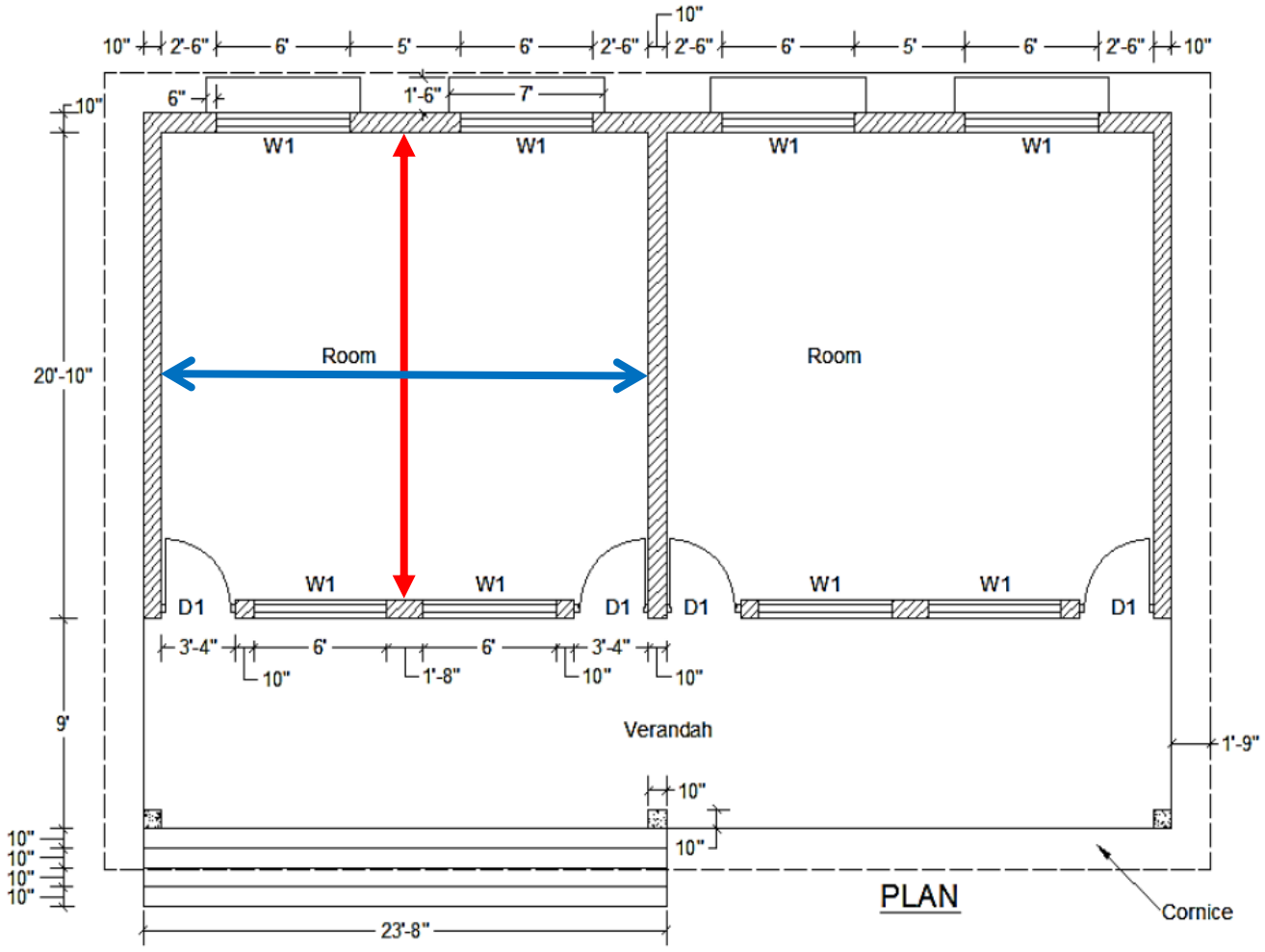
9. BFS (one layer) in Floors



Verandah:

- Length= (22’10”)*2-10”
- Width= 9’-10”

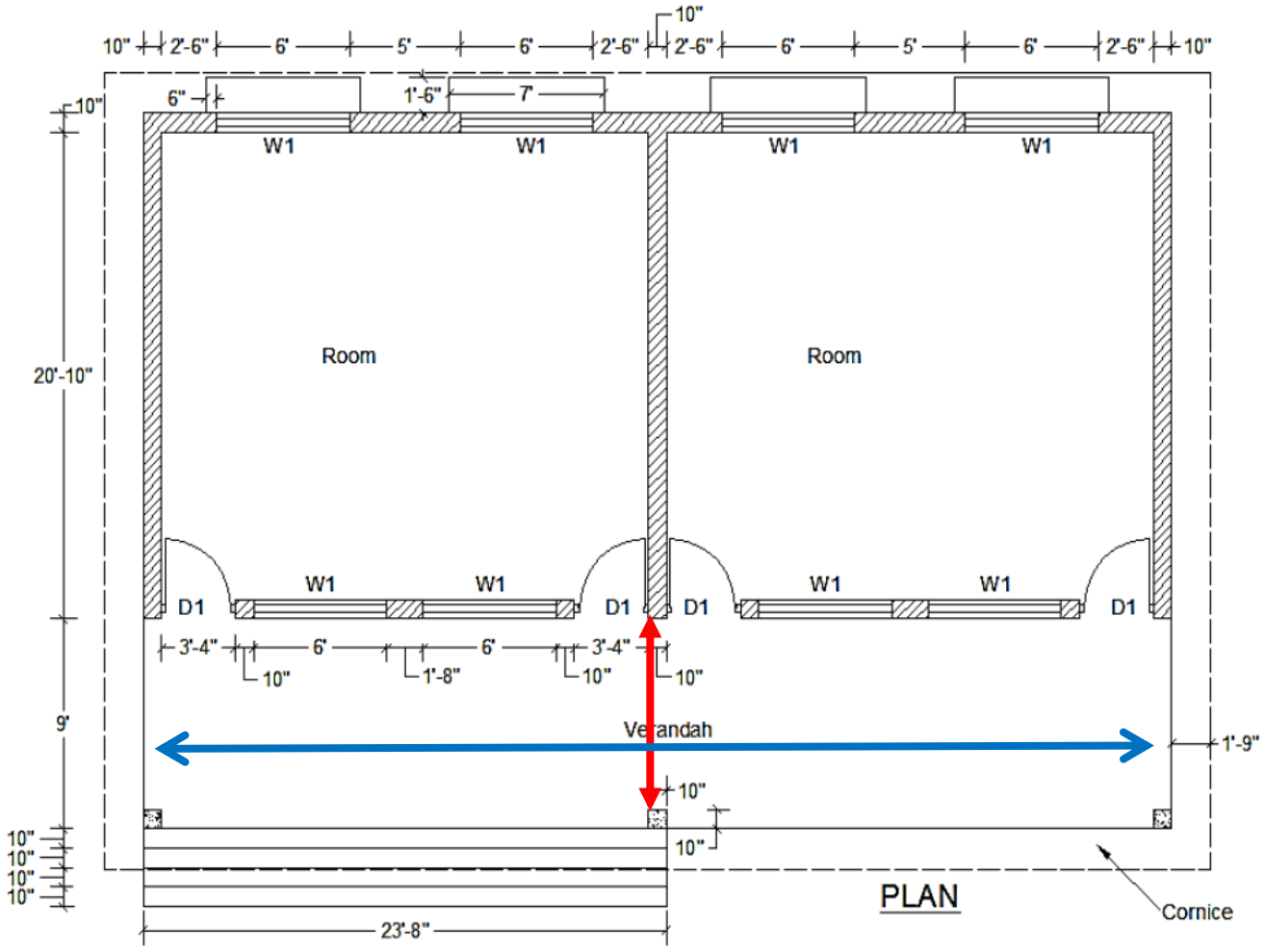
10. CC (3”) in Floors



Main Room:

- Length= 22’
- Width= 20’
- Thickness= 3”

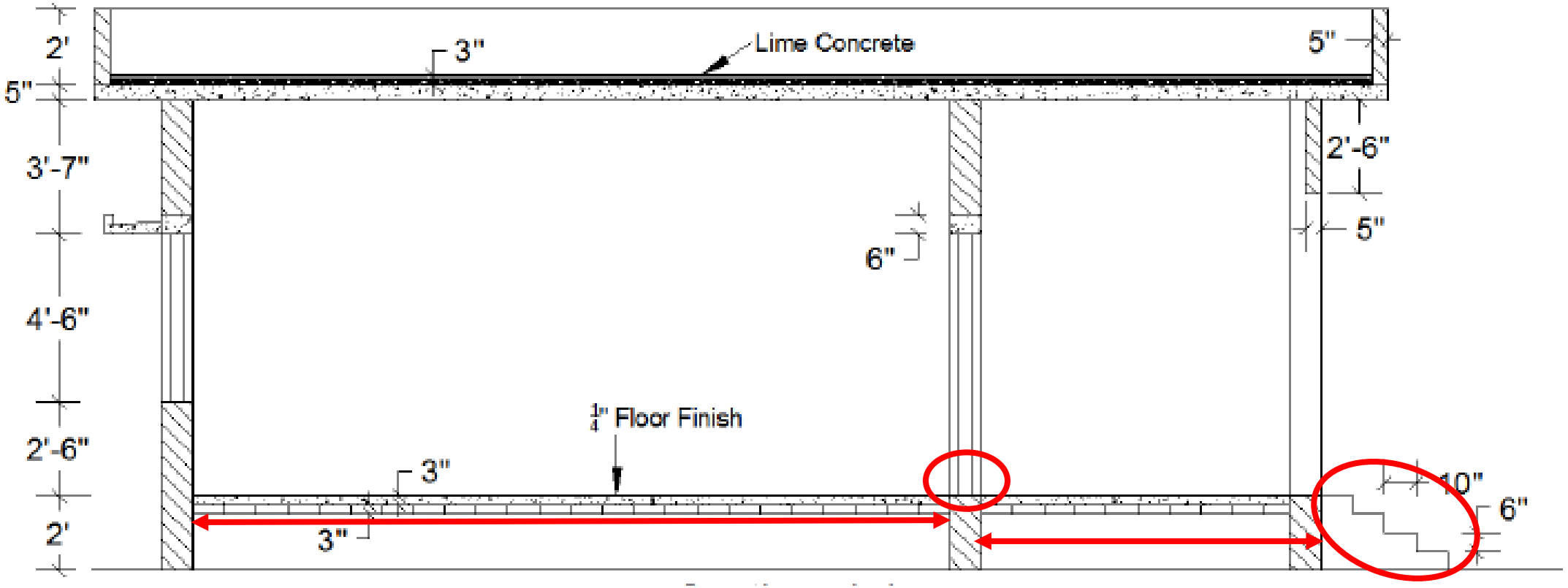
10. CC (3”) in Floors



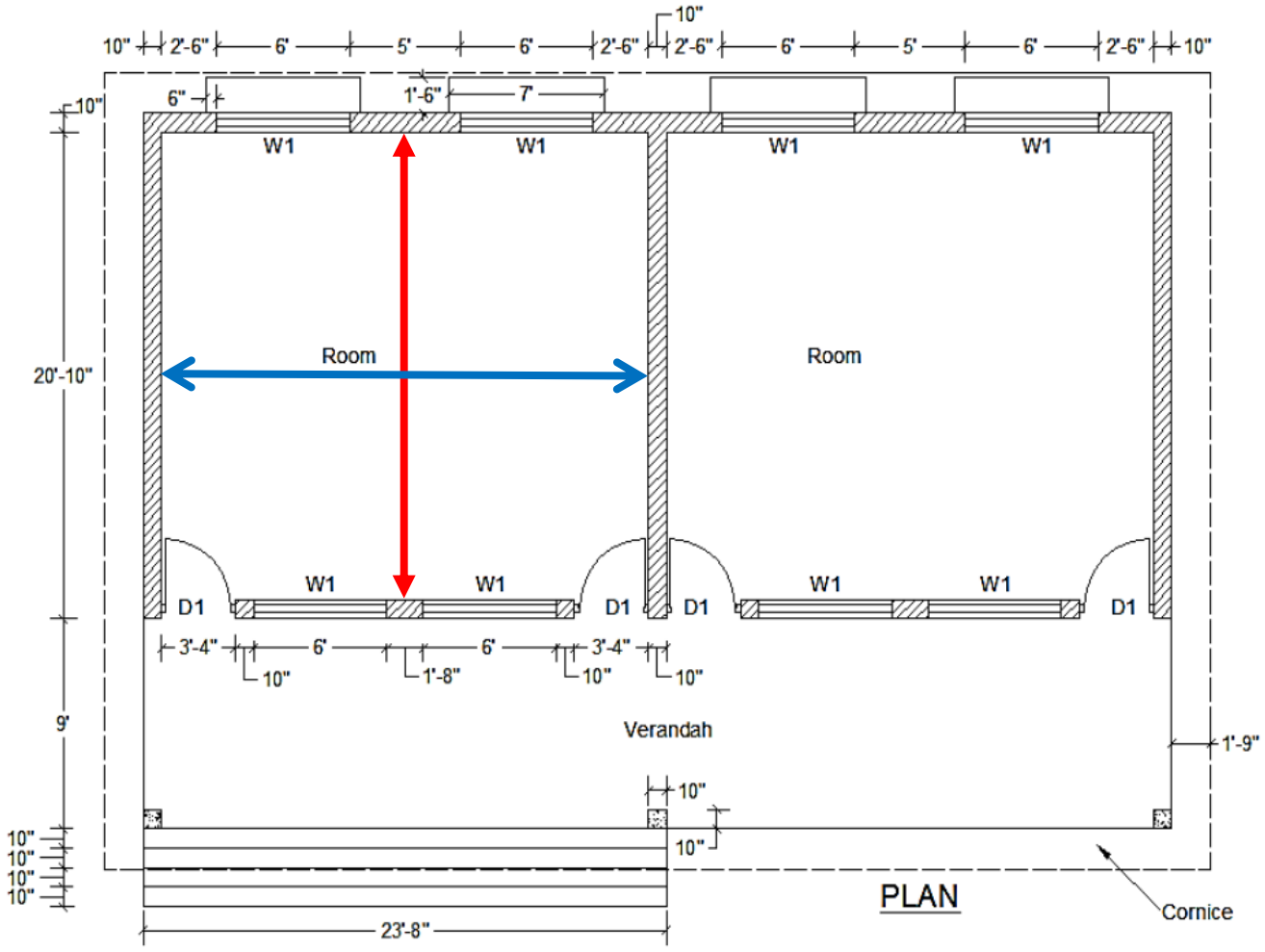
Verandah:

- Length= (22’10”)*2-10”
- Width= 9’-10”
- Thickness= 3”

11. Floor Finish, ¼” thick



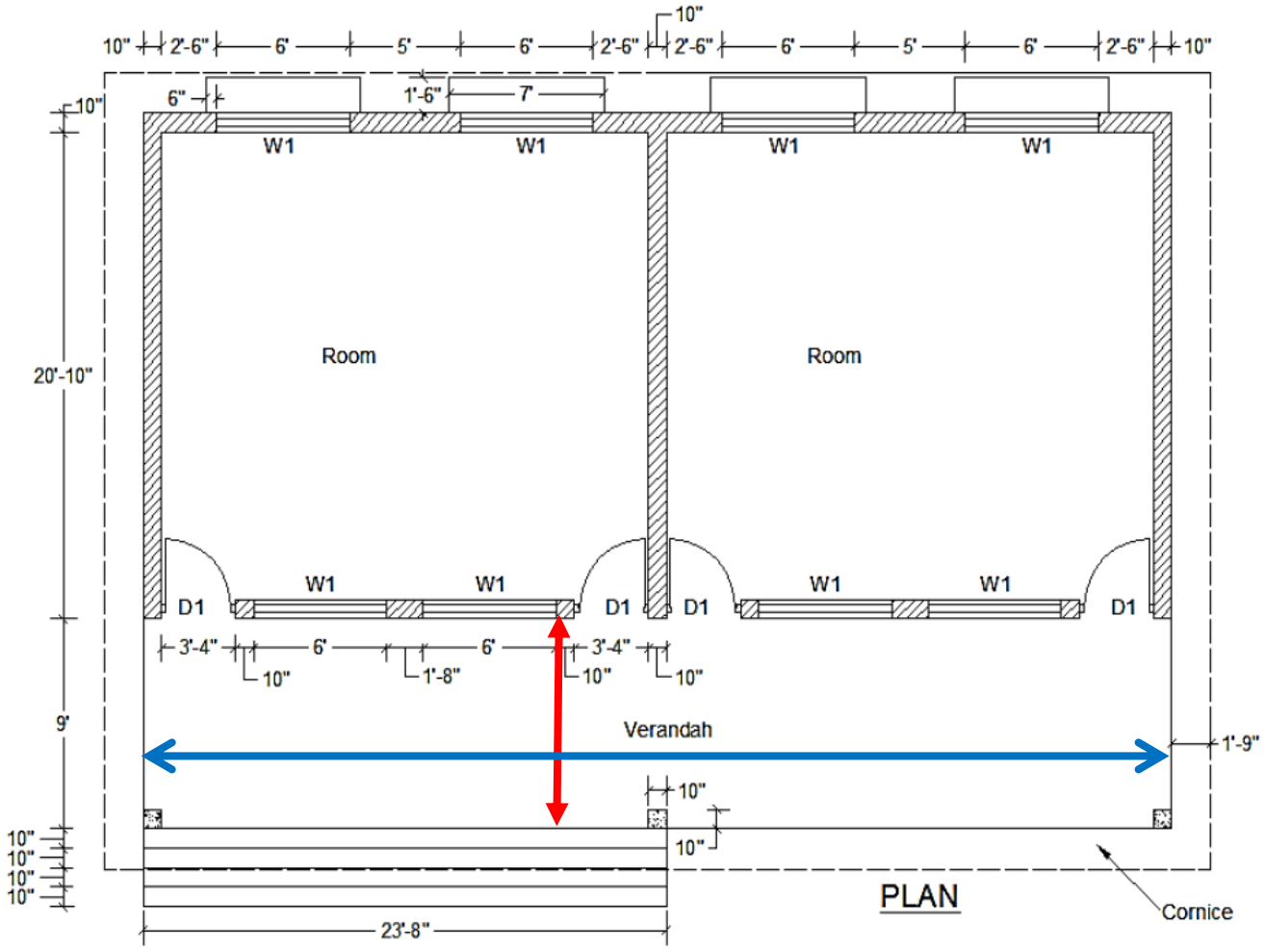
11. Floor Finish, ¼” thick



Main Floor:

- Length= 22’
- Width= 20’
- Thickness= 1/4”

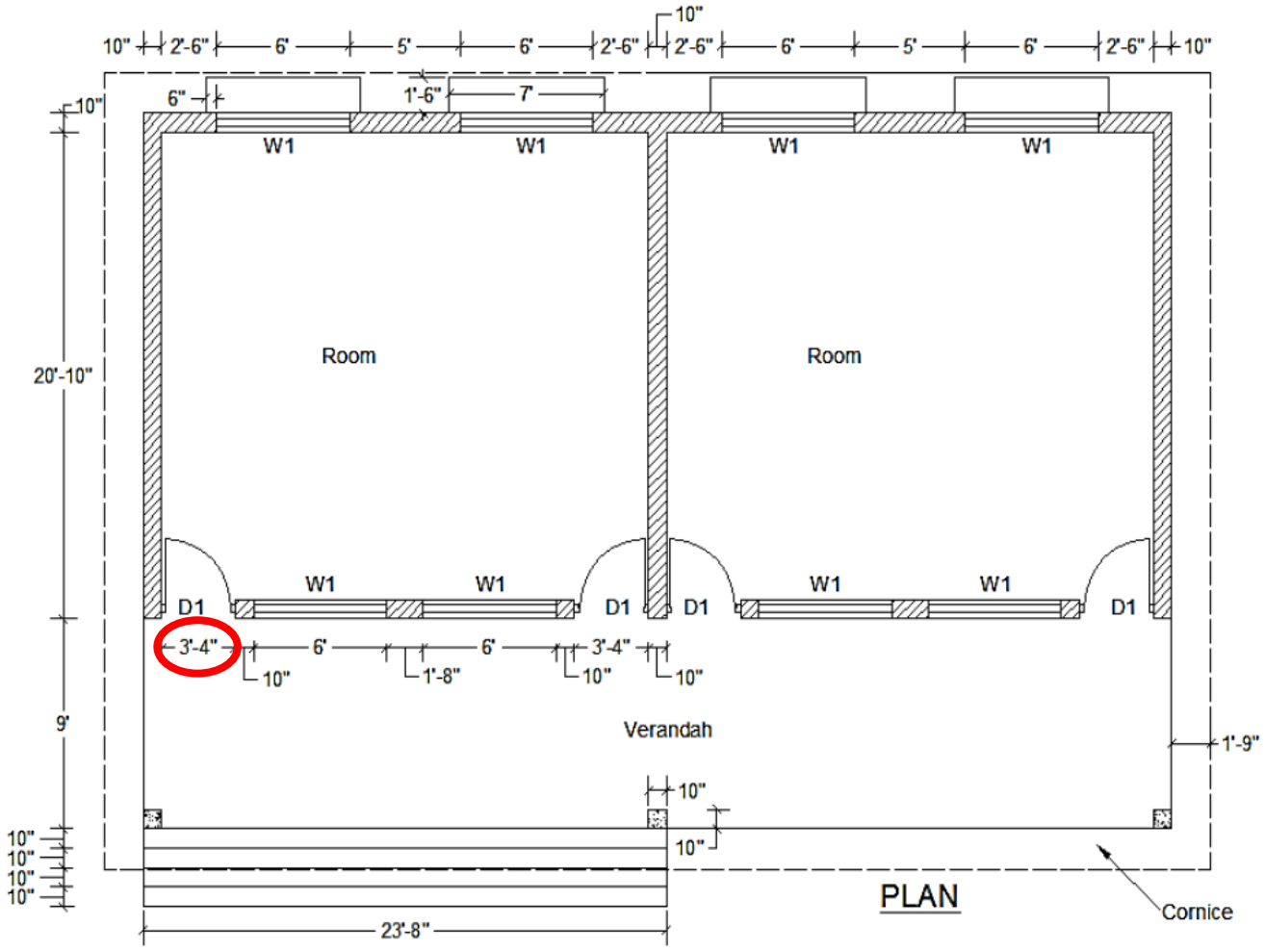
11. Floor Finish, ¼” thick



Verandah:

- Length= (22’10”)*2+10”
- Width= 9’
- Thickness= 1/4”

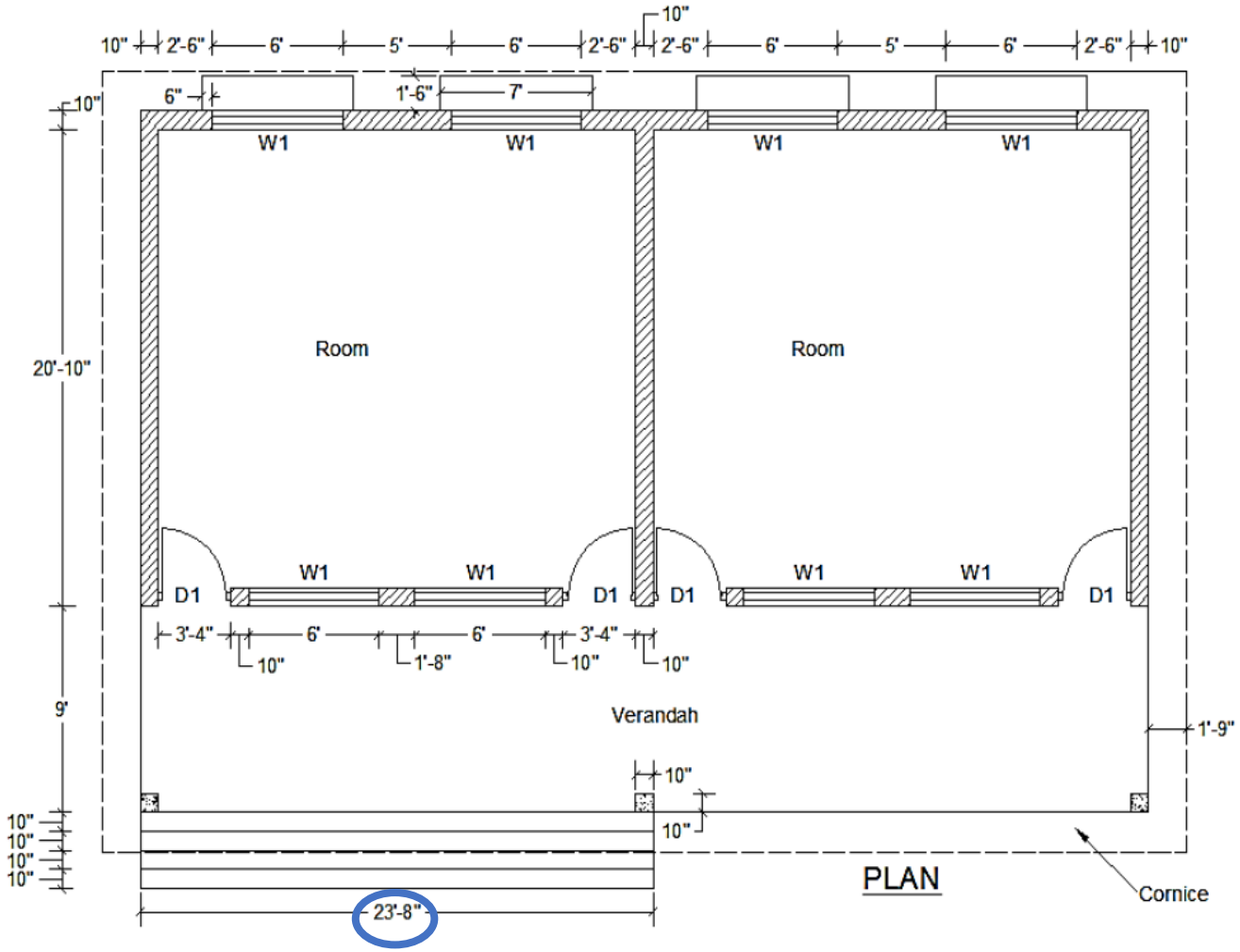
11. Floor Finish, ¼” thick



Door:

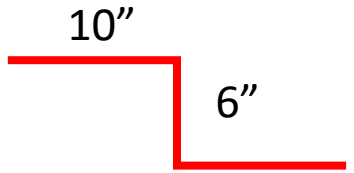
- Length= 3’4”
- Width= 10”
- Thickness= 1/4”

11. Floor Finish, ¼” thick

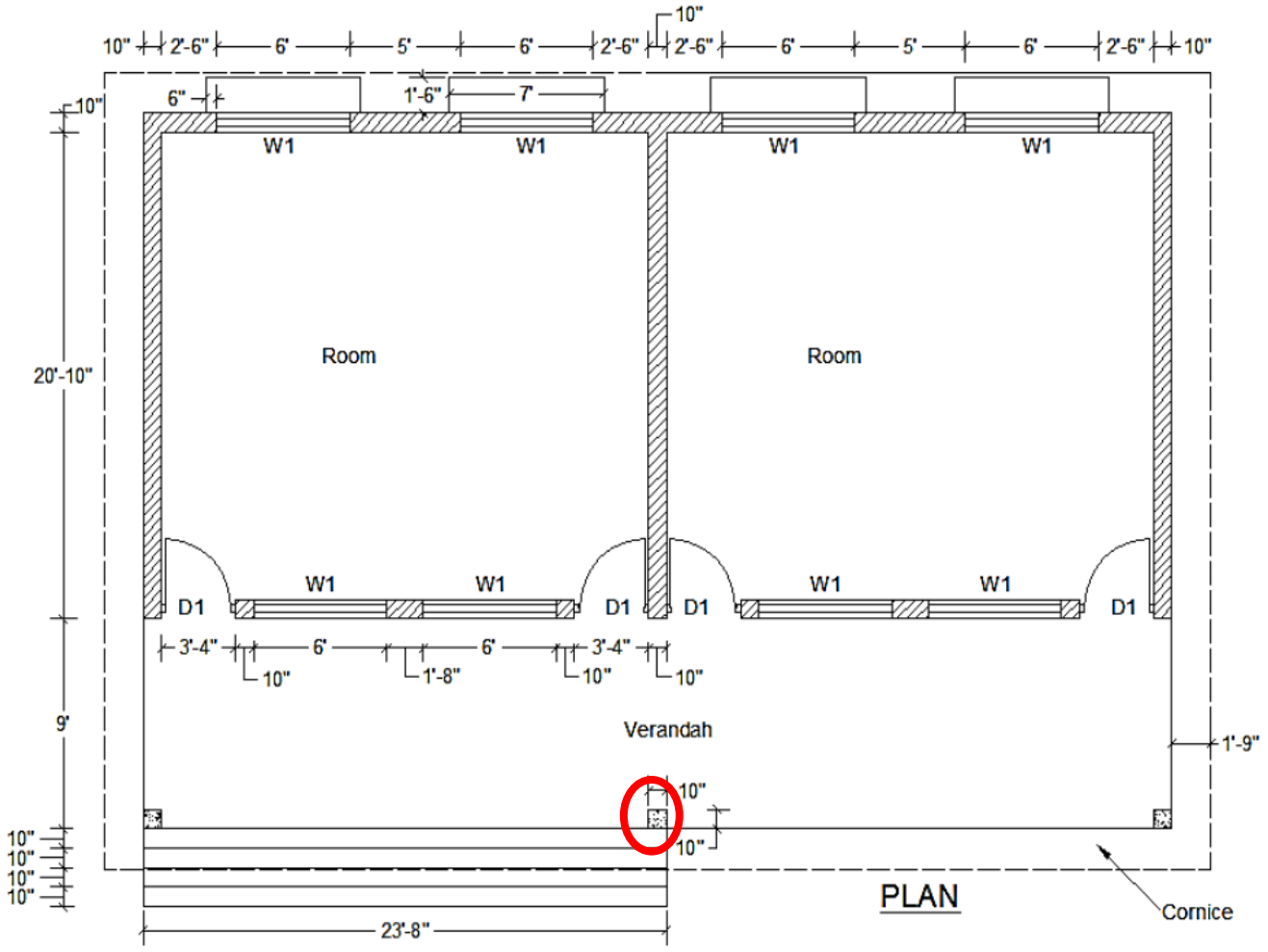


4 Steps:

- Length= 23’8”
- Width= 10”+6”
- Thickness= 1/4”



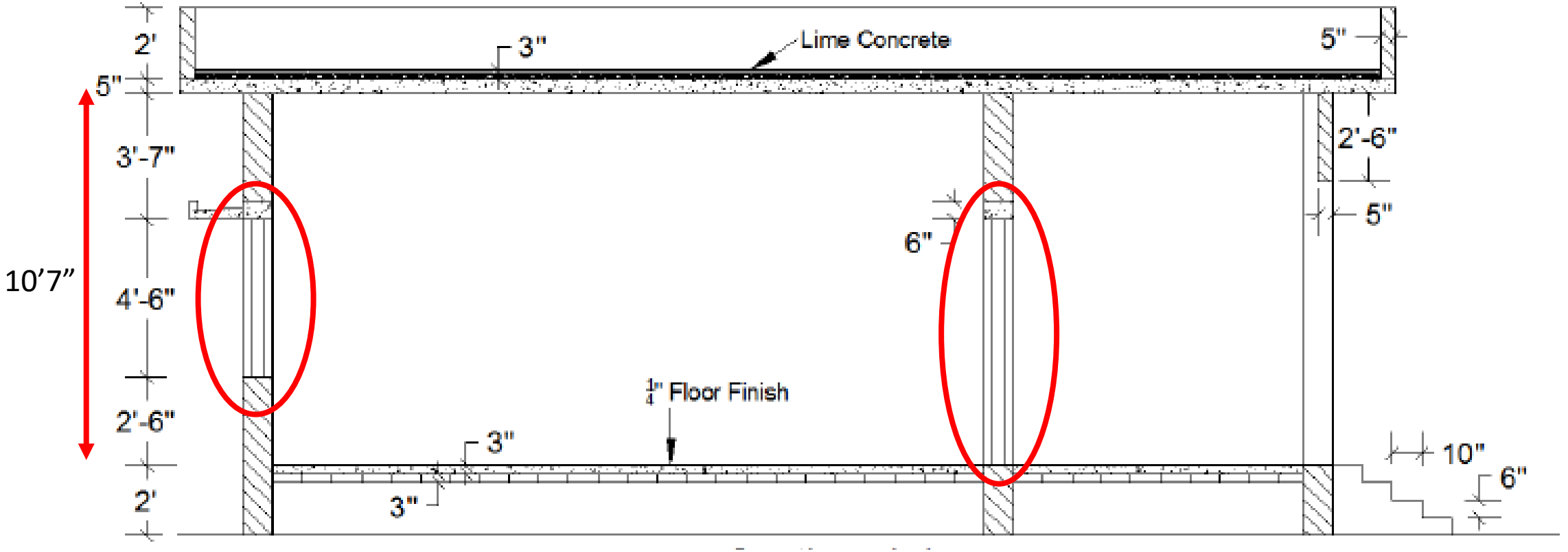
11. Floor Finish, ¼” thick



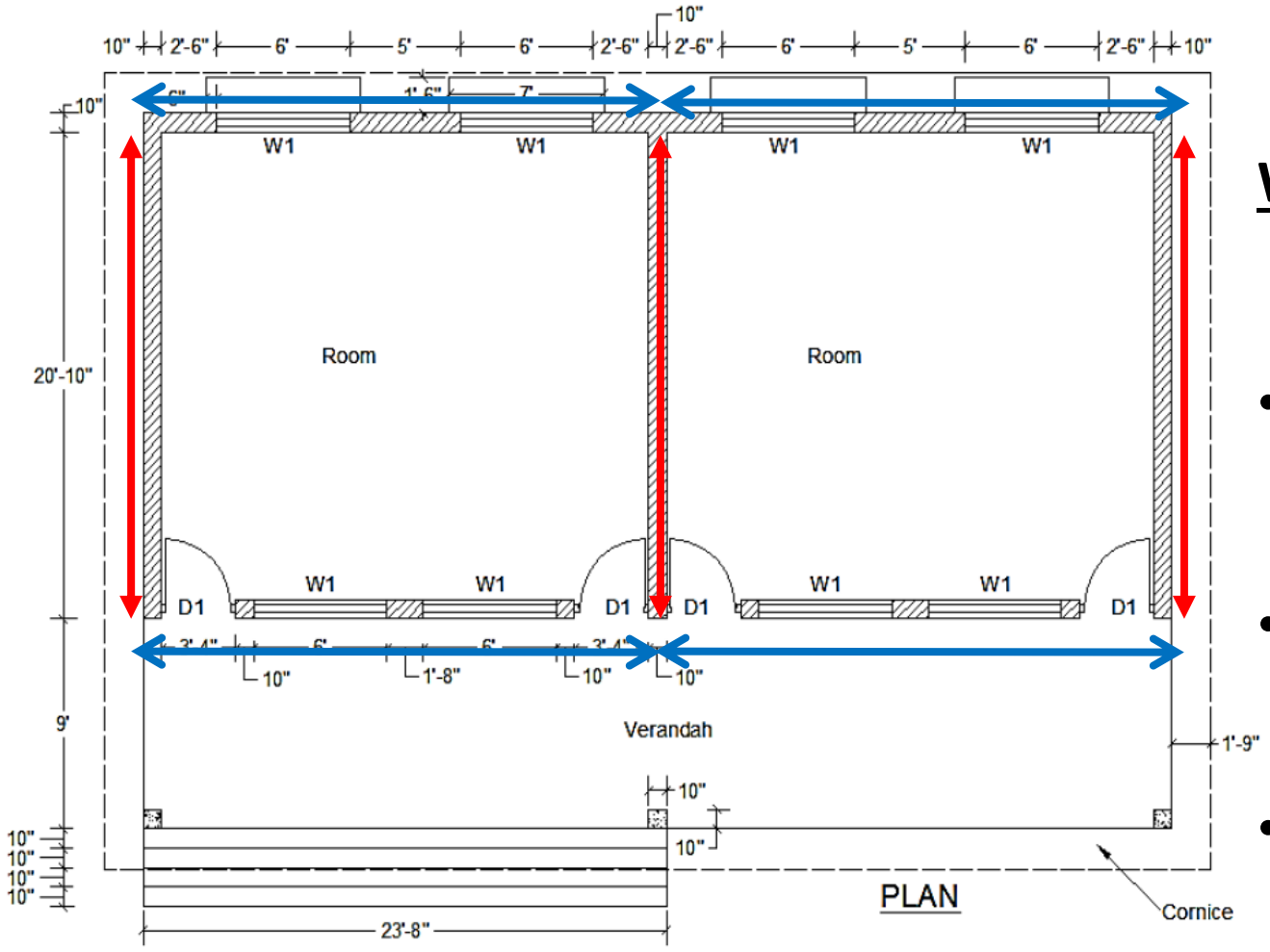
Deduct (Columns):

- Length= 10”
- Width= 10”

12. Brick walls in superstructure



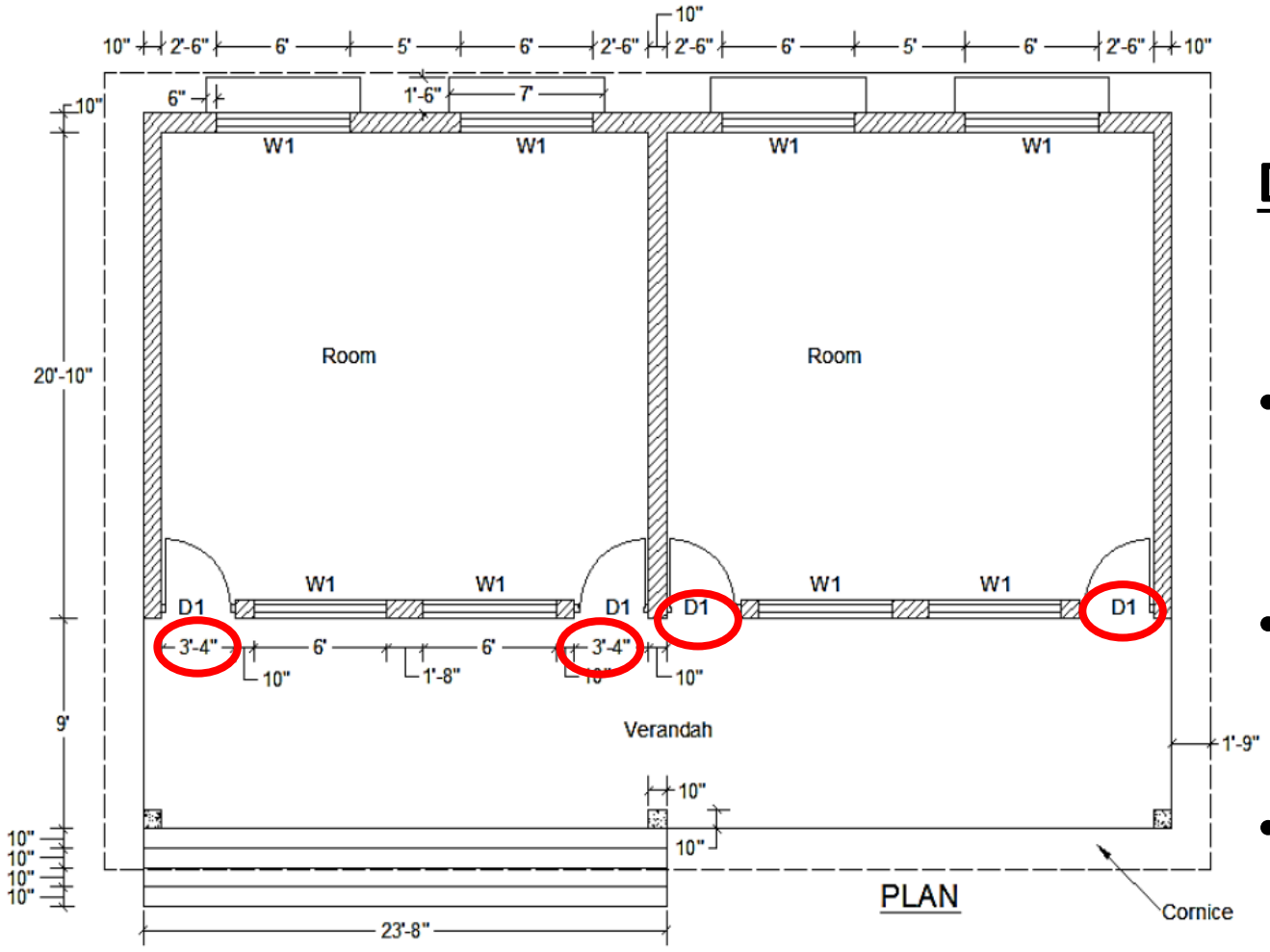
12. Brick walls in superstructure



Wall:

- Height= 10’7”
- Width= 10”
- Length= (20’10”-10”)*3 +(22’10”+5”)*4

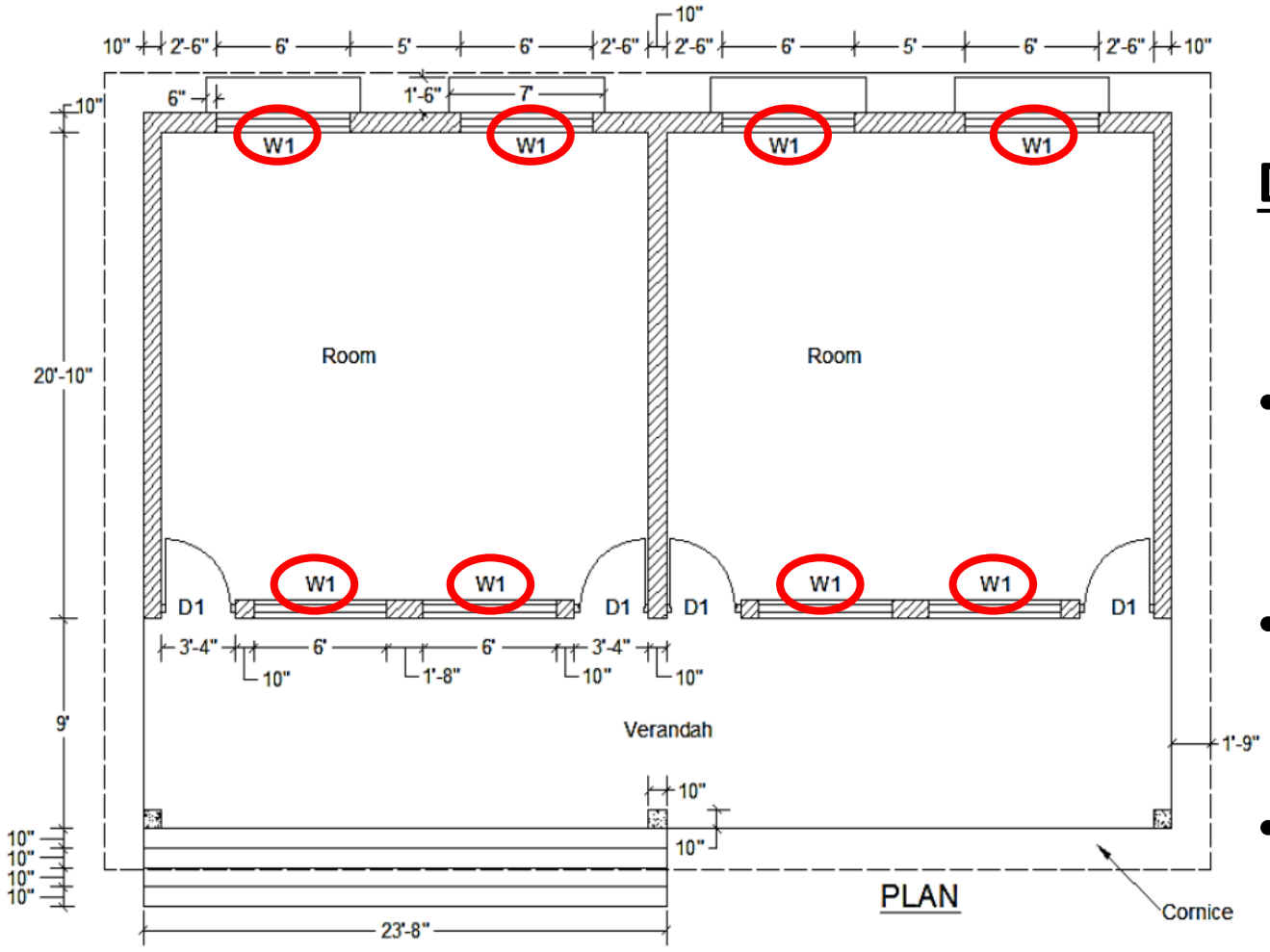
12. Brick walls in superstructure



Deduct (4 Doors):

- Height= 7’
- Width= 10”
- Length= 3’4”

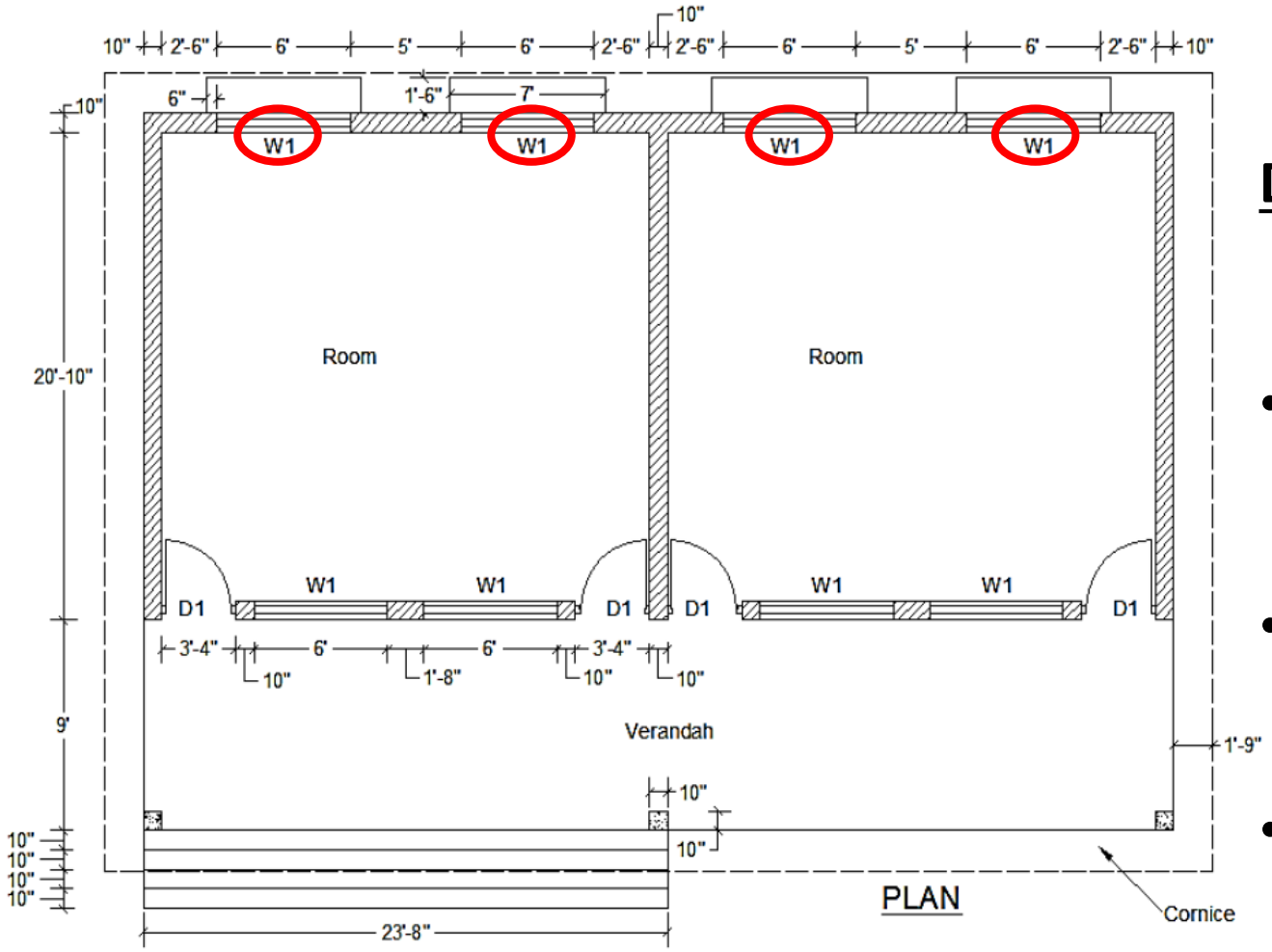
12. Brick walls in superstructure



Deduct (8 Windows):

- Height= 4’6”
- Width= 10”
- Length= 6’

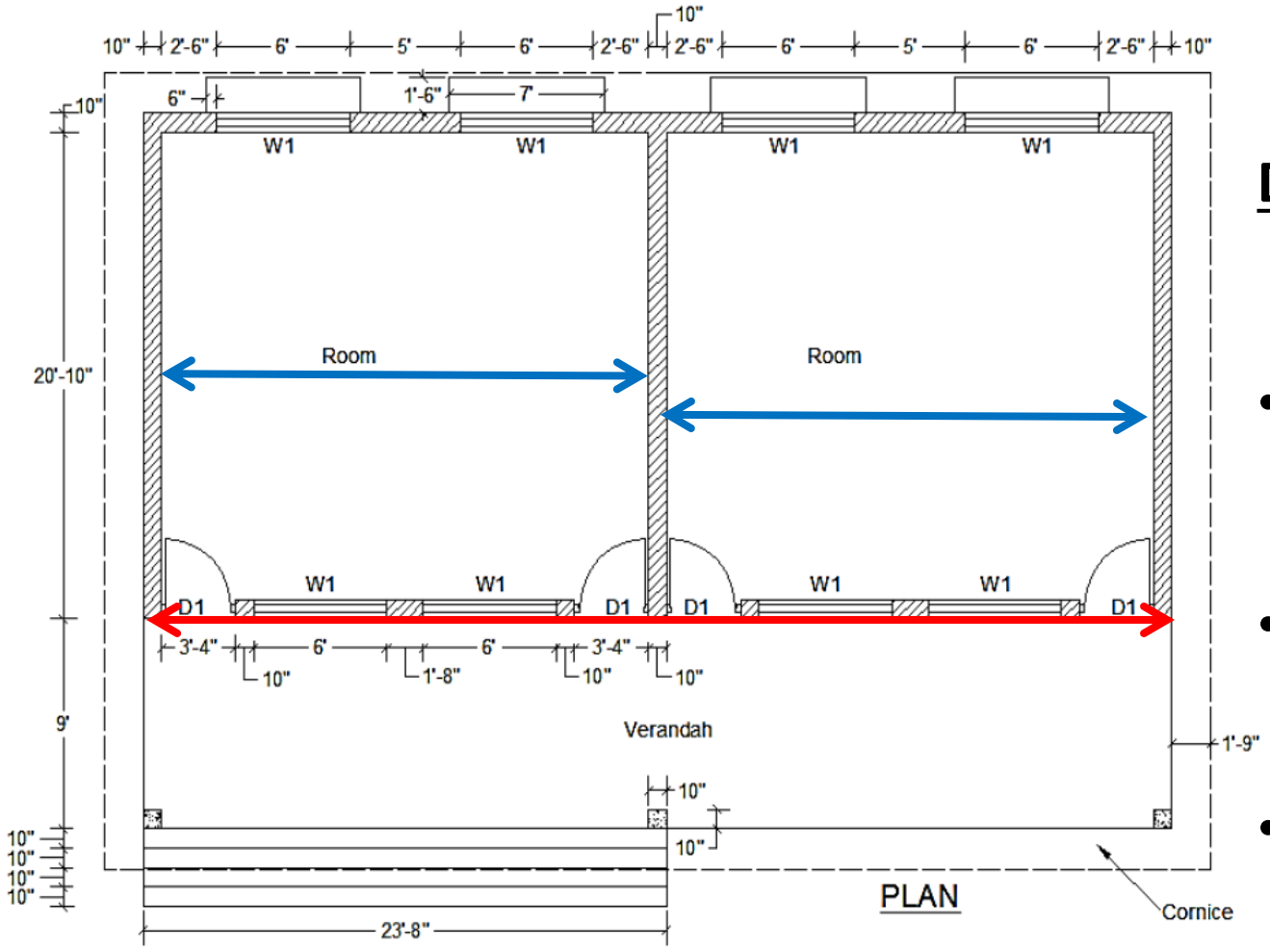
12. Brick walls in superstructure



Deduct (Lintel over Windows):

- Height= 6”
- Width= 10”
- Length= 7’

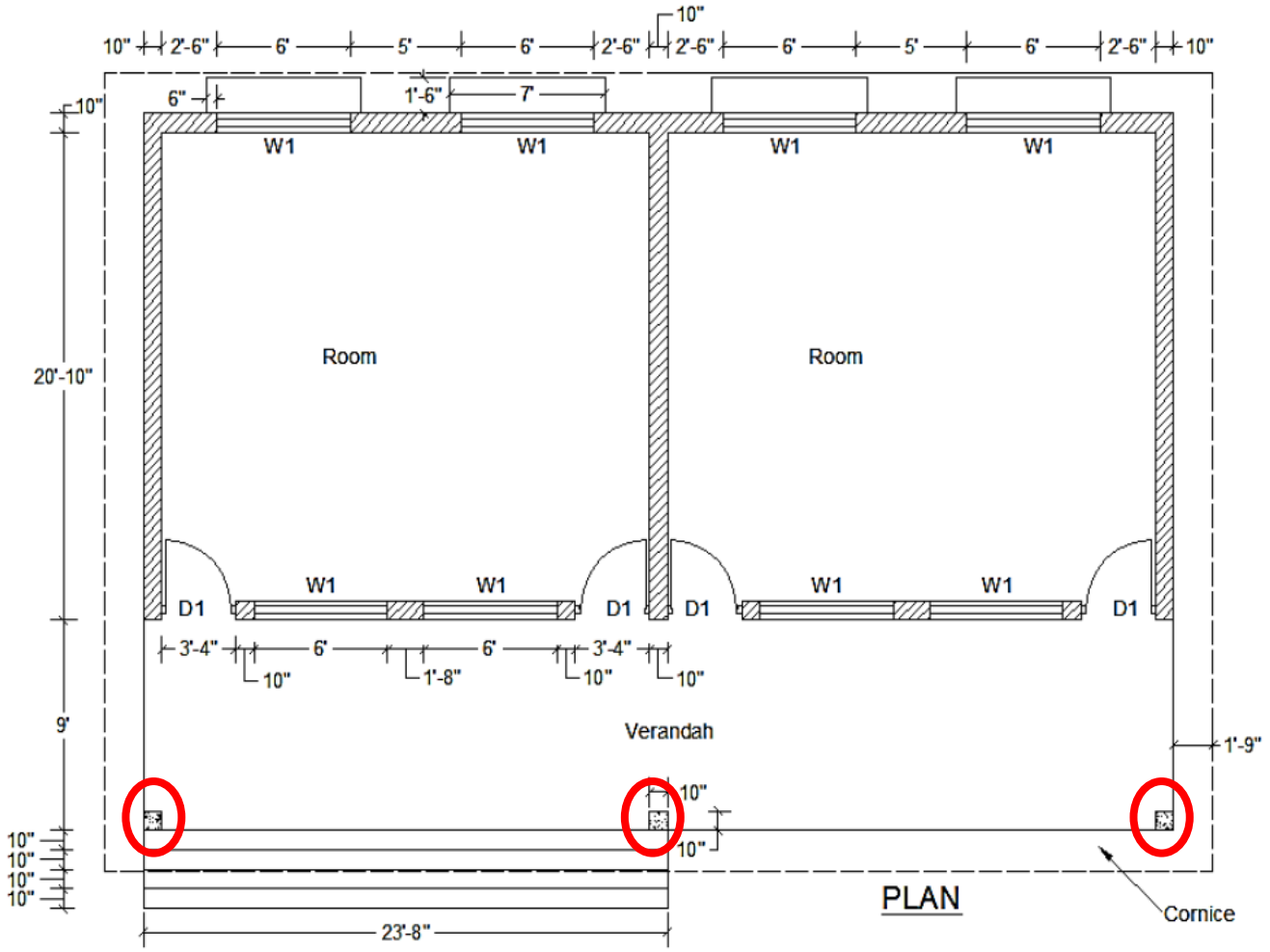
12. Brick walls in superstructure



Deduct (Lintel over W1+D1):

- Height= 6”
- Width= 10”
- Length= 2*22’+10”+2*6”

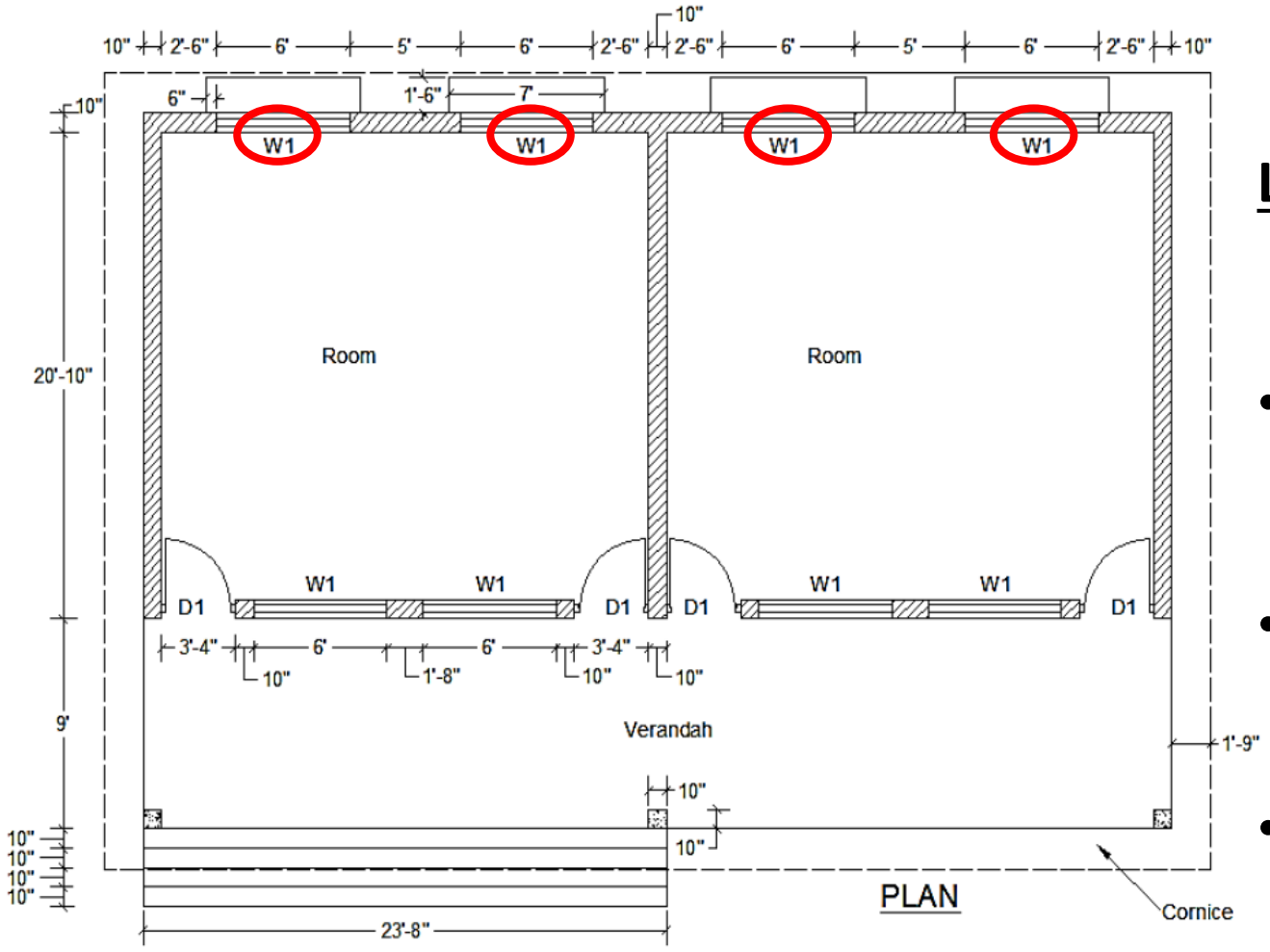
13. RCC in Columns



3 Columns:

- Length= 10”
- Width= 10”
- Height= 10’7”

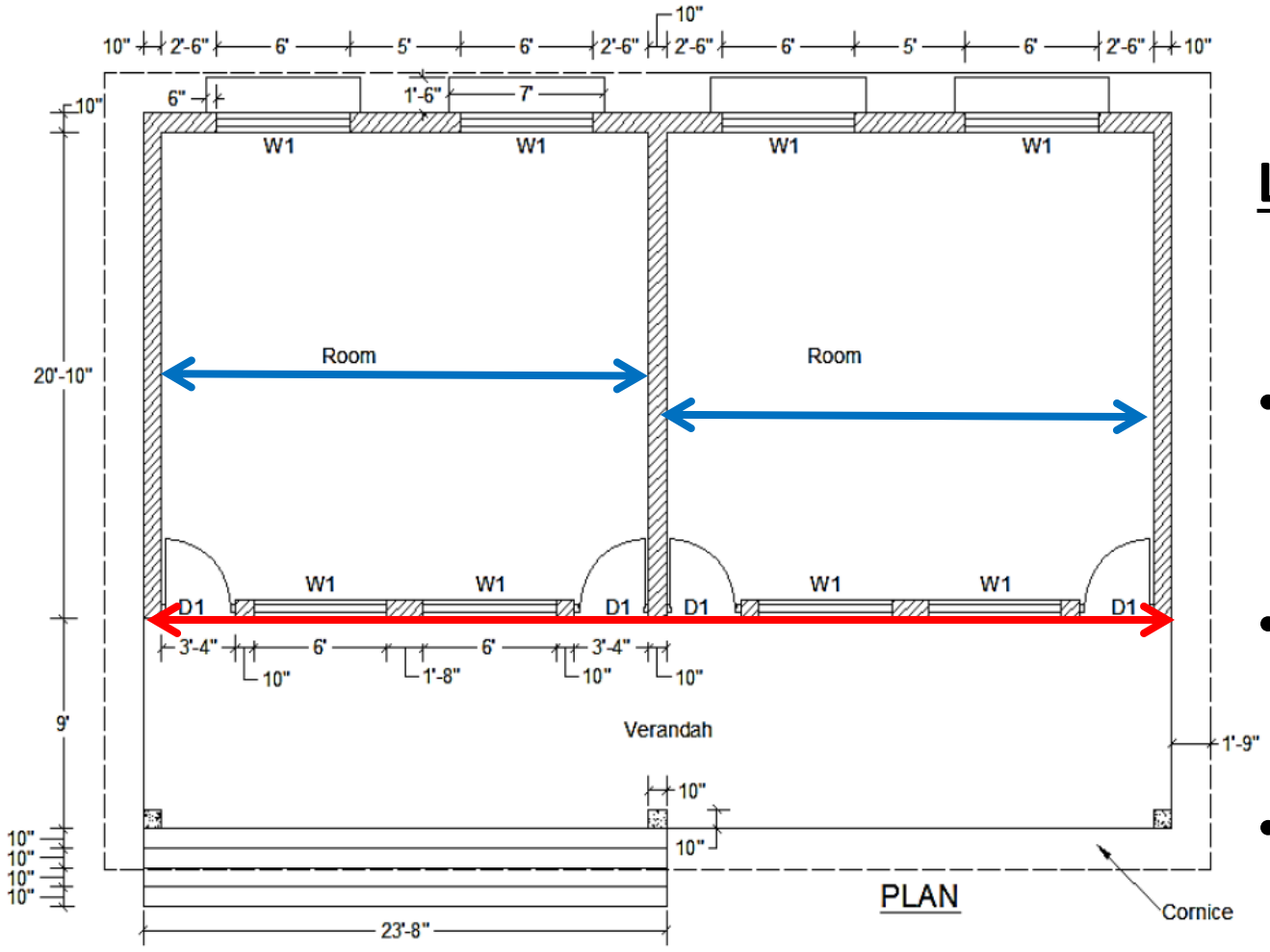
14. RCC in Lintel



Lintel over Windows:

- Height= 6”
- Width= 10”
- Length= 7’

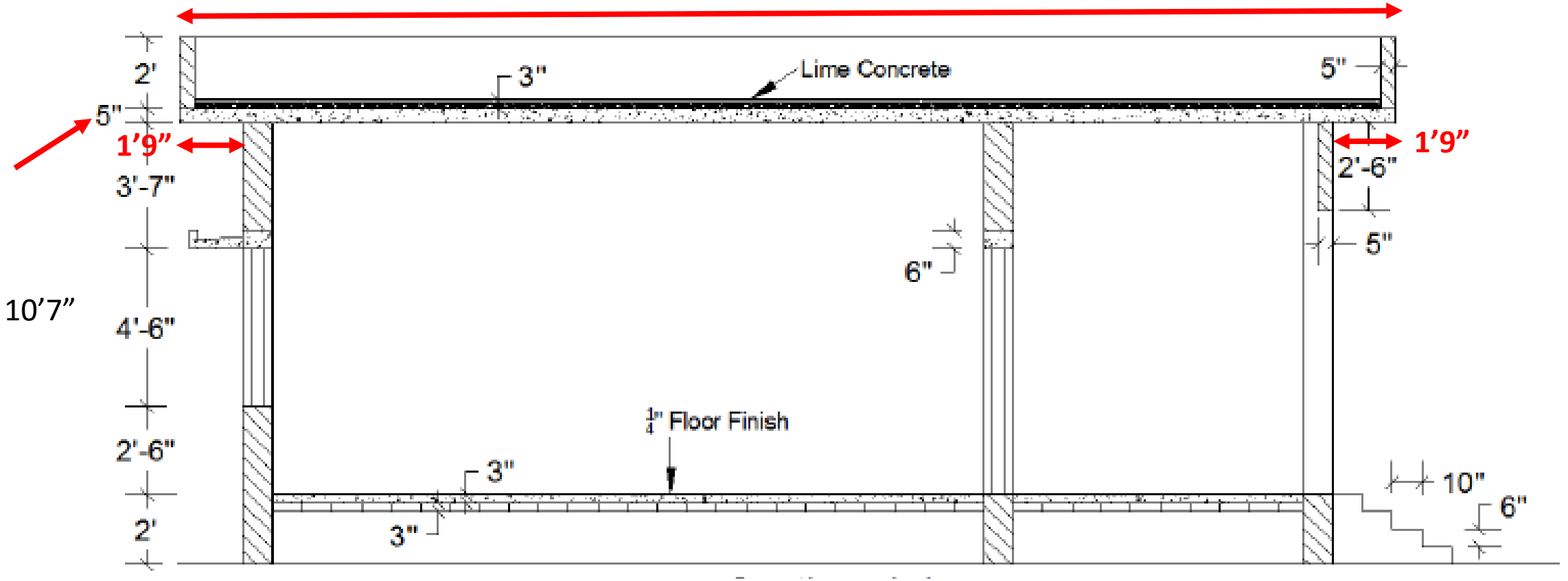
14. RCC in Lintel



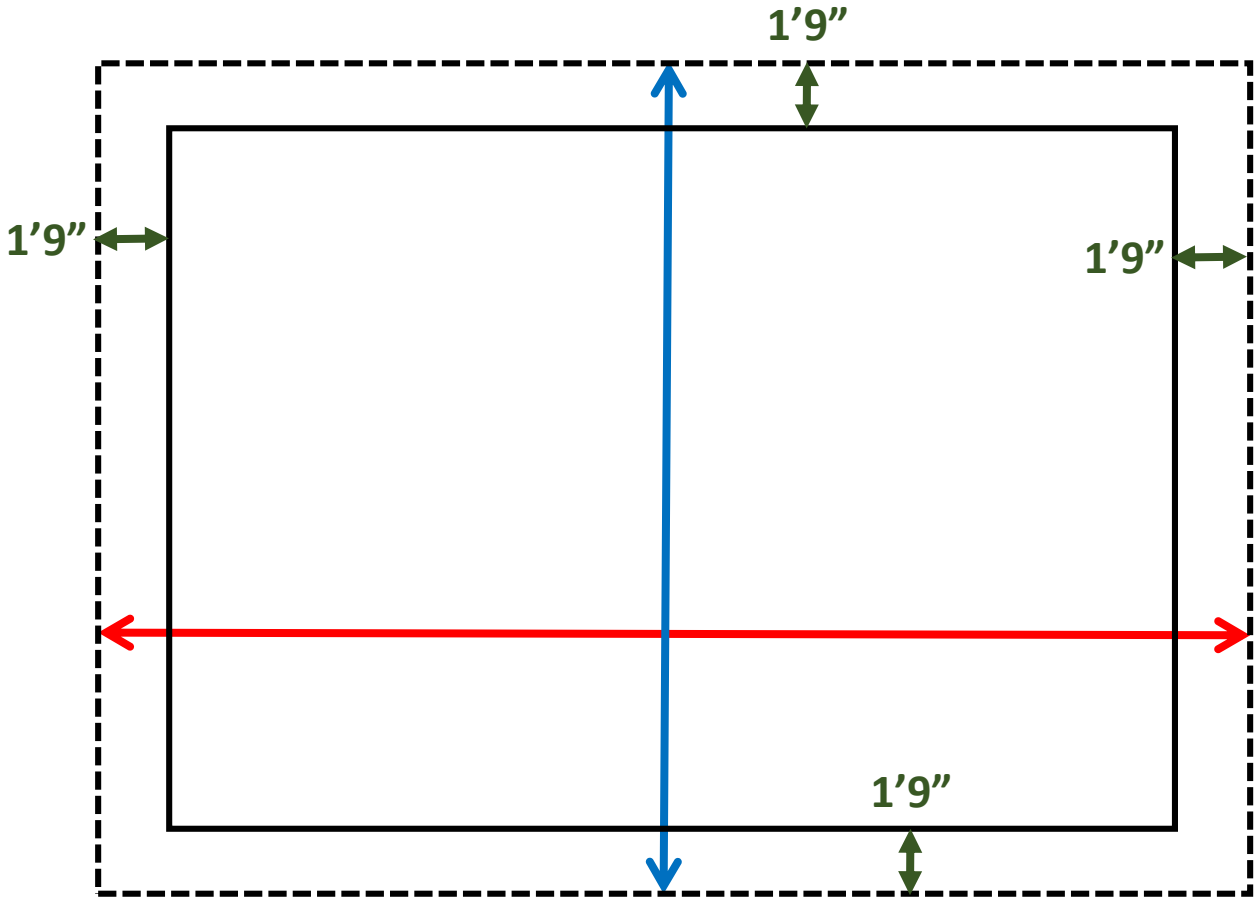
Lintel over W1+D1:

- Height= 6”
- Width= 10”
- Length= 2*22’+10”+2*6”

15. RCC in Roof



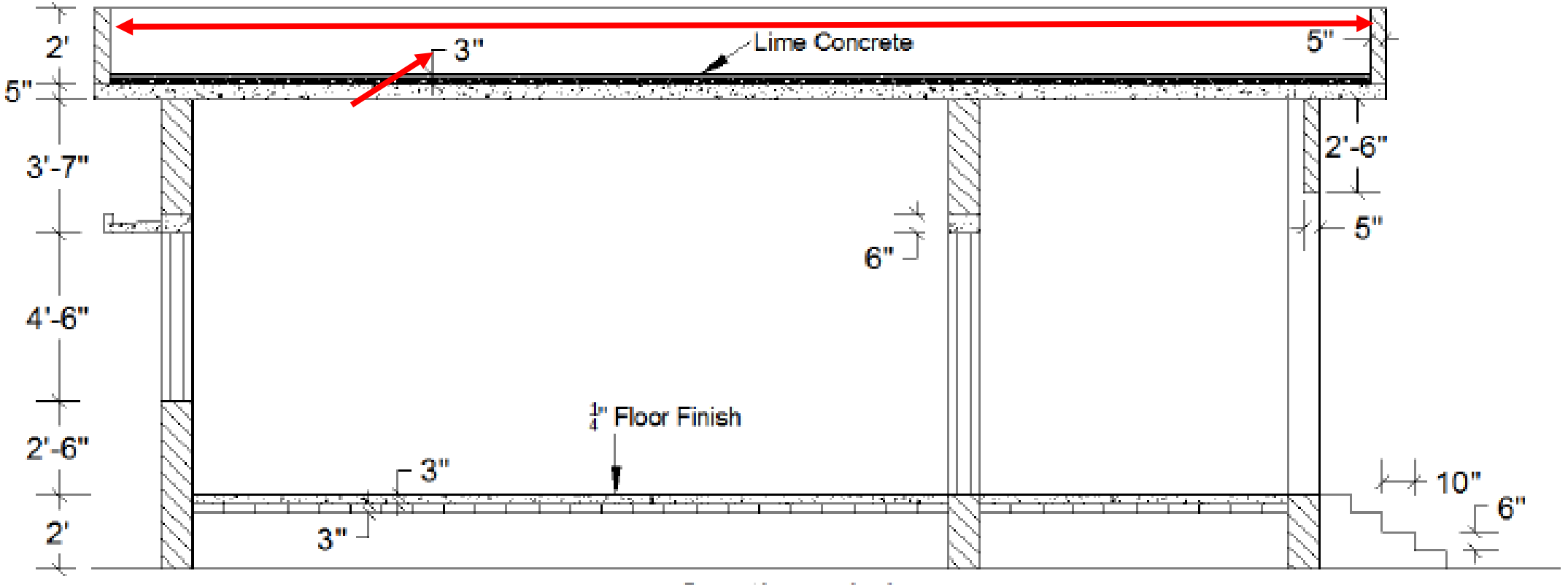
15. RCC in Roof



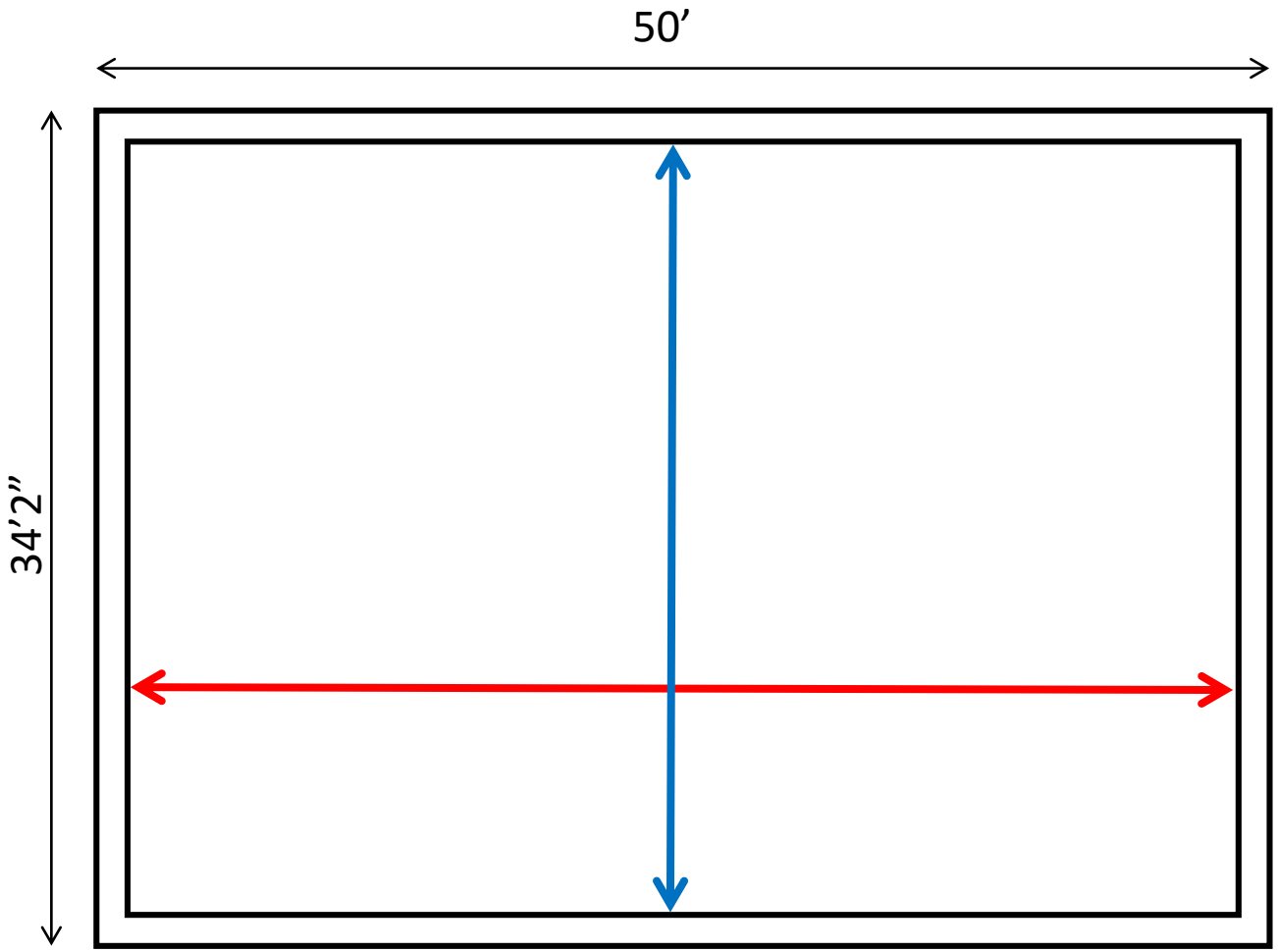
- Thickness= 5”
- Width= 30’8”+ 2*1’9”
- Length= 46’6”+2*1’9”

16. Lime Concrete in Roof

10’7”

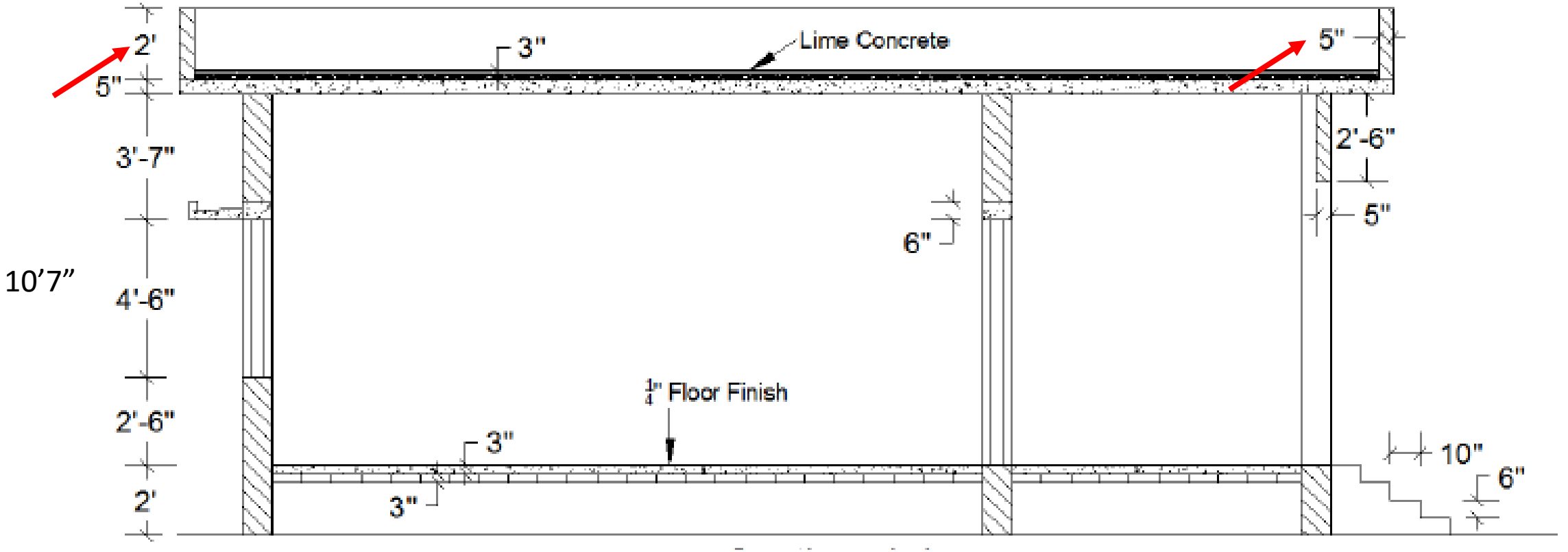


16. Lime Concrete in Roof

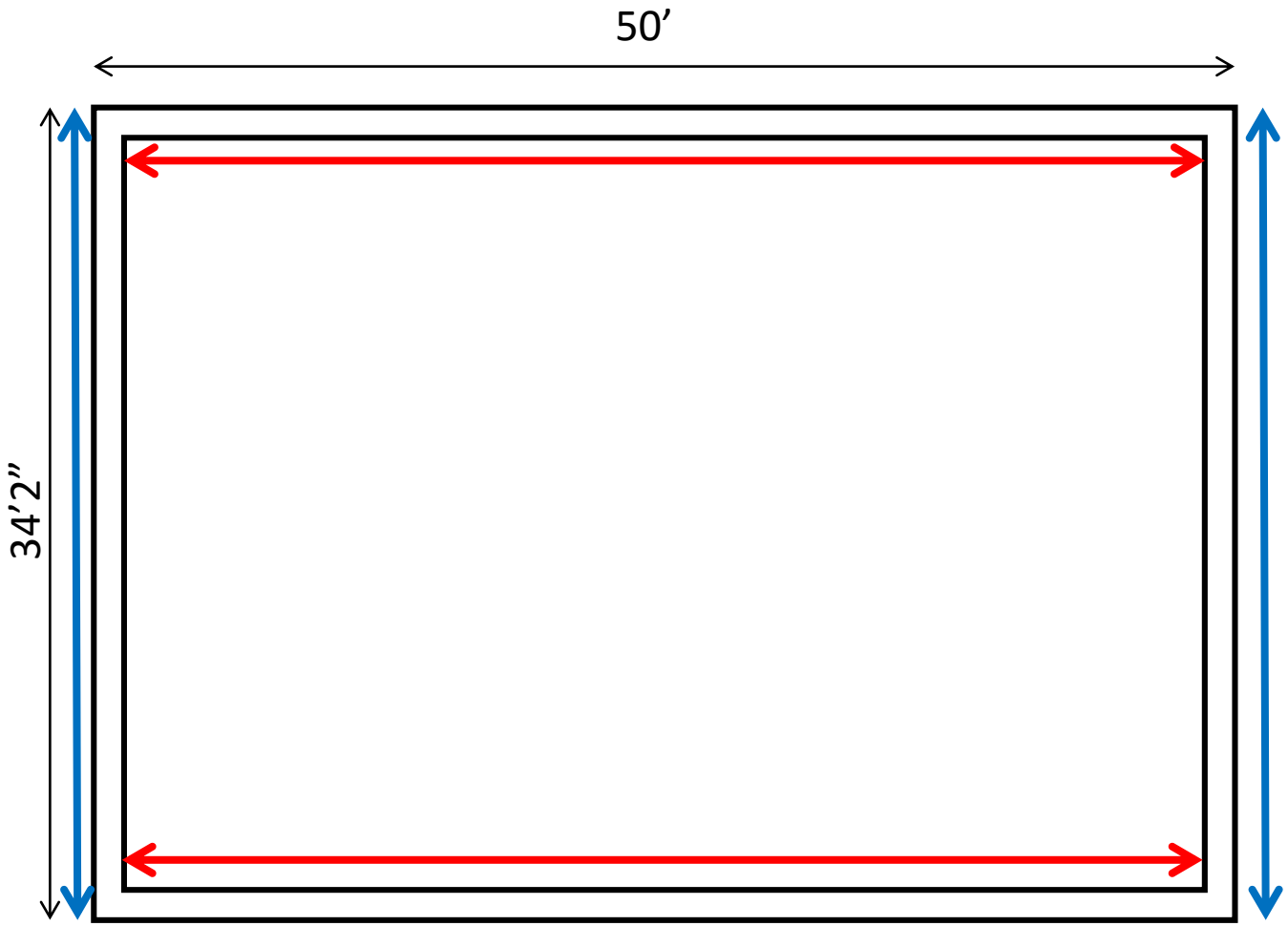


- Thickness= 3”
- Width= 34’2”-10”
- Length= 50’-10”

17. Bricks in Parapet

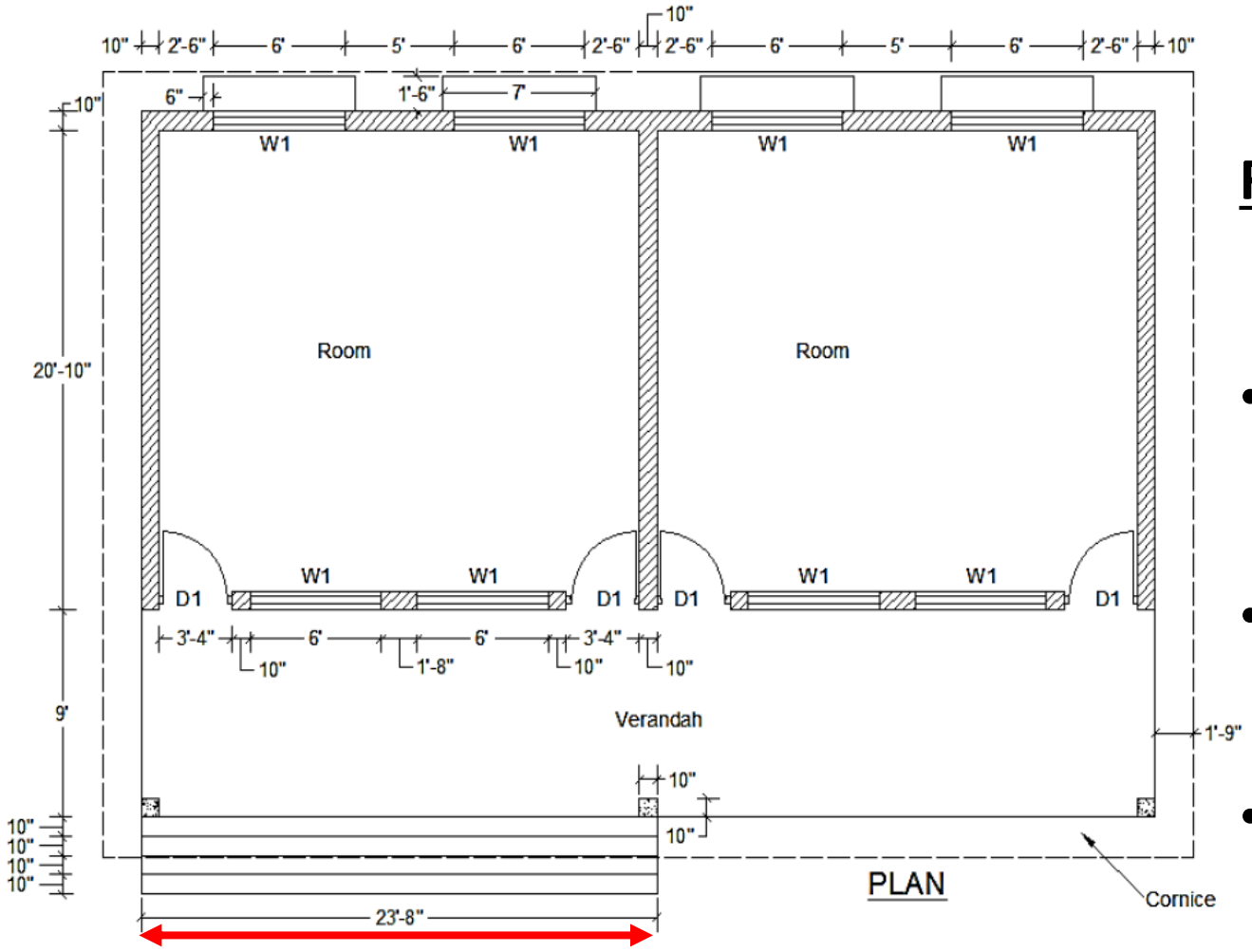


17. Bricks in Parapet



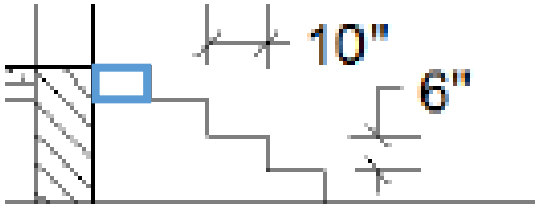
- Height= 2’
- Width= 5”
- Length= 2*((50’-10”)+34’2”)

18. Brickwork in Stairs

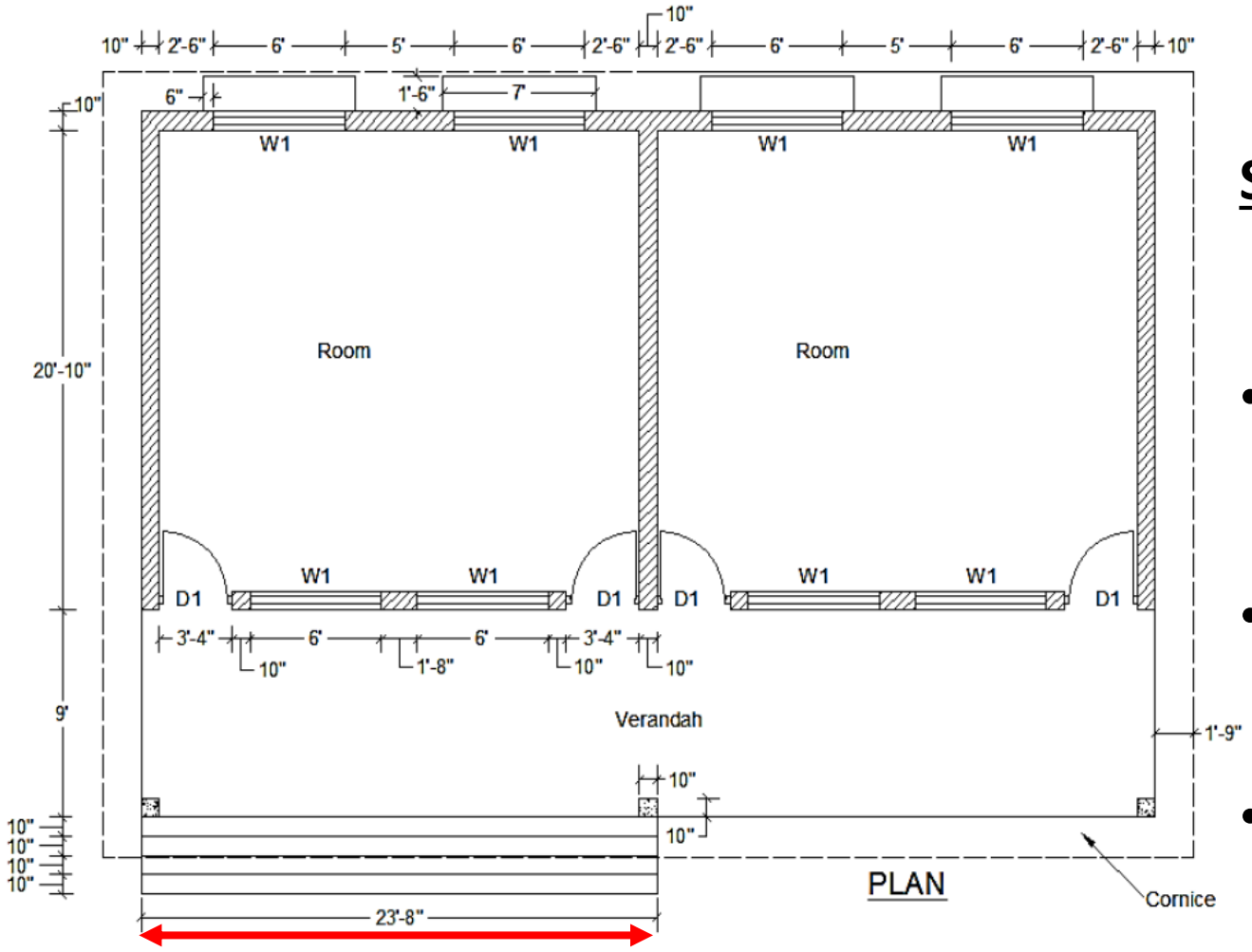


First Step:

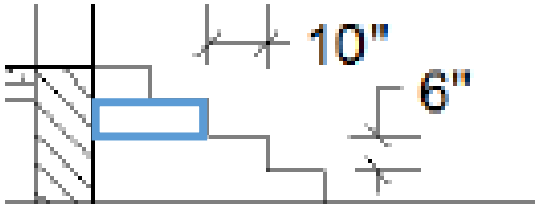
- Height= 6”
- Width= 10”
- Length= 23’8”



18. Brickwork in Stairs

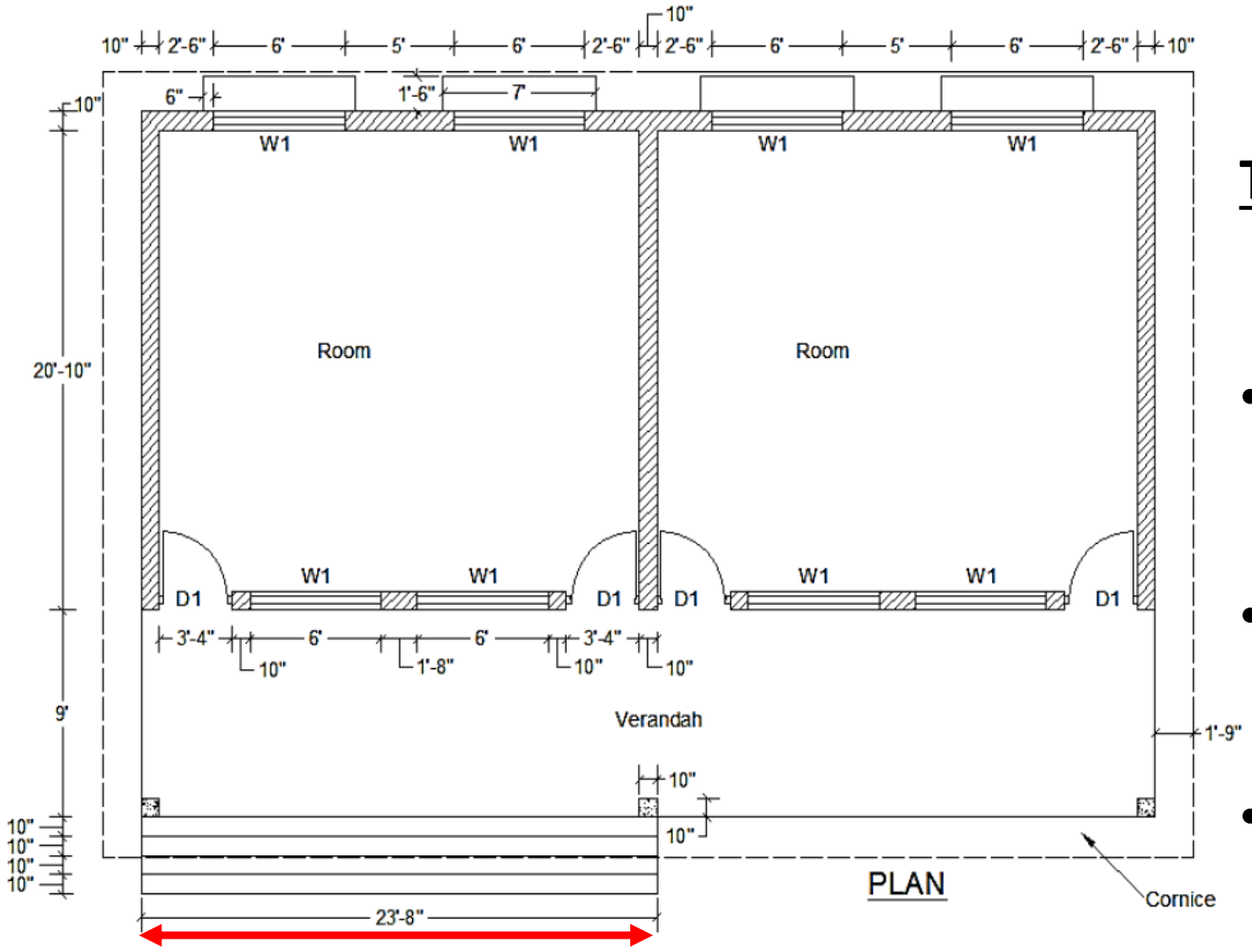


Second Step:



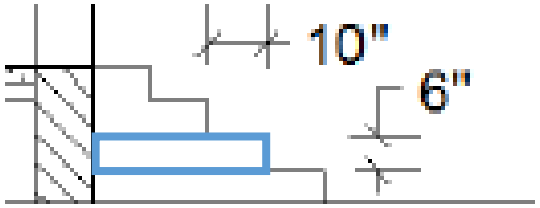
- Height= 6”
- Width= 20”
- Length= 23’8”

18. Brickwork in Stairs

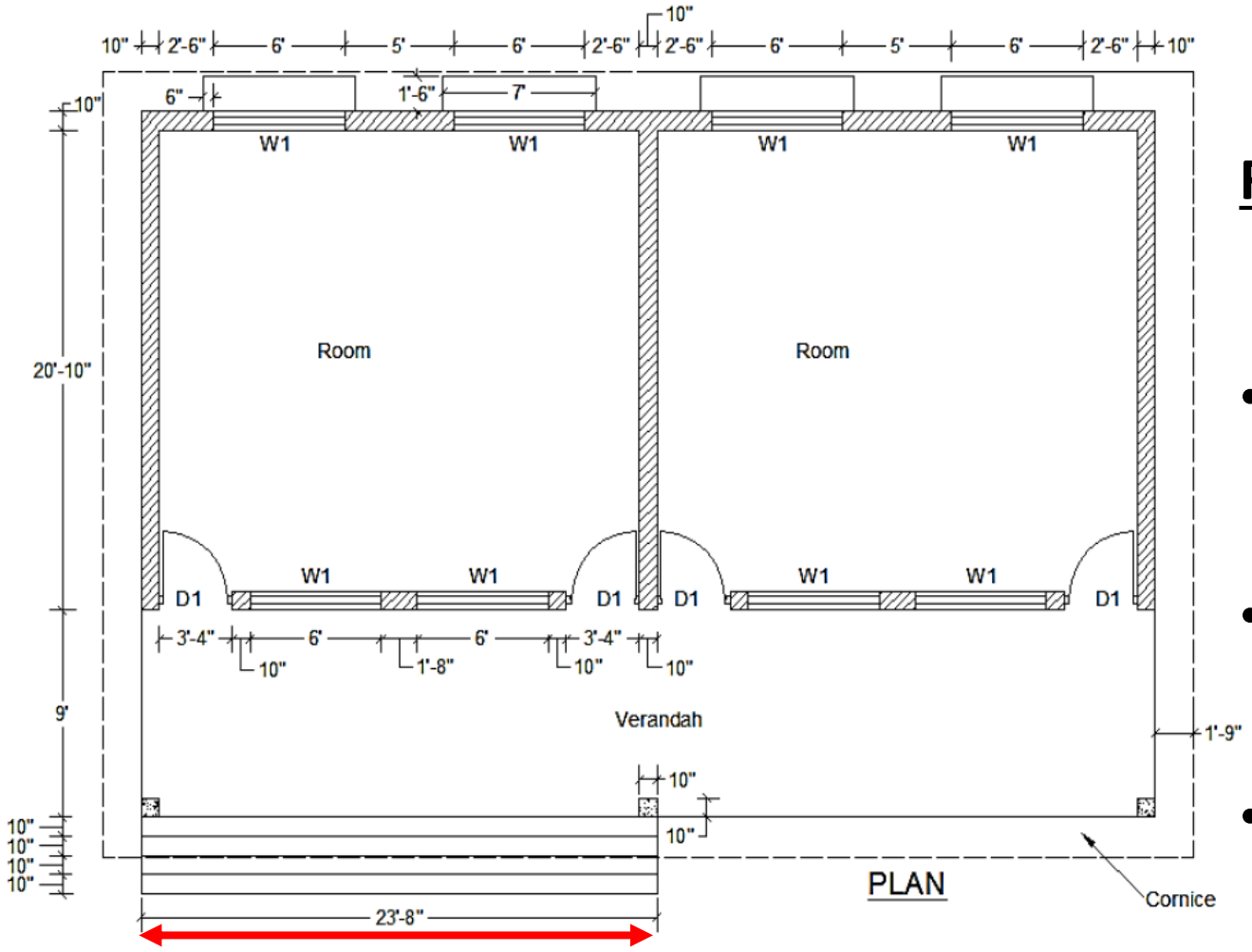


Third Step:

- Height= 6”
- Width= 30”
- Length= 23’8”

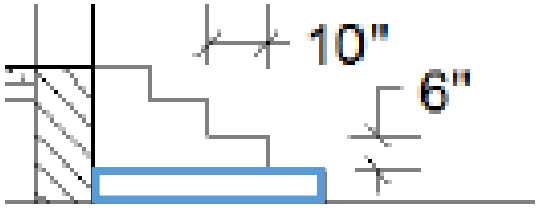


18. Brickwork in Stairs



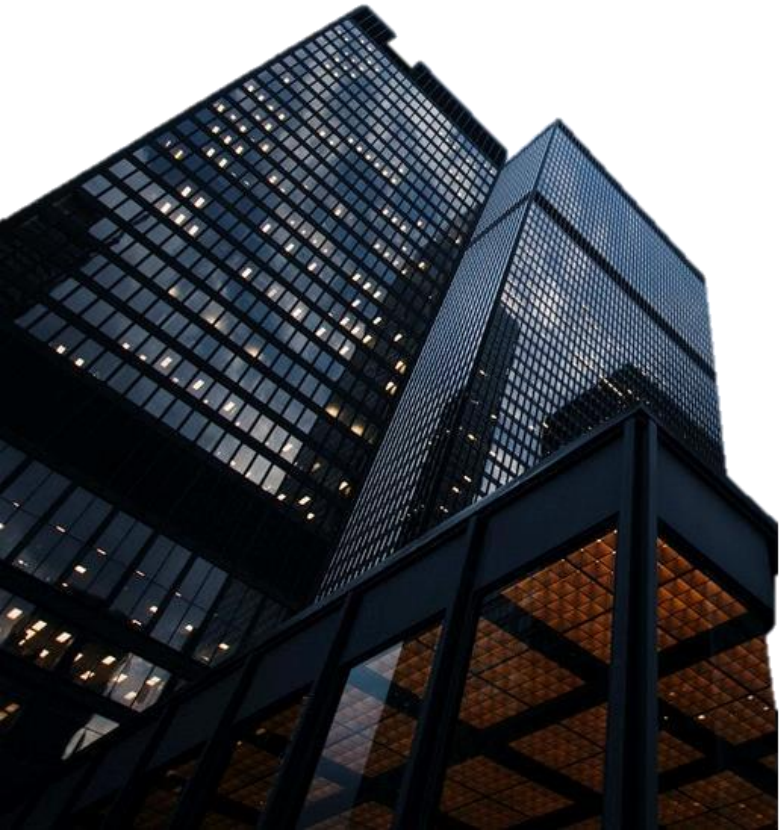
Fourth Step:

- Height= 6”
- Width= 40”
- Length= 23’8”



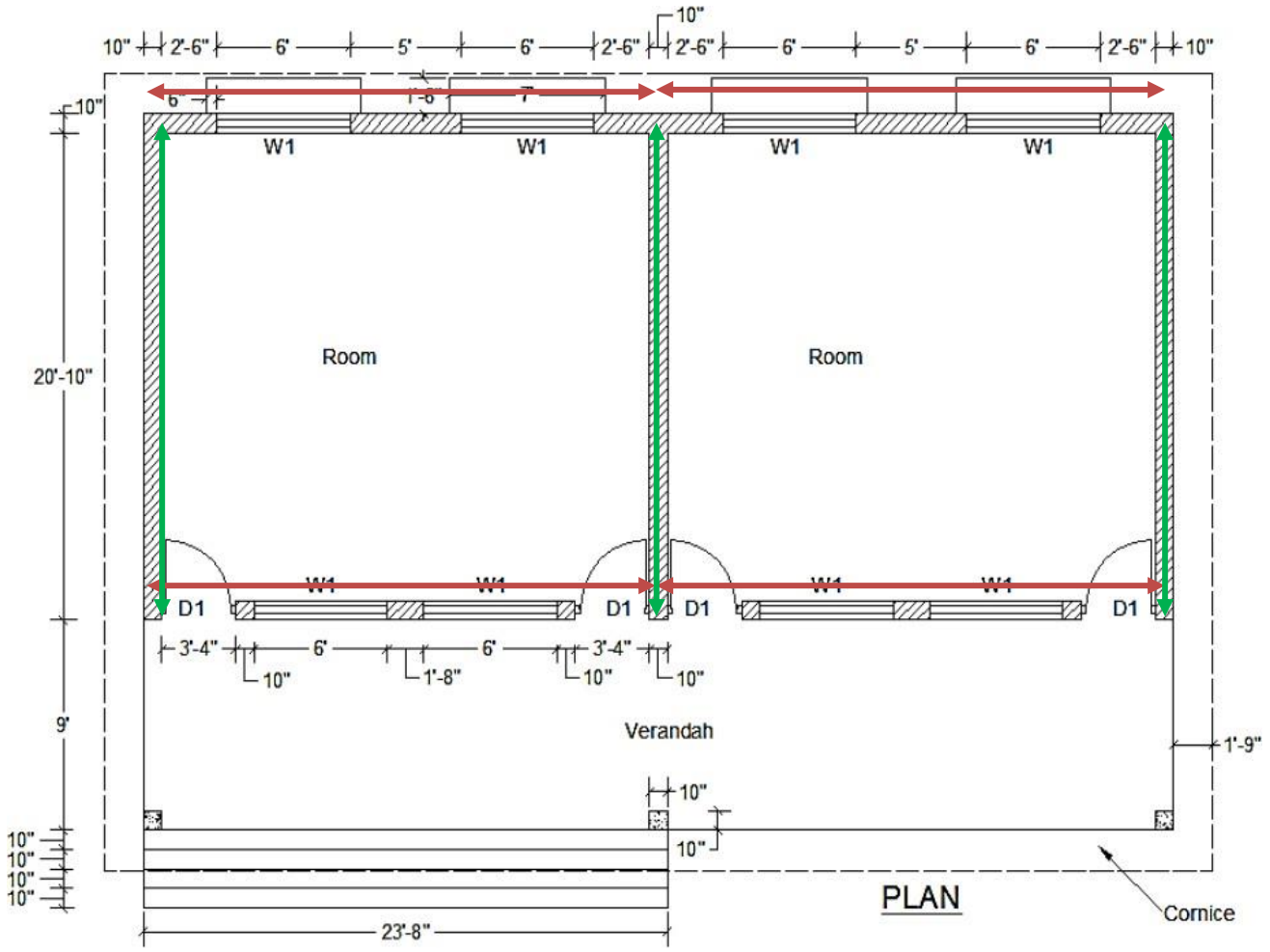
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19. Inside Plastering (Thickness-0.25”)



1) Inside wall

a) **Main room** = 2* {(22’+22’+20’+20’)* (10’7”-10” (skirting)) * 0.25”} = 34.12 cft

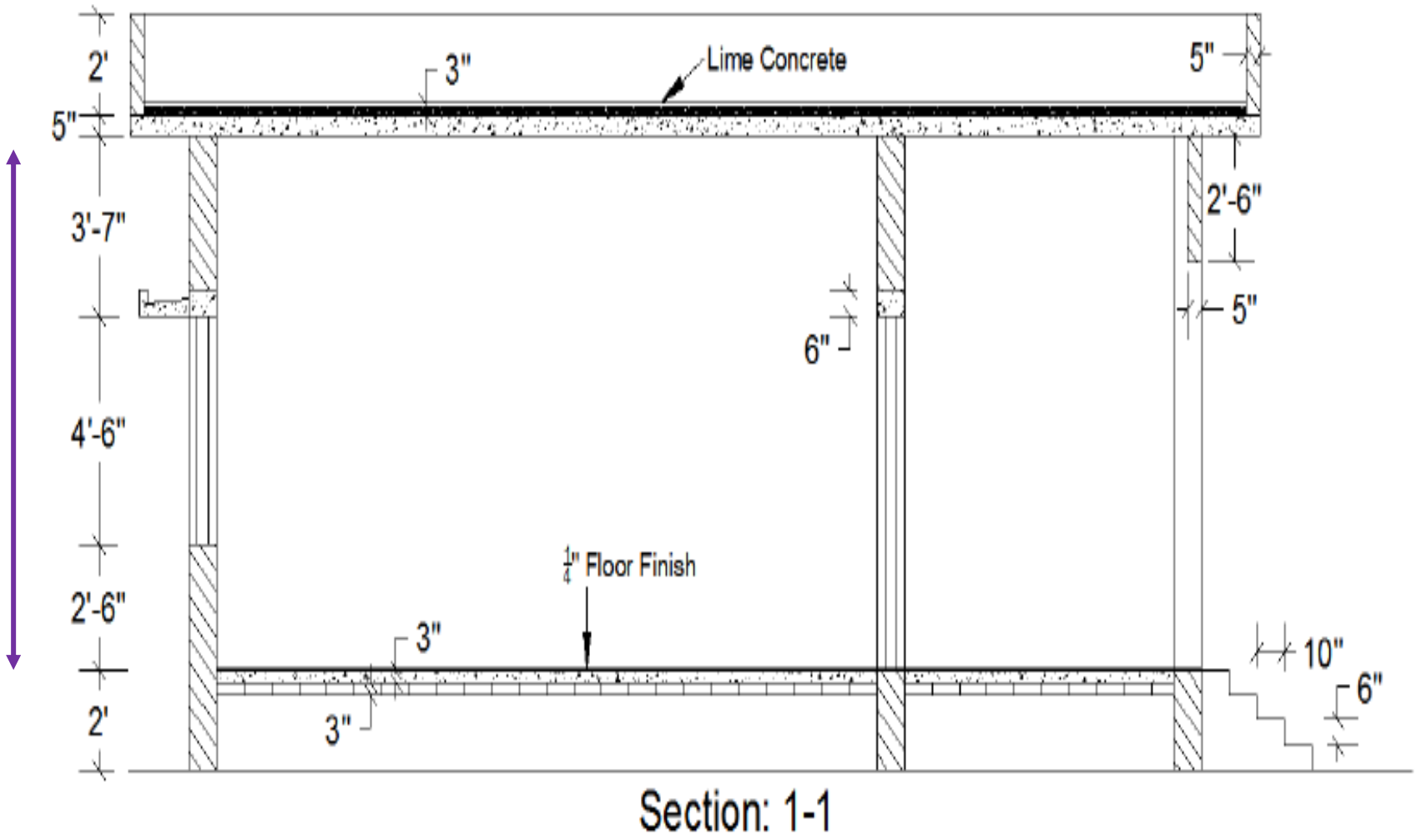
Deduction:

i. **Door** = 4 * (3’4”) * (7’-10” (skirting)) * 0.25 = 1.71 cft

ii. **Window** = 8 * 6’ * (4’6”) * 0.25” = 4.5 cft

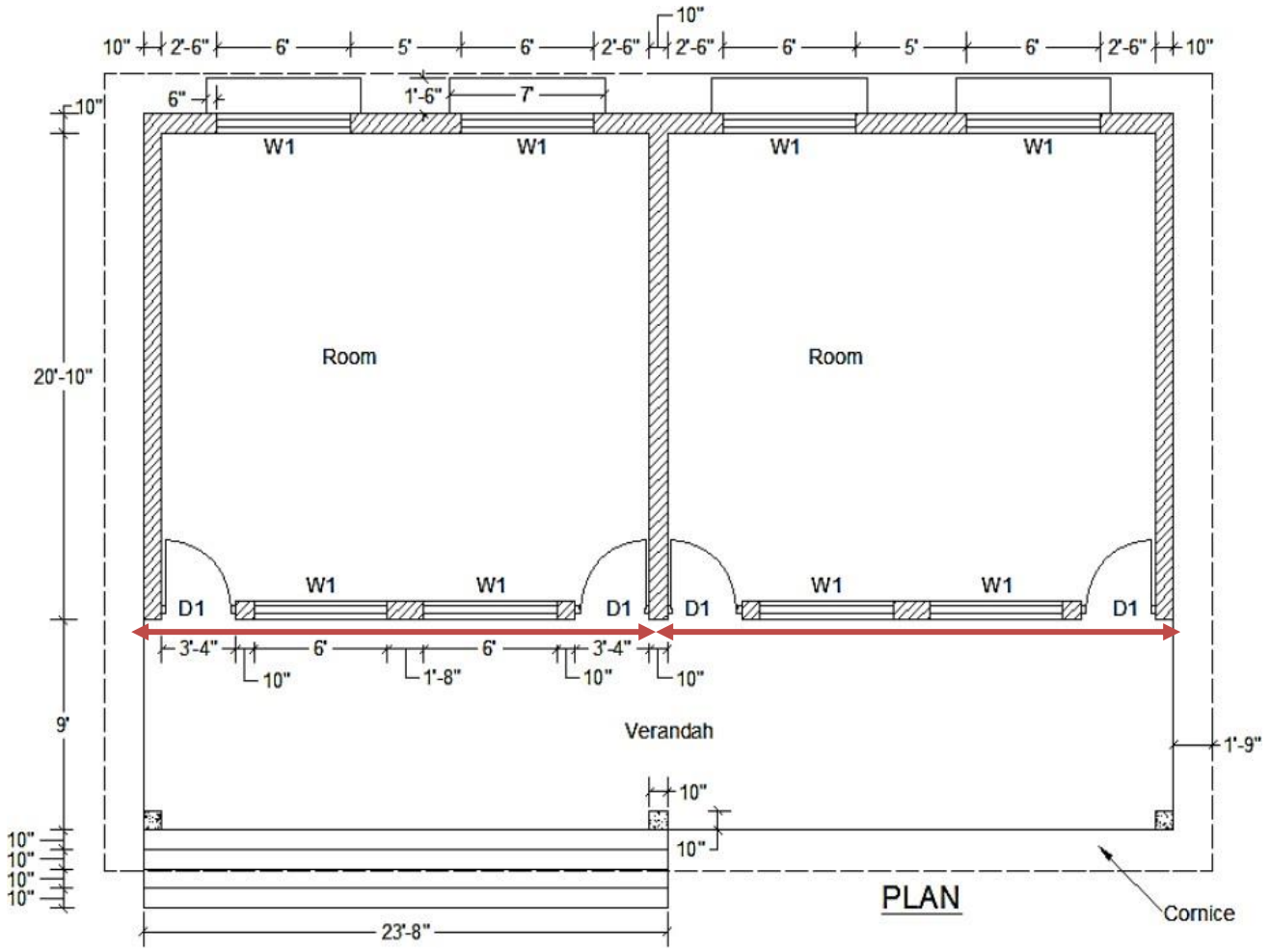
Total = (34.12 – 1.71 – 4.5) cft = 27.91 cft

19. Inside Plastering (Thickness-0.25”)



Section: 1-1

19. Inside Plastering (Thickness-0.25”)



1) Inside wall

b) Verandah= (23’3”+23’3”)* (10’7”-10” (skirting)) * 0.25”} = 9.45 cft

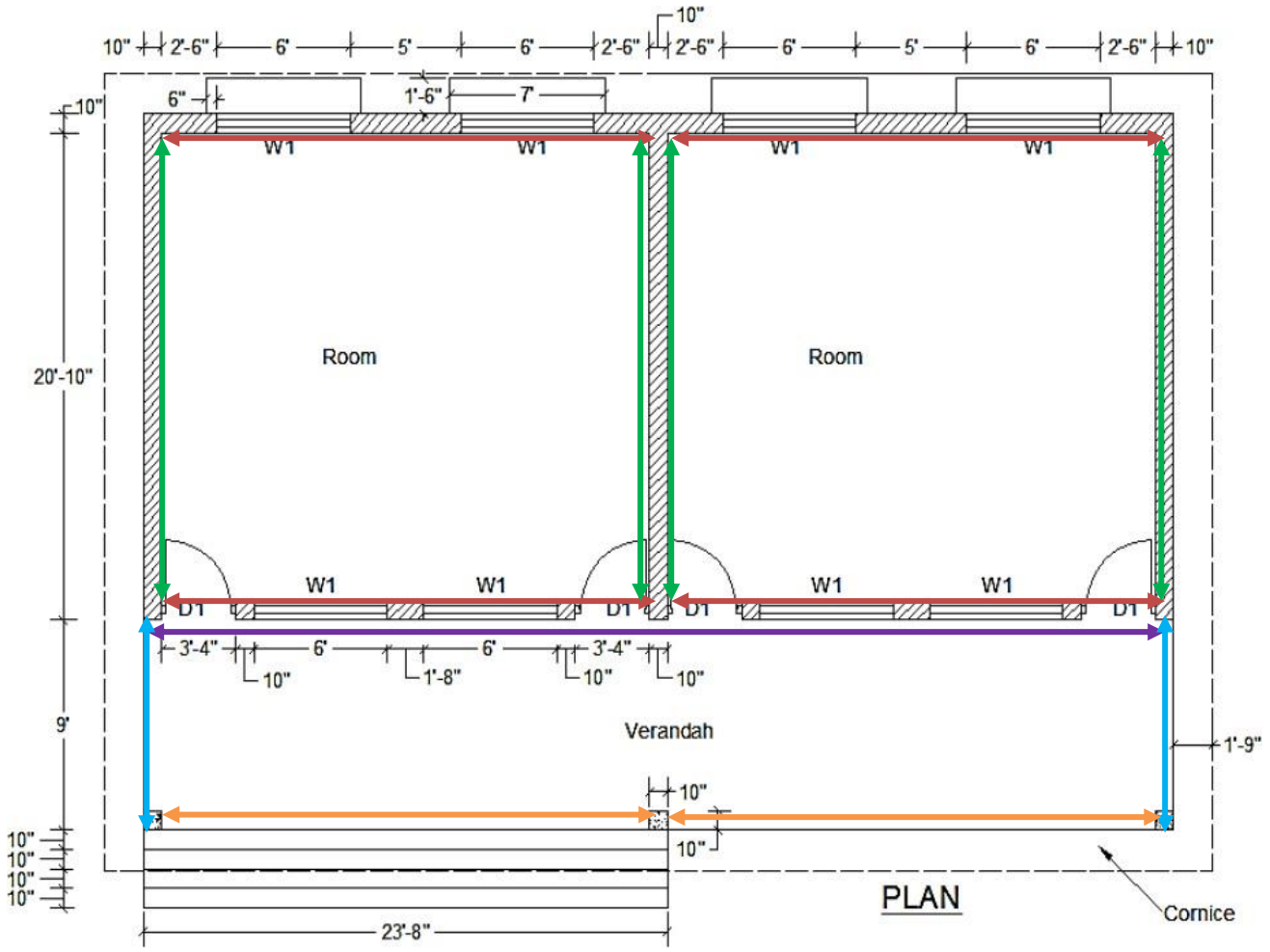
Deduction:

i. **Door** = 4 * (3’4”) * (7’-10” (skirting)) * 0.25 = 1.71 cft

ii. **Window** = 4 * 6’ * (4’6”) * 0.25” = 2.25 cft

Total = (9.45 – 1.71 – 2.25) cft = 5.49 cft

19. Inside Plastering (Thickness-0.25”)



2) Ceiling

a) Main room= 2*(22’*20’)*0.25” =

18.33 cft

b) Verandah= 46.5’*9’*0.25” = 8.72 cft

Deduction:

I. Column=

3*(10”*10”*0.25”)=0.04cft

II. Drop wall= (9’+9’+22’+22’) * 5”

*0.25” = 0.54 cft

Total = 18.33+8.72-0.04-0.54 = 26.47

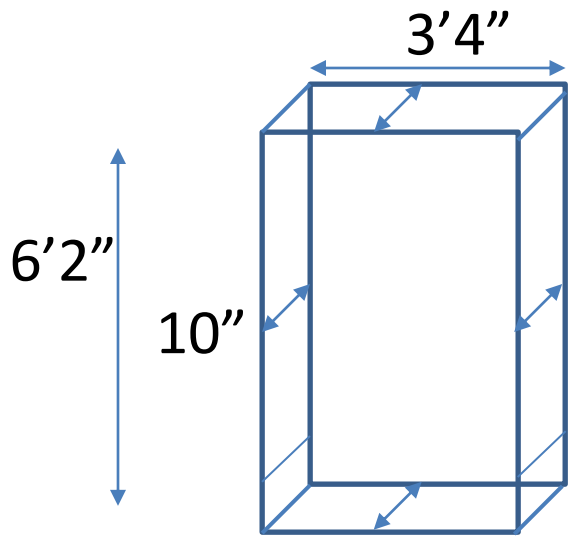
cft

19. Inside Plastering (Thickness-0.25”)

3) Edges

a) Door = 4* (6’2”+3’4”+6’2”) * 10” * 0.25” = 1.09 cft

b) Window = 8 * (4’6”*2+6’*2) * 10” * 0.25” = 2.92 cft



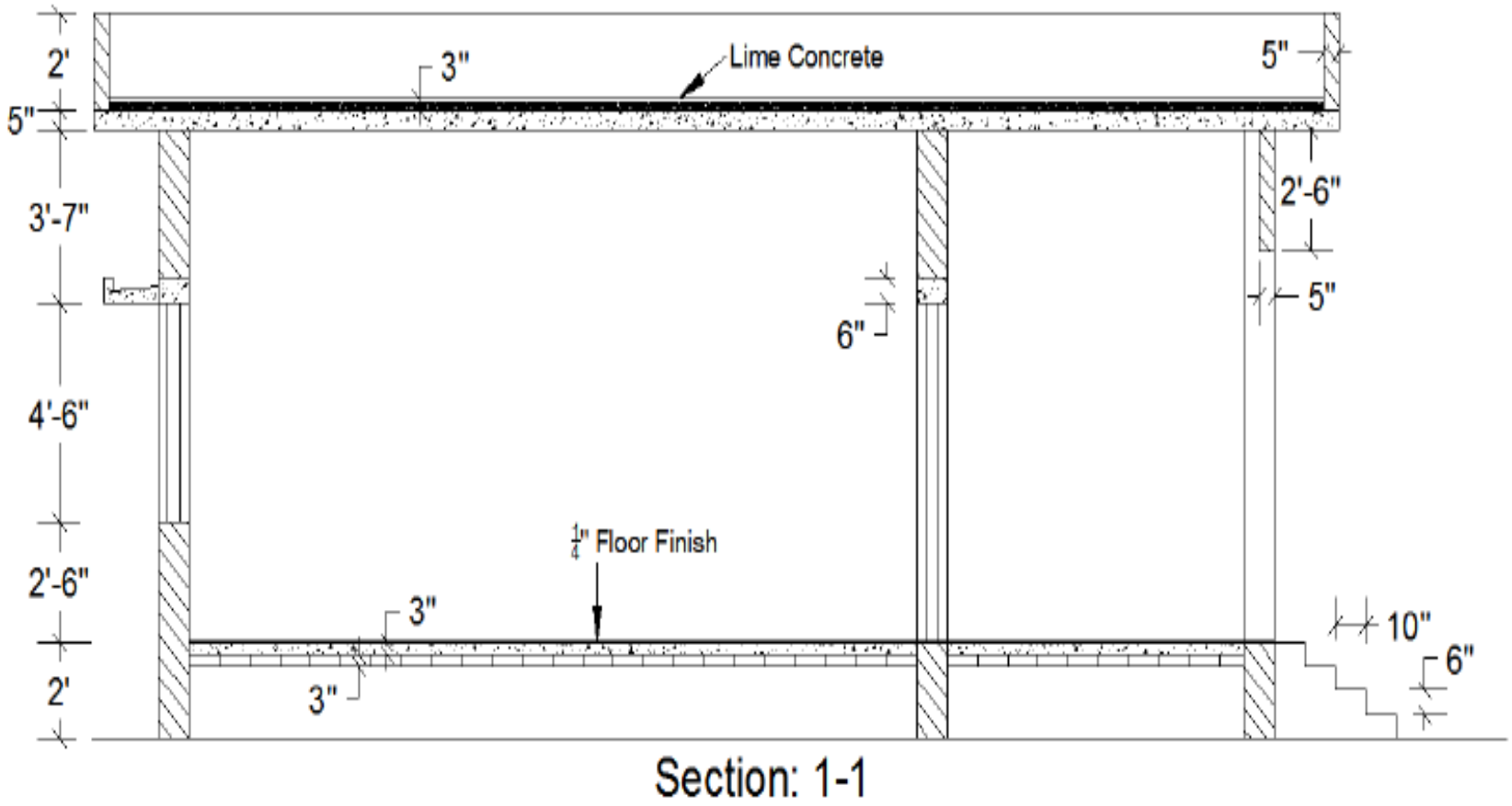
Total = 1.09+2.92 = **4.01** cft

19. Inside Plastering (Thickness-0.25”)

4) Drop wall Inside face

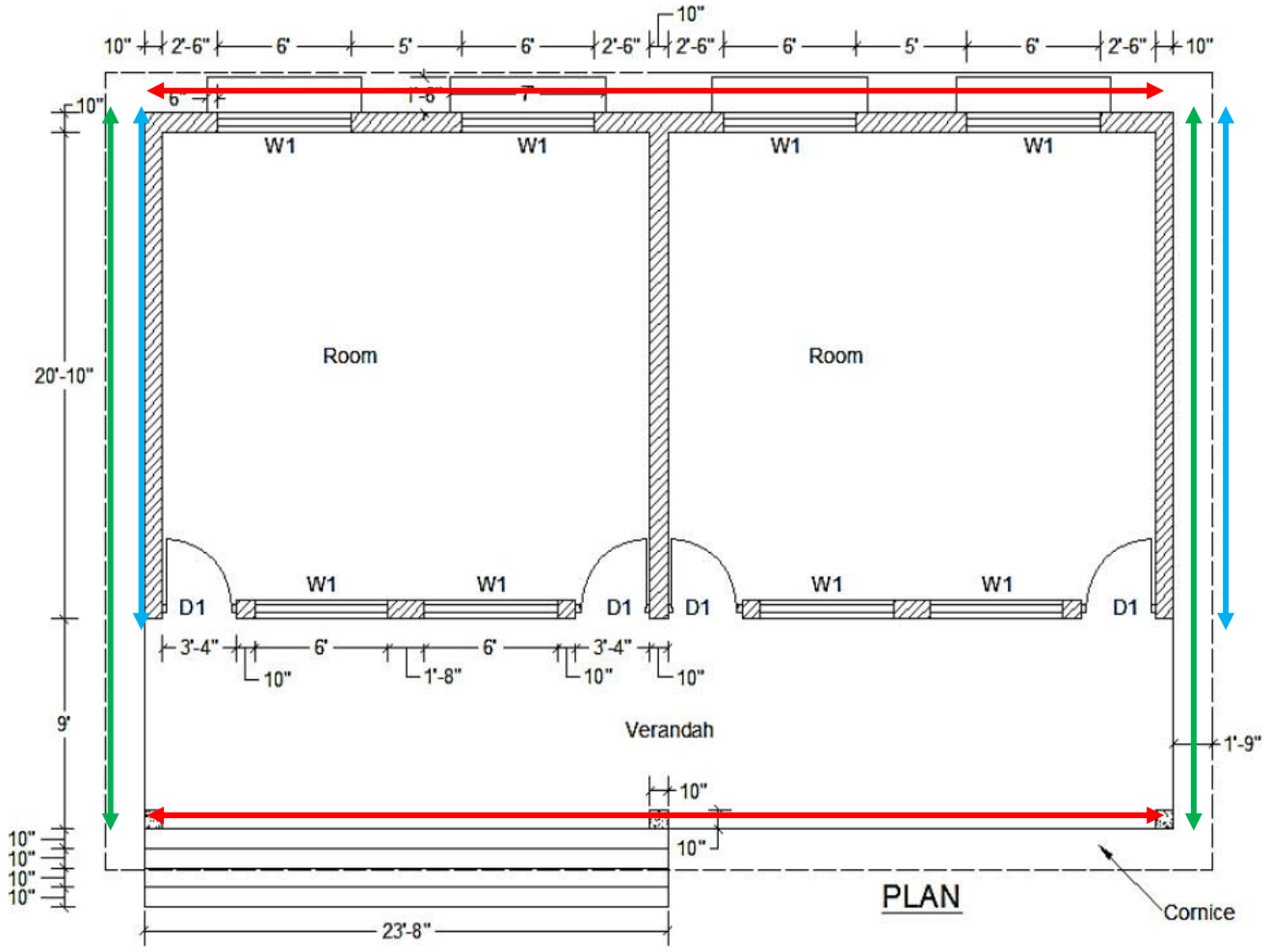
a) **Inside face =** 62’ * 2.5’ * 0.25” = 3.23 cft

b) **Bottom edge** = 62’ * 5” * 0.25” = 0.54 cft



Section: 1-1

20. Outside Plastering (Thickness-0.5”)

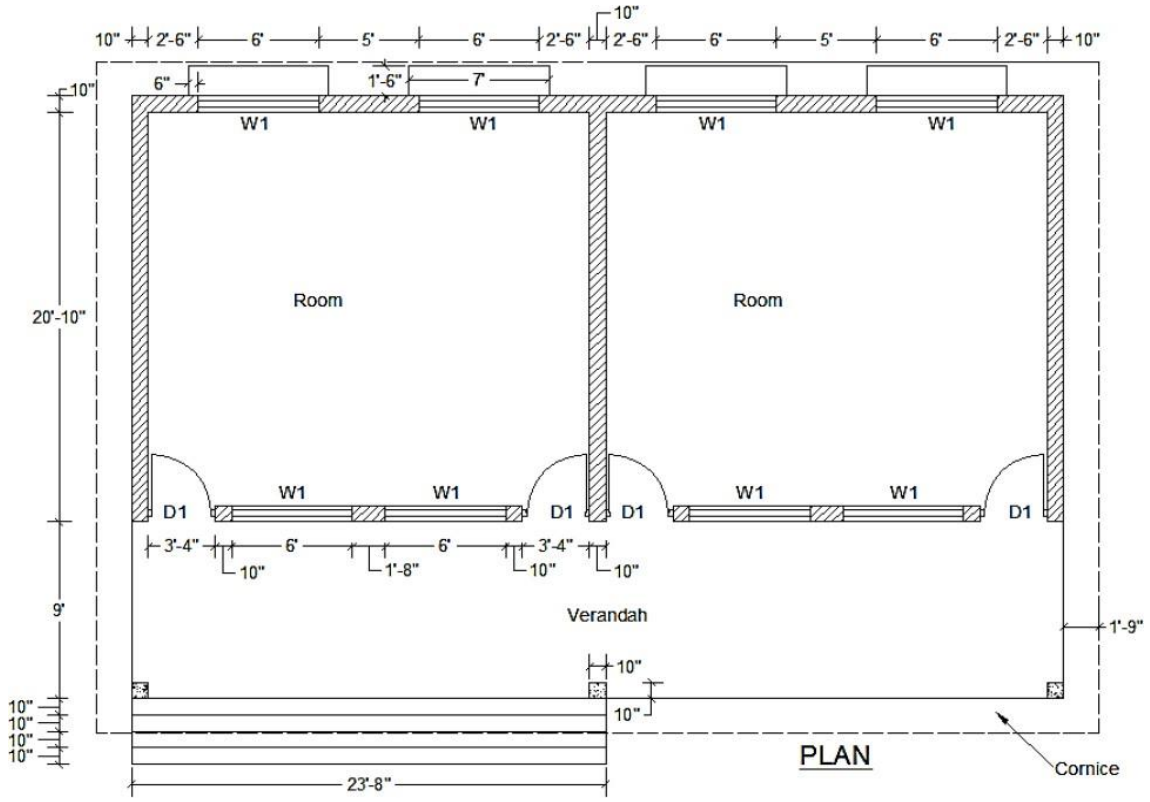


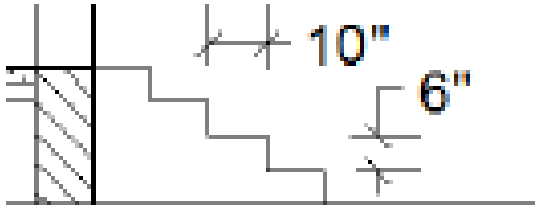
1) **GL to PL** = (46.5*2 + 30’8” *2) * 2’ * 0.5”
= 12.86 cft
Deduction:
Stair= 23’8” * 2’ * 0.5” = 1.97 cft

Total = 12.86-1.97 = 10.89 cft

2) **Outside wall** = (46.5+ 21’8”*2) * 10’
7” * 0.5” = 39.61 cft
Deduction
i. Window= 4 *4.5’* 6’*0.5”=4.5 cft
ii. Sunshade= 4 * 7’* 4”* 0.5 = 0.39 cft
Total = 34.72 cft

20. Outside Plastering (Thickness-0.5”)



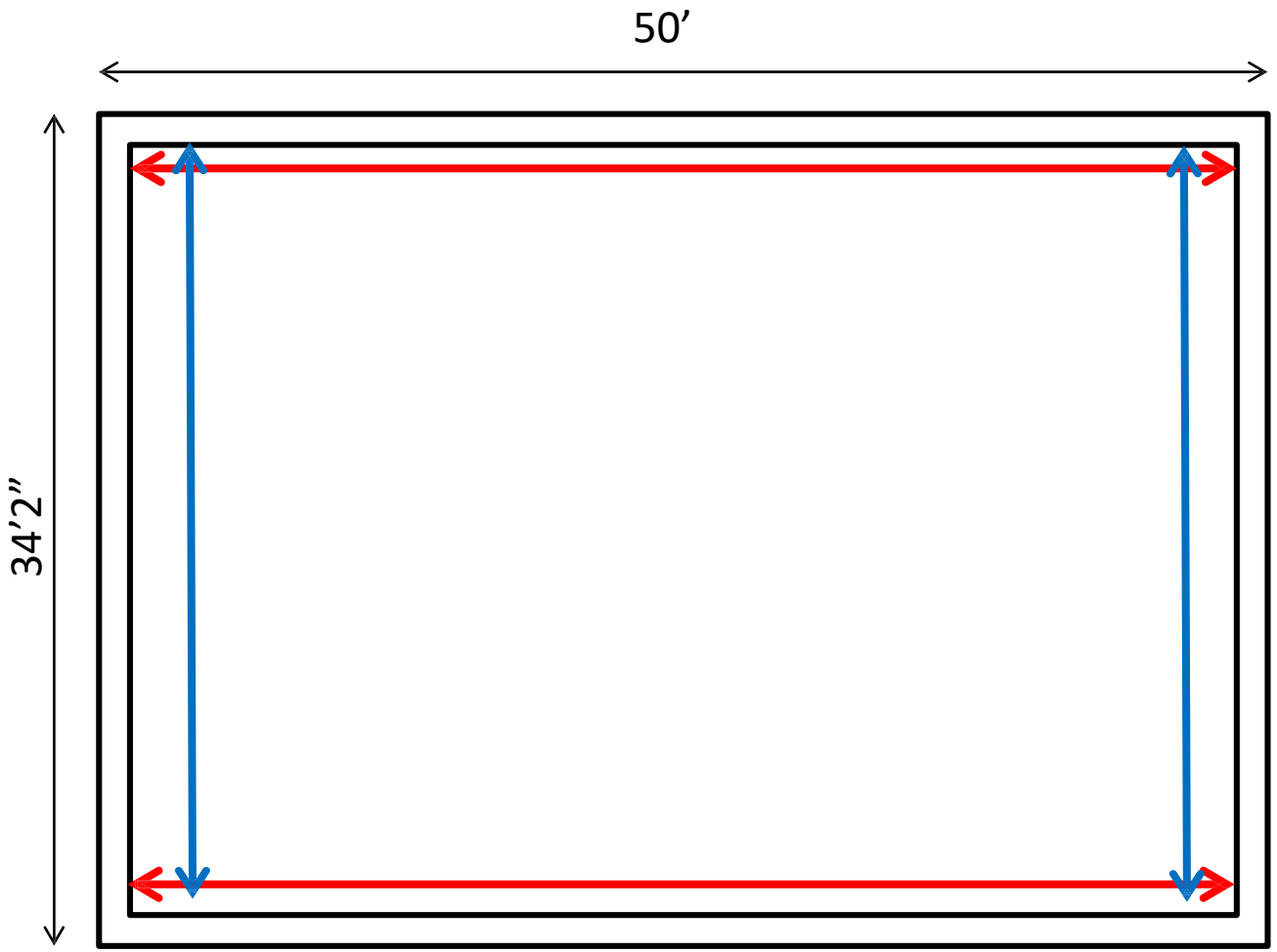


3) Column = 3* (10”*4)*(10’7”-10”) * 0.5”=
4.06 cft
Deduction:
Dropwall= 6* 2’6”*5”*0.5” = 0.26 cft
Total = 3.80 cft

4) **Stairs**

i. 1st step = 2*10”*6”*0.5” = 0.035 cft
ii. Similar for other steps.

20. Outside Plastering (Thickness-0.5”)



5) Parapet

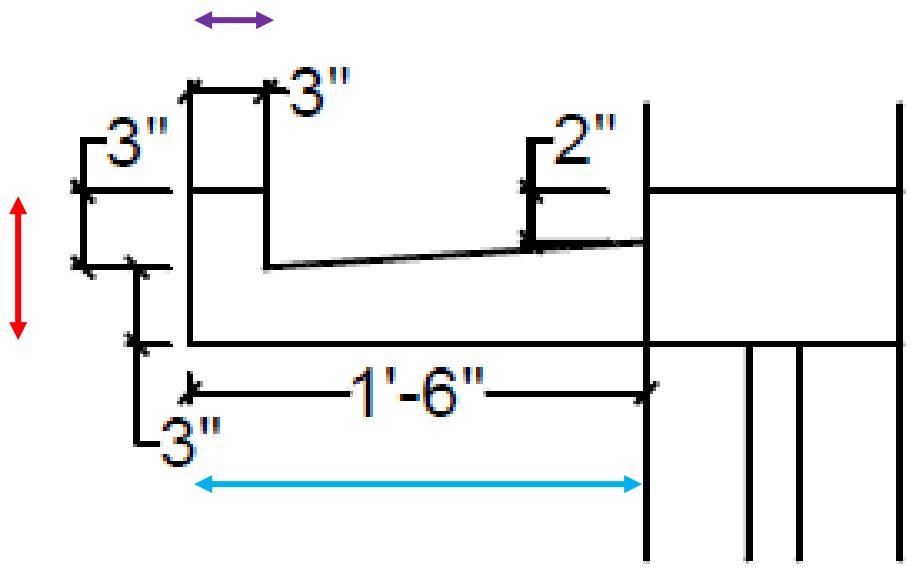
i. **Inside =** (49’2”*2+ 33’4”*2)*2’ *

0.5” = 13.75 cft

ii. **Outside = Similar**

iii. **Top =** Similar

20. Outside Plastering (Thickness-0.5”)



6) Sunshade

i. Bottom face = 4* 7’ * 1’6” * 0.5” = 1.75 cft

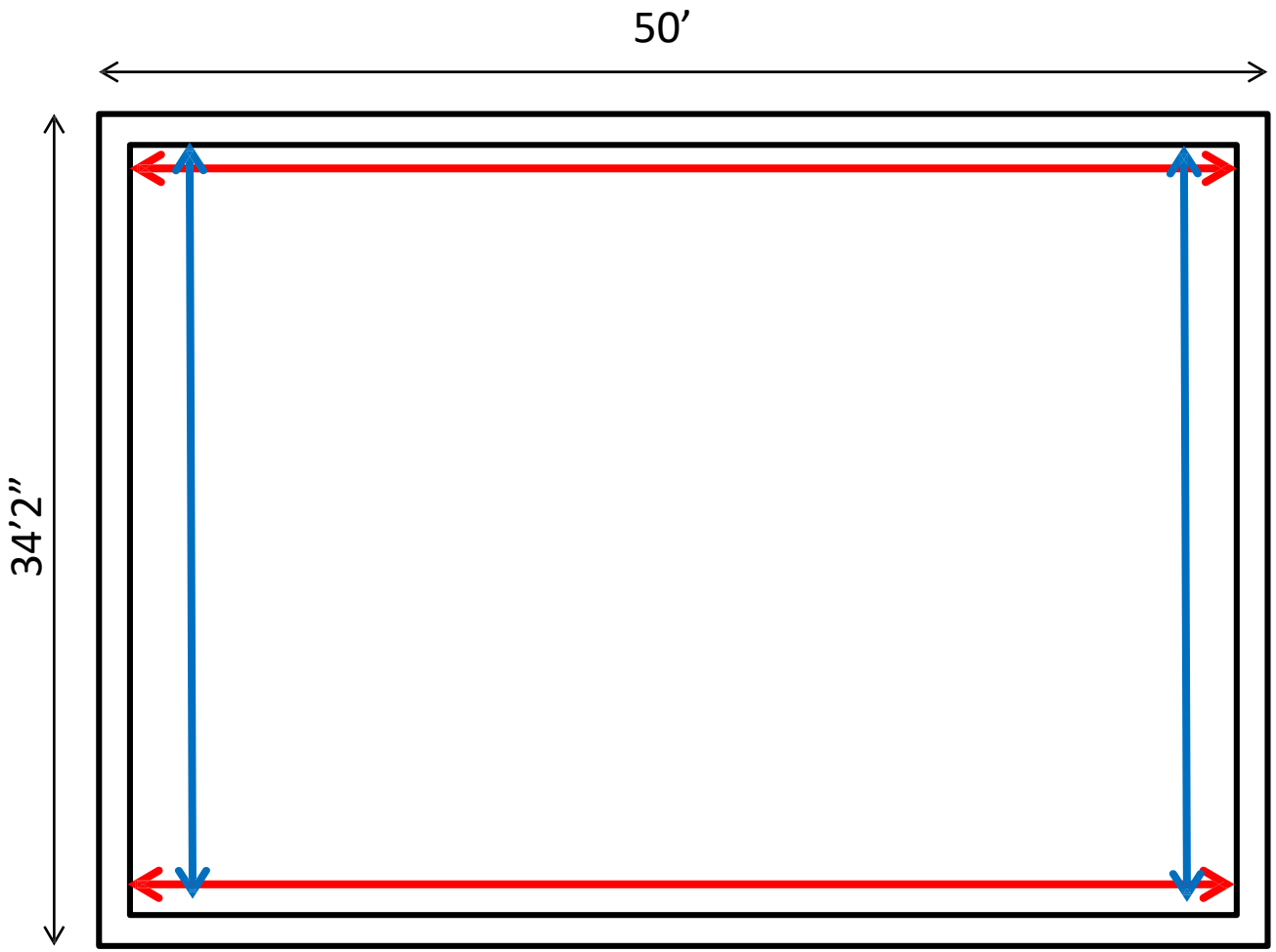
ii. Front face = 4* 7’* 6” * 0.5” = 0.58 cft

iii. Top face = 4* 7’ * 3” * 0.5” = 0.29 cft

iv. Inside face =

v. Side edge =

20. Outside Plastering (Thickness-0.5”)

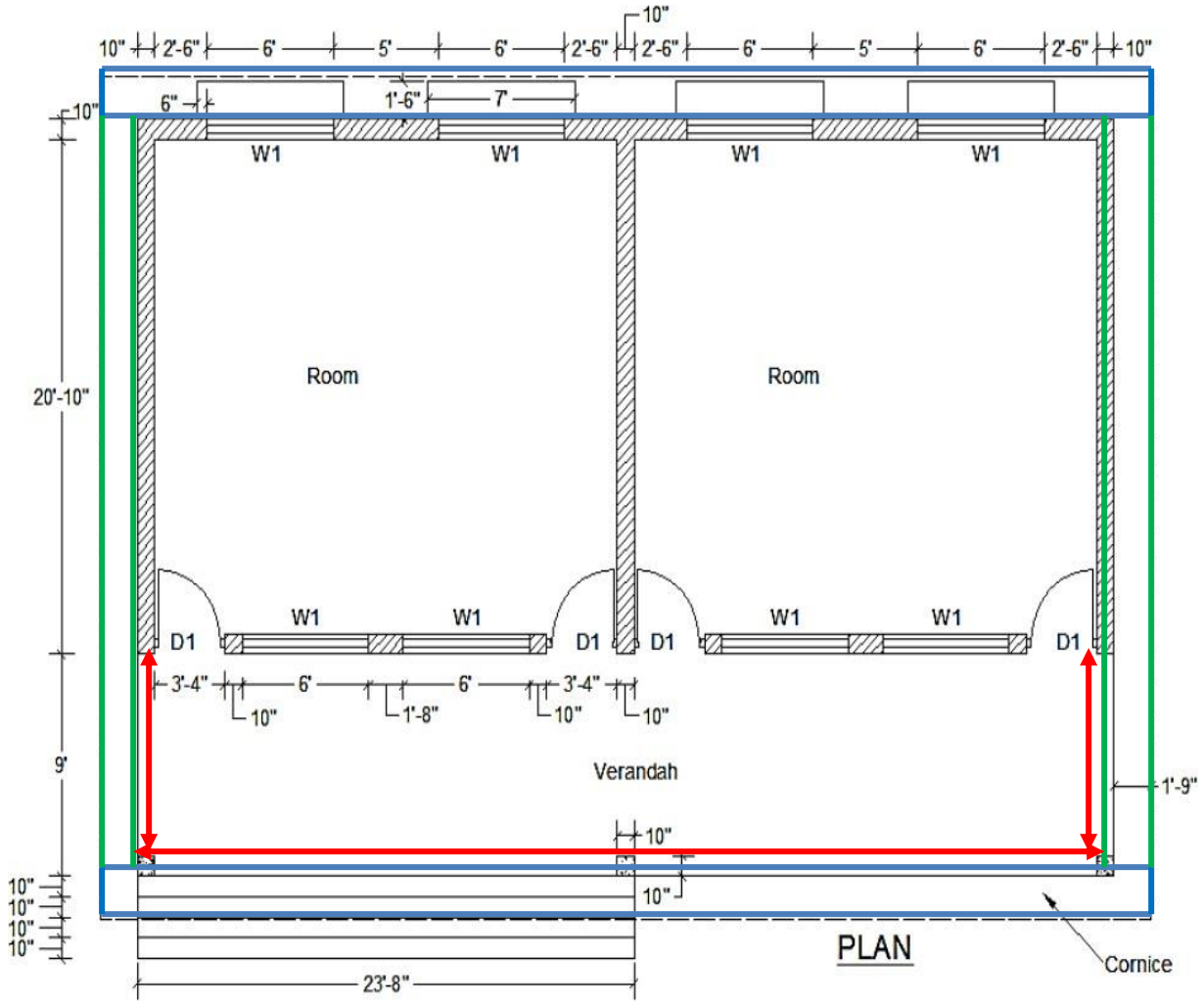


7) Cornice

i. **Side edge =**

(50’*2+34’2”*2)*0.5”*5”= 2.92 cft

20. Outside Plastering (Thickness-0.5”)



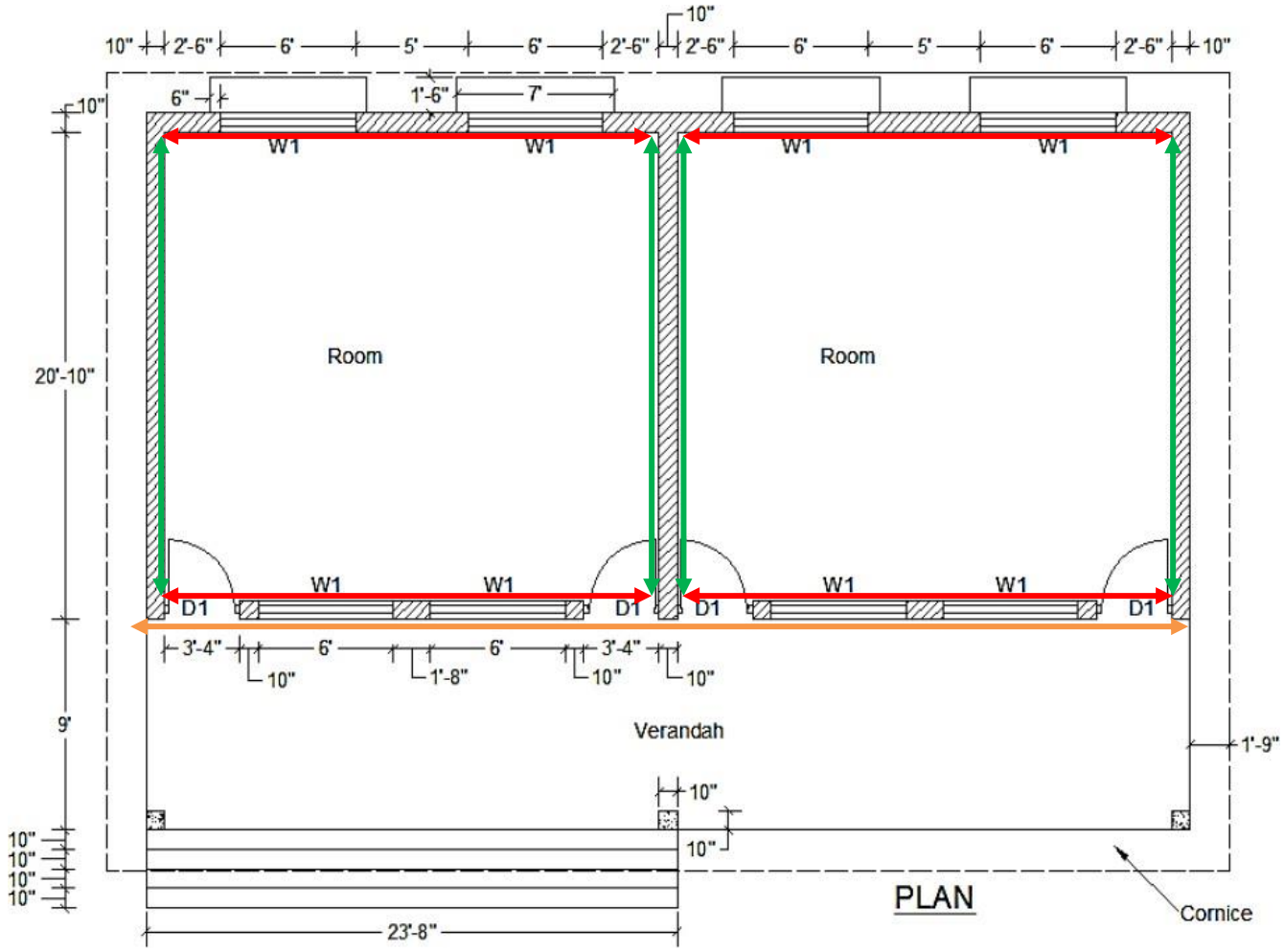
7) Cornice

i. Bottom edge = {(**46.5’+2* 1’9”)*1’9”**}*2*0.5” + {(**30’8”*1’9”)}***2*0.5” = 11.76 cft

8) Drop Wall

i. **Outside face =** 62’*2’6”*0.5” = 6.45 cft

21. Skirting (Thickness 0.75”)

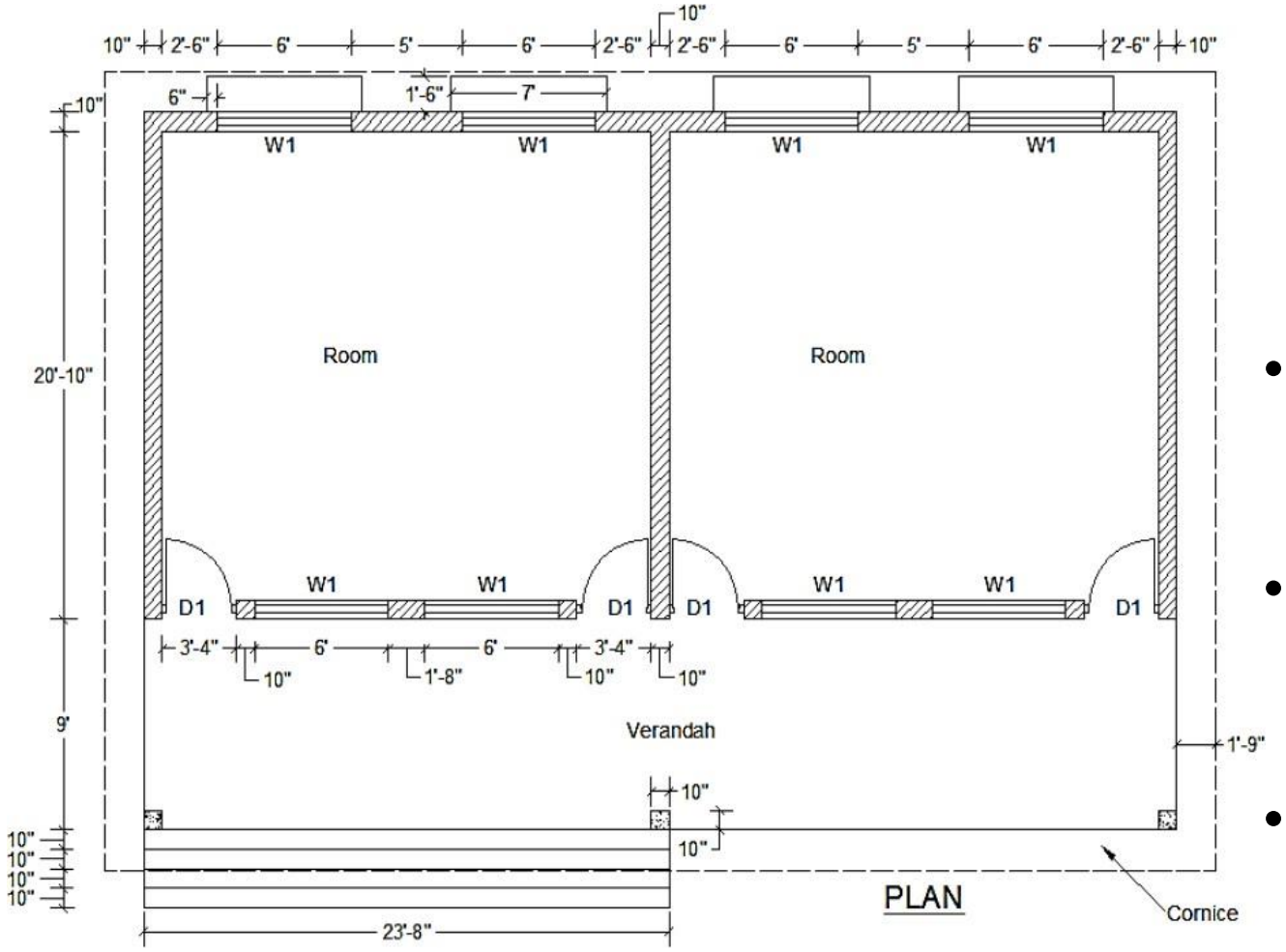


1. **Main room** = (**22’*4**+ **20’*4**)*10”*0.75”= 8.75 cft
   Deduction:
   i. Door = 4* 3’4”* 10”*0.75” = 0.69 cft

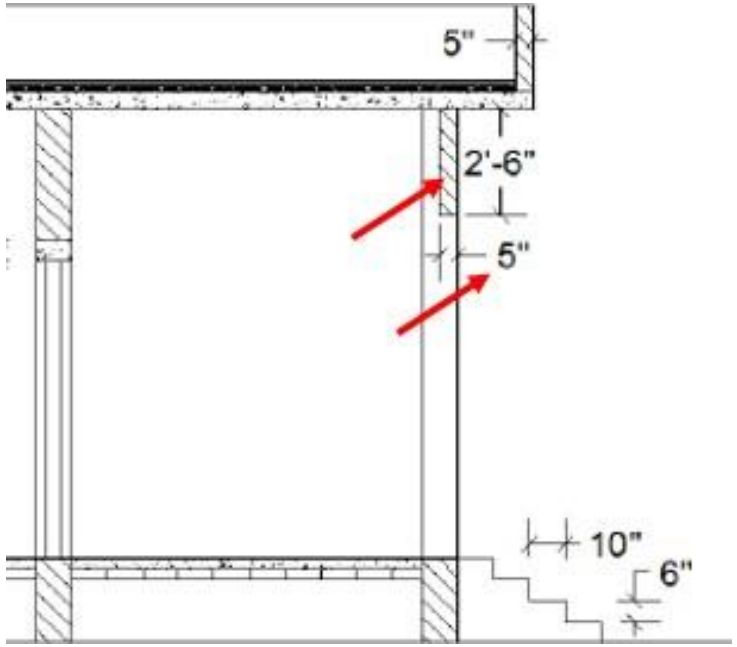
2. **Verandah** = **46’6”** * 10”* 0.75” = 2.42 cft
   Deduction:
   i. Door = 4* 3’4”* 10”*0.75” = 0.69 cft

3. **Column** = 3*(10”*4)*10”*0.75” = 0.52 cft

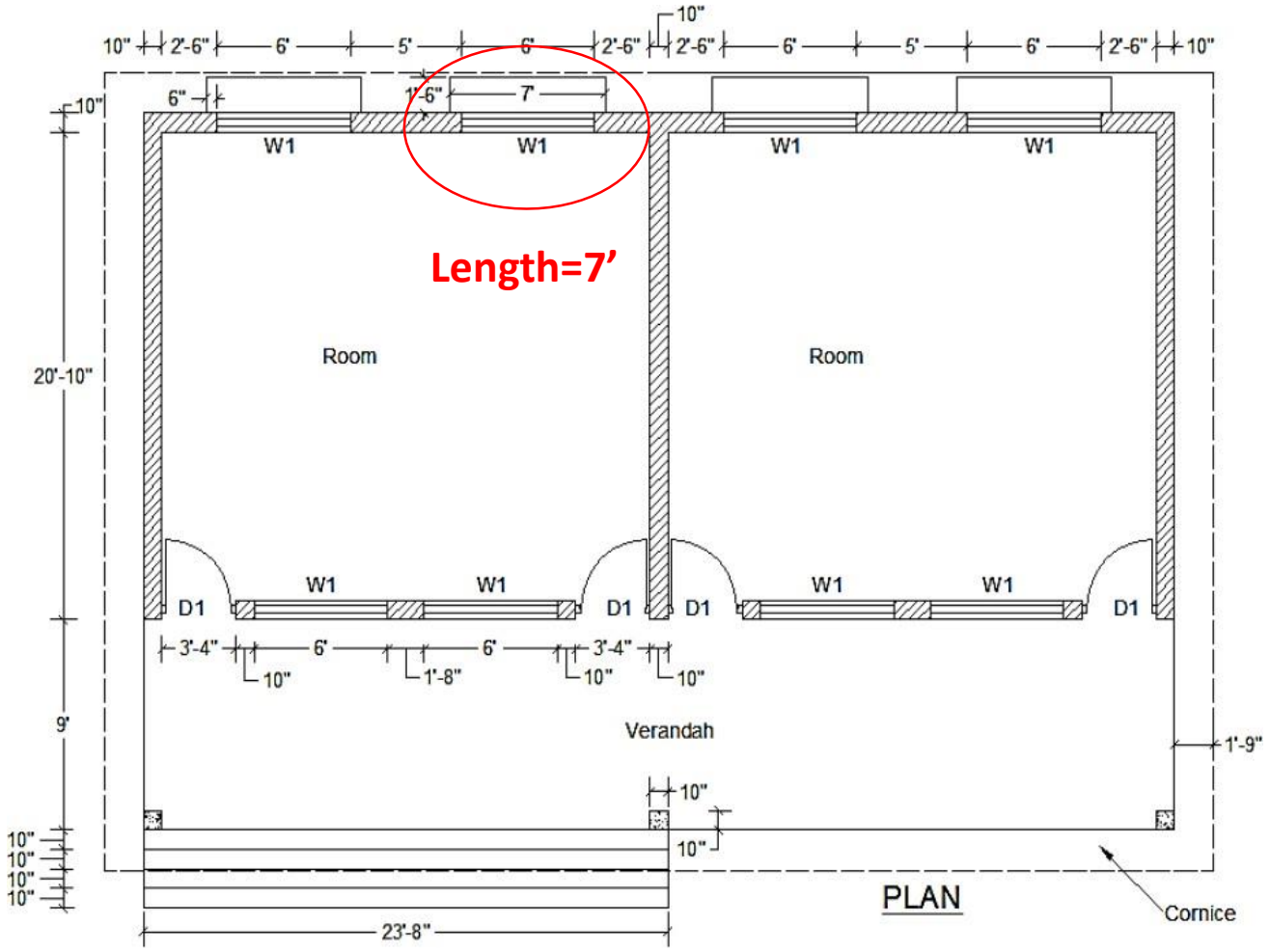
22. RCC in Drop walls



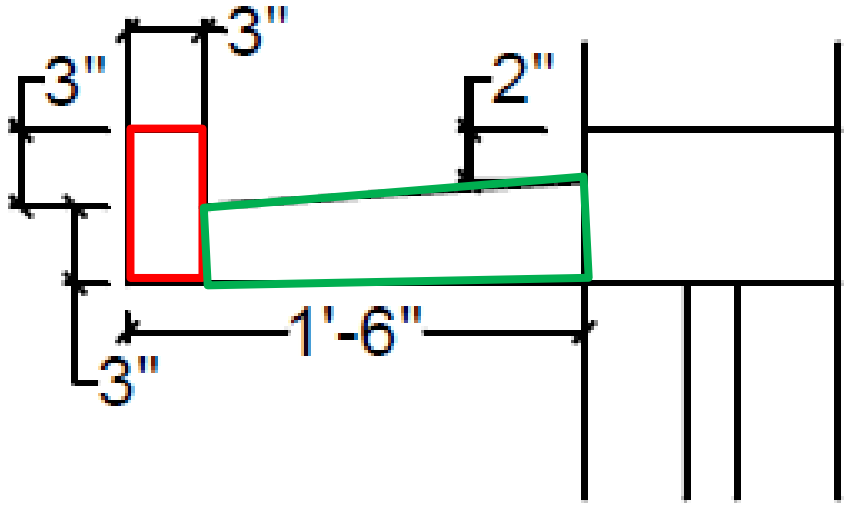
- Height= 2’6”
- Width= 5”
- Length= 62’



23. RCC in Sunshades



23. RCC in Sunshades



Volume= 4*((3”x6”) + (0.5x (3”+4”) x15”)) x7’